Documentation

Marshmallo Enterprise MSSP Infrastructure

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Project Overview & Concept

Project Name: Marshmallo Enterprise MSSP Infrastructure

Project Scope: MCSA

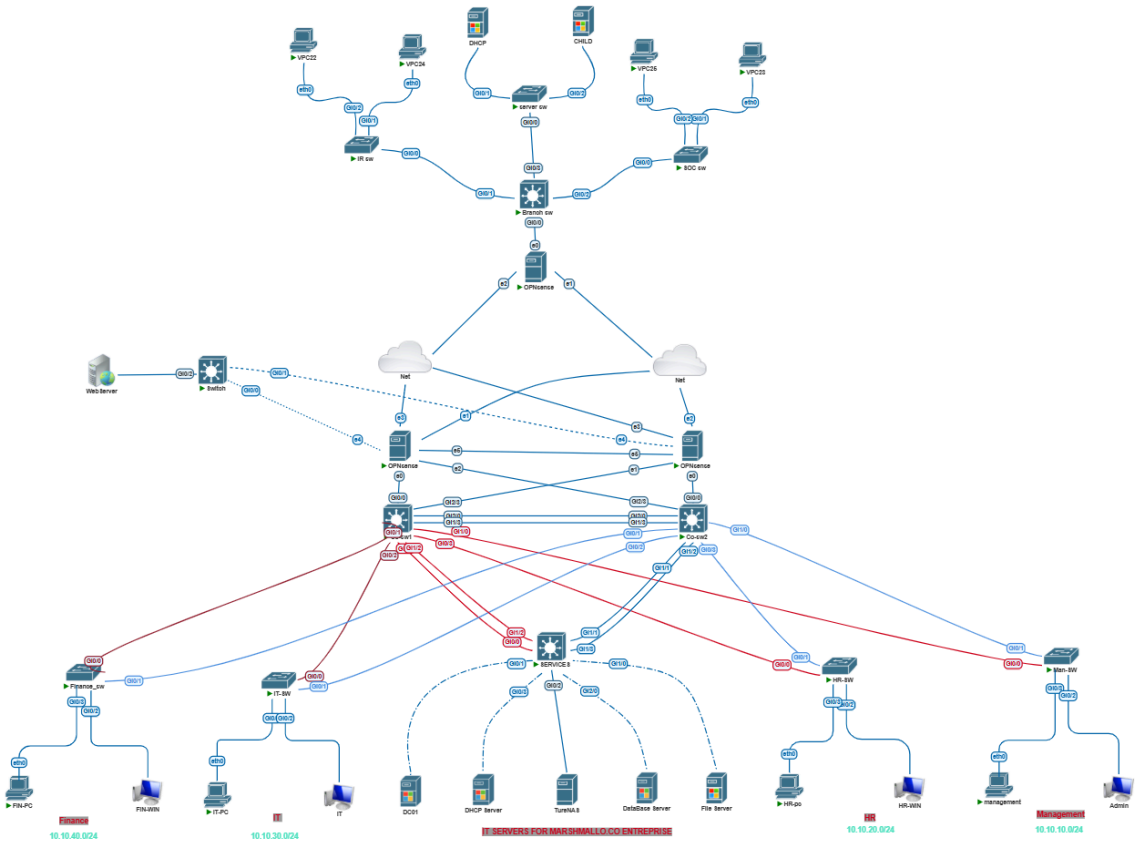
1. Executive Summary:

The Marshmallo project represents an integrated infrastructure for a company specialized in providing **MSSP (Managed Security Service Provider)** services. The core role of this enterprise is to assist other organizations in fully securing their networks, monitoring traffic, and updating security and protection policies centrally and efficiently. This project was practically designed and implemented to simulate a real-world enterprise environment featuring a Headquarters (HQ) and a dependent Branch, while applying modern standards of network and operating system administration based on **MCSA** principles.

2. Infrastructure Architecture:

The project was architecturally structured and divided into the following key components:

- **Headquarters (HQ):**
  - **Core Servers:** Houses primary identity and network management servers, including DC01 (Active Directory), a DHCP server, a centralized TrueNAS storage server, a Database Server, and a File Server.
  - **DMZ Zone:** Dedicated to securing external-facing services, featuring an isolated Web Server to protect the internal network from direct threats.
  - **Internal VLANs:** The internal network at the headquarters is segmented into organizational sectors, including HR, IT, Management, and Finance, ensuring traffic isolation and strict security policy enforcement.
- **Branch:**
  - **Branch Servers:** Features a Child domain controller integrated with the main domain, alongside a local DHCP server to manage branch devices.
  - **Branch VLANs:** Includes subnets dedicated to security operations departments, specifically IR (Incident Response) and SOC (Security Operations Center), for network monitoring and cyber incident response.


VPC22
VPC24
DHCP
CHILD
VPC25
VPC23
server sw
IR sw
SOC sw
Branch sw
OPNsense
Net
Net
Web Server
Switch
OPNsense
OPNsense
Co-sw1
Co-sw2
Finance_sw
IT-SW
SERVICES
HR-SW
Main-SW
FIN-PC
FIN-WIN
IT-PC
IT
DC01
DHCP Server
TuneNAS
DataBase Server
File Server
HR-po
HR-WIN
management
Admin
Finance
10.10.40.0/24
IT
10.10.30.0/24
IT SERVERS FOR MARSHMALLO.CO ENTREPRISE
HR
10.10.20.0/24
Management
10.10.10.0/24

Here is the complete network documentation for the Marshmallo Enterprise project

Marshmallo Enterprise Network Documentation

Part I: Headquarters (HQ) Infrastructure Documentation

1. VLAN Segmentation (VLANs 10 to 40)

- Details: Created independent virtual local area networks for the company's administrative layout (from VLAN 10 to VLAN 40), covering departments such as HR, IT, Management, and Finance.
- Significance & Value:
  - Achieves traffic isolation, enhances security, and prevents data overlap between different departments.
  - Reduces broadcast domain sizes and improves overall network performance.

2. Trunking Links

- Details: Configured trunk links between switches to carry traffic for all VLANs across a single physical connection efficiently.
- Significance & Value: Enables smooth multi-department traffic flow between core switches and minimizes structural cabling requirements.

3. Rapid Spanning Tree Protocol (RSTP)

- Details: Implemented RSTP to protect the network from loops and provide fast backup paths upon link failures.
- Significance & Value: Ensures high availability with ultra-fast network convergence times (within seconds) compared to traditional STP.

4. Switch Virtual Interfaces (SVI) on Core 1 & Core 2

- Details: Configured virtual interfaces on core switches to act as the default gateways for each VLAN, enabling inter-VLAN routing.
- Significance & Value: Empowers switches to operate as Layer 3 switches, routing data between different VLANs at wire speed.

5. Port Security Features (Access Ports, PortFast, & BPDU Guard)

- Details: Configured user-facing ports as Access, enabled PortFast to bypass connection delay states on startup, and activated BPDU Guard for security enhancement.

- Significance & Value: Protects the network from accidental connection of unauthorized switches and prevents topology loops at edge ports.

6 & 9. Virtual Router Redundancy Protocol (VRRP & VRRP Transit)

- Details: Combined Core 1 and Core 2 using VRRP to provide a single virtual gateway for clients, alongside configuring a transit path for routing traffic between them.
- Significance & Value: Delivers ultimate redundancy. If the primary switch fails, traffic seamlessly shifts to the backup switch without any noticeable interruption for users.

7. Network Testing and Client IP Assignment

- Details: Tested network connectivity by assigning IP addresses and executing ping tests between various departments.
- Significance & Value: Validates the physical layer integrity and routing logic prior to deploying live services.

8. Transit Routing between Core Sw 1 & Core Sw 2

- Details: Set up routing pathways and transit links between the two core switches to ensure smooth packet exchanges.
- Significance & Value: Guarantees stable and high-speed data flow across the core network layers.

10. OPNsense Firewall Configuration

- Details: Configured the OPNsense firewall as the primary perimeter defense and gateway for the enterprise network.
- Significance & Value: Handles traffic filtering, blocks cyber threats, and enforces advanced perimeter security policies.

11. Server Farm VLAN Isolation (VLAN 60)

- Details: Isolated all core enterprise servers within a dedicated virtual network (VLAN 60).
- Significance & Value: Protects critical servers by entirely separating them from general user traffic and simplifying security policy enforcement.

12. DHCP Relay Agent

- Details: Activated the DHCP Relay Agent feature on routing devices to forward DHCP requests coming from various VLANs to the centralized

DHCP server.
- Significance & Value: Allows a single centralized DHCP server to distribute IP configurations across multiple company departments and subnets without needing local servers everywhere.

13 & 14. Centralized Servers Deployment & Domain Joining (DC01, DHCP, TrueNAS, Database, File & Web Servers)

- Details: Deployed core servers including DC01 for Active Directory identity management, TrueNAS for centralized storage, Database, File, and an isolated DMZ Web Server, with all machines successfully joined to the domain.
- Significance & Value: Builds the enterprise core architecture to supply centralized authentication, secure file sharing, and robust service hosting.

Part II: Branch Infrastructure & Topology Analysis

Based on the provided branch network topology (Branch Topology), the architecture was designed to accurately support security operations departments through the following components:

1. Branch Connectivity & Topology Breakdown:

- Branch Gateway (OPNsense / Router): Connects via its primary interface (e0) to the Branch sw, serving as the central interconnection point and routing nexus for the branch.
- Branch Central Switch (Branch sw): Links to three primary directions to distribute workload and segment traffic:
  1. IR Path (Incident Response): Connects to the IR sw, which links simulation security endpoints (VPC22 and VPC24) dedicated to the incident response team via a dedicated VLAN.
  2. SOC Path (Security Operations Center): Connects to the SOC sw, linking security operation monitoring workstations (VPC23 and VPC25) for threat tracking and real-time oversight.
  3. Server Path (server sw): Interconnects the branch core servers (CHILD and DHCP) to guarantee local delivery of domain and IP addressing services.

2. Branch Servers Configuration (CHILD & DHCP):

- CHILD DC Server: Configured as a child domain controller tied to the main headquarters (HQ) domain, designed to provide low-latency authentication and minimize WAN link dependency by maintaining a local

replica of the Active Directory.

- Local Branch DHCP Server: Configured to dynamically assign IP addresses across branch subnets (such as IR and SOC zones), ensuring complete operational independence and rapid deployment for branch client devices (VPC22 through VPC25).

Inter-VLAN Routing, Perimeter Security & Web Server Access

1. Internal Routing via Core Switches (Core 1 & Core 2) vs. External Routing via OPNsense

- **Details:**
  - **Inter-VLAN Routing:** All internal routing between the corporate VLANs (VLAN 10 to VLAN 40) is handled locally at Layer 3 by the core switches (Core 1 and Core 2) utilizing SVIs (Switch Virtual Interfaces).
  - **Internet Routing (Gateway to Outside):** Traffic bound for external networks or the internet is directed outward through the perimeter firewall (OPNsense) rather than the core switches.
- **Significance & Value:**
  - Offloading internal inter-VLAN traffic to the core switches preserves high-speed wire-rate switching and reduces latency across departmental boundaries.
  - Delegating external routing to the firewall enforces strict inspection and perimeter security policies before traffic exits or enters the corporate enterprise.

2. Selection and Importance of OPNsense Firewall in the MSSP Project

- **Details:** OPNsense was chosen as the enterprise perimeter firewall and gateway for the **Marshmallo MSSP** infrastructure.
- **Significance & Value:**
  - Provides robust, open-source enterprise-grade security features including advanced stateful packet inspection, traffic shaping, and intrusion detection/prevention.
  - Acts as the central security enforcement point, perfectly aligning with the MSSP objective of monitoring, filtering, and securing client and corporate network perimeters against external threats.

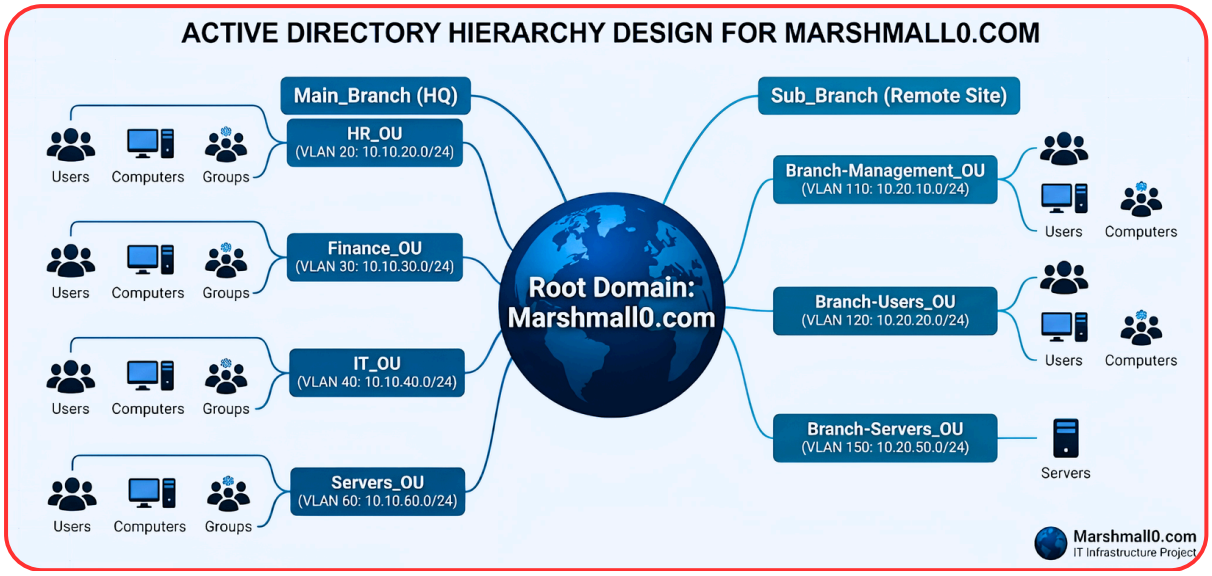
3. OPNsense & Web Server Security Configuration (DMZ Integration)

- **Details:** The perimeter firewall was configured alongside an isolated Web Server placed within the dedicated DMZ zone.
- **Significance & Value:**
  - Isolating the Web Server in a DMZ ensures that public-facing services are completely segregated from the sensitive internal corporate network (HR, Finance, IT, etc.).
  - Configured specific firewall rules and NAT (Network Address Translation) policies on OPNsense to control incoming and outgoing web traffic securely.

4. Enabling Internal VLAN Access to the DMZ Web Server

- **Details:** To allow internal clients residing across various corporate VLANs to access the isolated Web Server securely, specific configurations were implemented:
  - **Inter-Zone Firewall Rules:** Created dedicated firewall rules on OPNsense and core routing policies allowing authorized internal subnets to communicate with the Web Server IP/Port.
  - **Routing & NAT Policies:** Configured appropriate routing paths and port forwarding/NAT rules so that internal users can query and reach the web services without compromising internal security boundaries.
- **Significance & Value:** Guarantees that internal employees can seamlessly interact with enterprise web services while maintaining a heavily monitored and tightly controlled boundary enforced by the OPNsense firewall.

Active Directory (DC01) Infrastructure & OU Architecture



1. Theoretical Concept: Active Directory Structure & Design

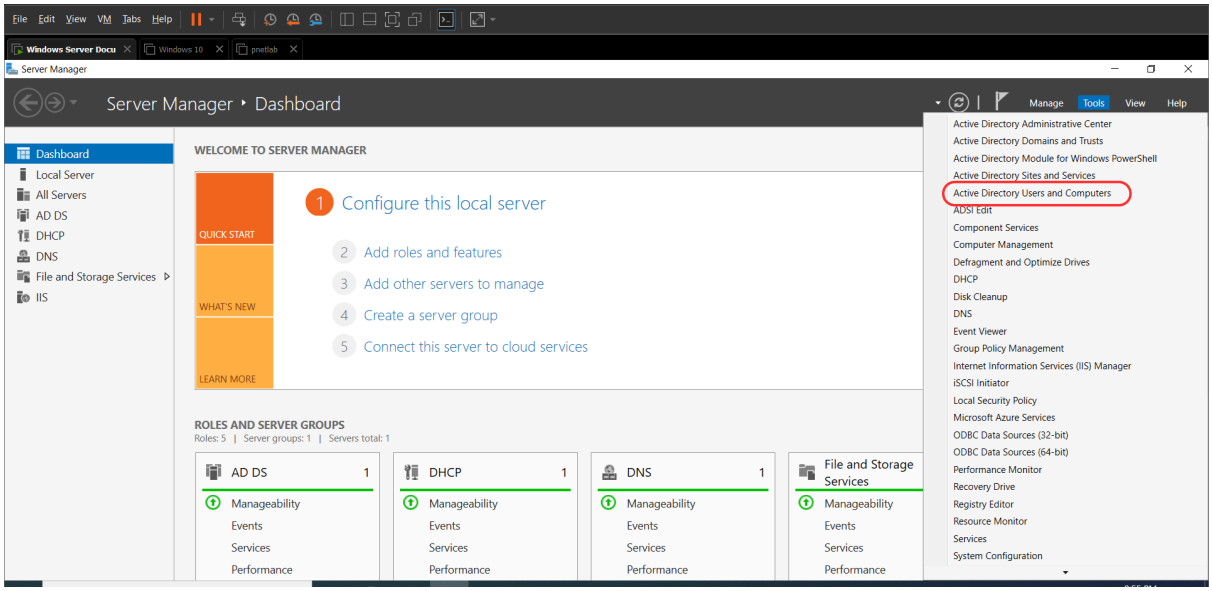
The **Active Directory** serves as the backbone for Identity and Access Management within the **Marshmallo Enterprise** infrastructure. The Organizational Unit (OU) hierarchy was designed based on enterprise-grade standards and MCSA principles to achieve the following:

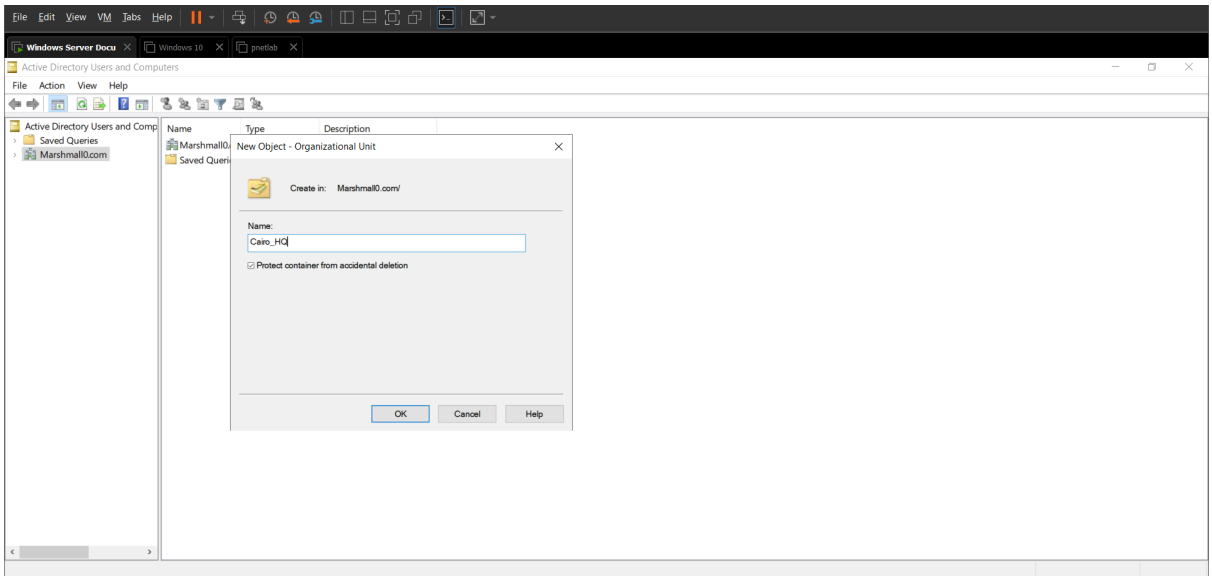
- **Geographical & Administrative Separation:** The root domain (Marshmall0.com) was divided into primary branches (Main_Branch) and secondary sites (Sub_Branch) to ensure seamless enforcement of centralized security policies (Group Policies - GPOs) based on physical location.
- **Departmental Segmentation:** Within the headquarters, independent OUs were established for each department (HR, Finance, IT, Management, and Servers) to align directly with the network VLAN architecture.
- **Precise Account & Device Management:** Inside each departmental OU, dedicated sub-containers were created, categorized into:
  - **Users:** For managing employee user accounts.
  - **Computers:** For organizing and tracking department-joined endpoints.
  - **Groups:** For establishing security groups to streamline mass permission assignments.

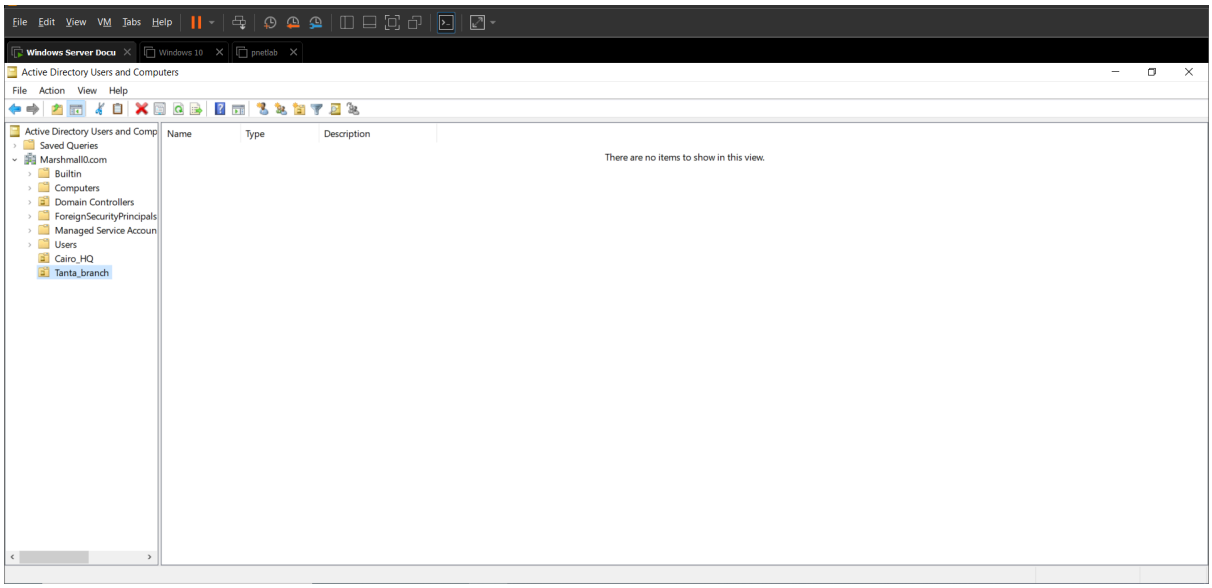
2. Practical Implementation: Step-by-Step Build in AD

Phase 1: Building the Parent OUs

1. Open the **Active Directory Users and Computers (ADUC)** console from the main server DC01.
2. Right-click the root domain name Marshmall0.com and select **New** -> **Organizational Unit**.
3. Name the first unit: Main_Branch.
4. Repeat the process to create the second unit named: Sub_Branch.

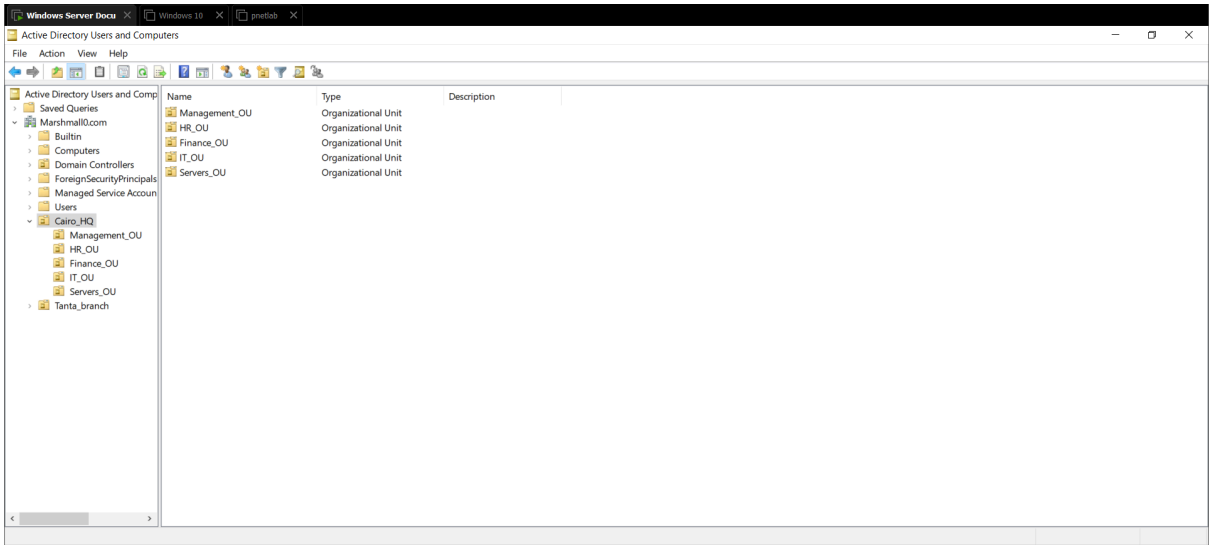


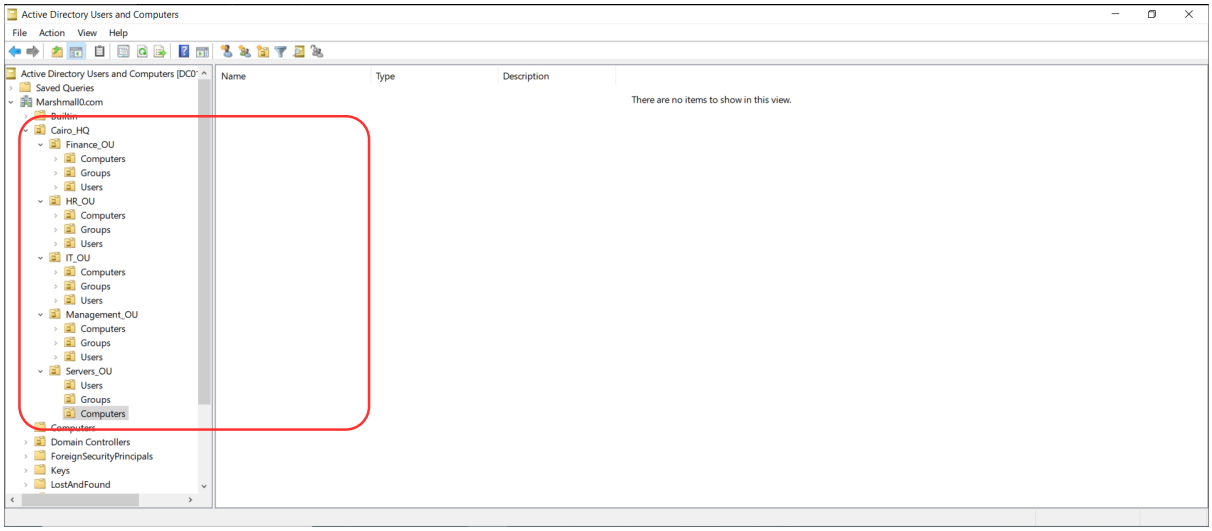




Phase 2: Building Main_Branch Departmental OUs

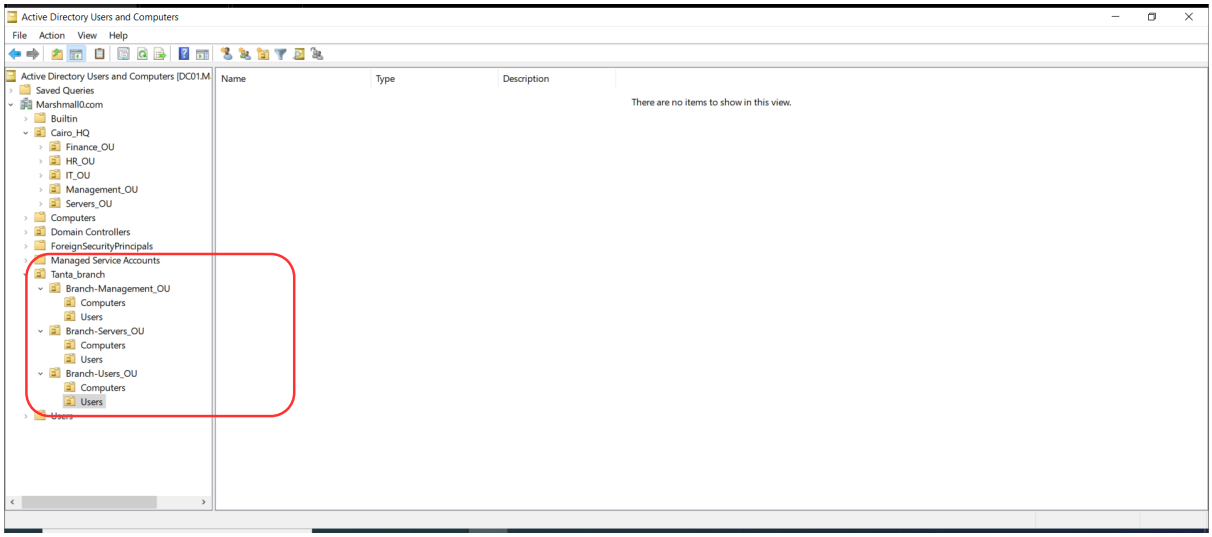
1. Right-click Main_Branch and select **New** -> **Organizational Unit**.
2. Create the department OUs sequentially:
    - Management_OU
    - HR_OU
    - Finance_OU
    - IT_OU
    - Servers_OU





Phase 3: Building Sub-Containers

- Inside each of the five department units (e.g., Management_OU):
    1. Right-click in the whitespace and select **New** -> **Organizational Unit**.
    2. Create the three sub-folders: Users, Computers, and Groups.
    3. Repeat this step meticulously across all departments within the Main_Branch.



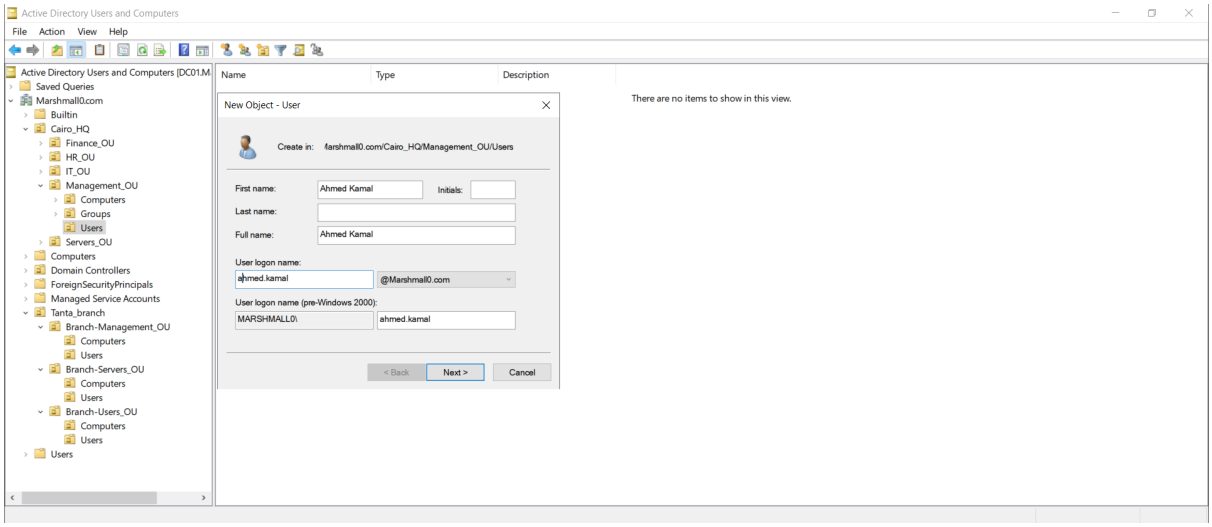
Phase 4: Building Sub_Branch OUs

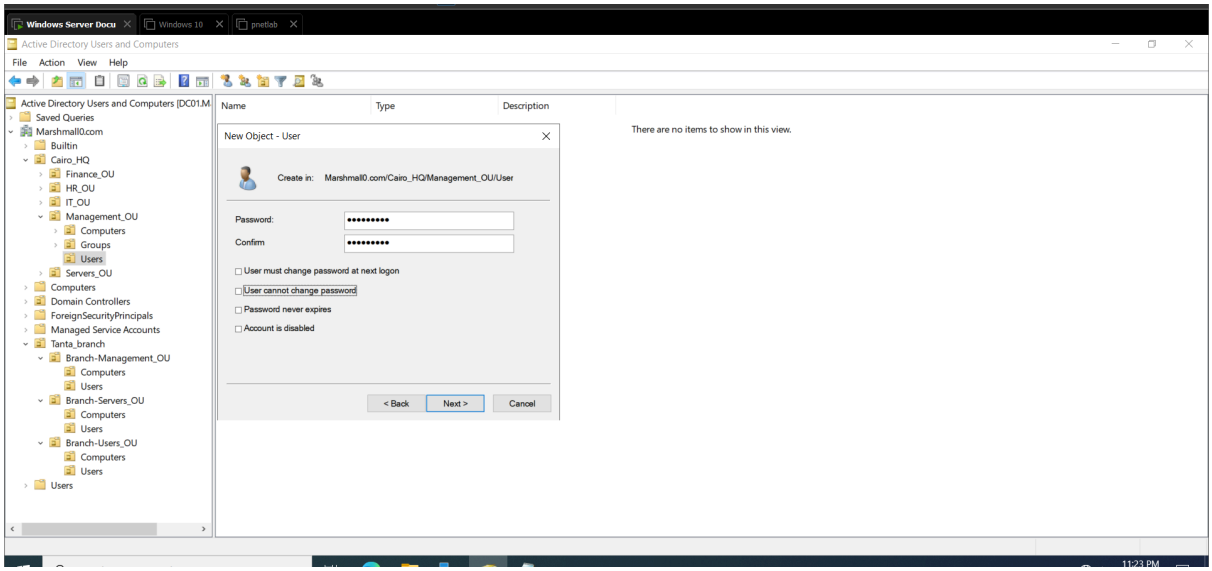
1. Navigate inside the Sub_Branch root OU.
2. Create the sub-site OUs:
   - Branch-Management_OU
   - Branch-Users_OU
   - Branch-Servers_OU
3. Inside each unit, create the respective sub-containers (Users and Computers).

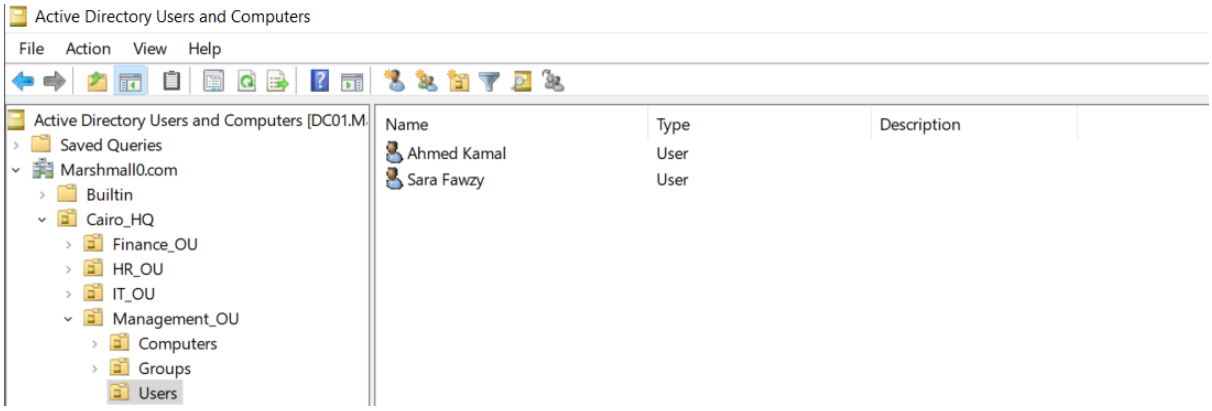
3. Enterprise Real-World Users & Groups Table

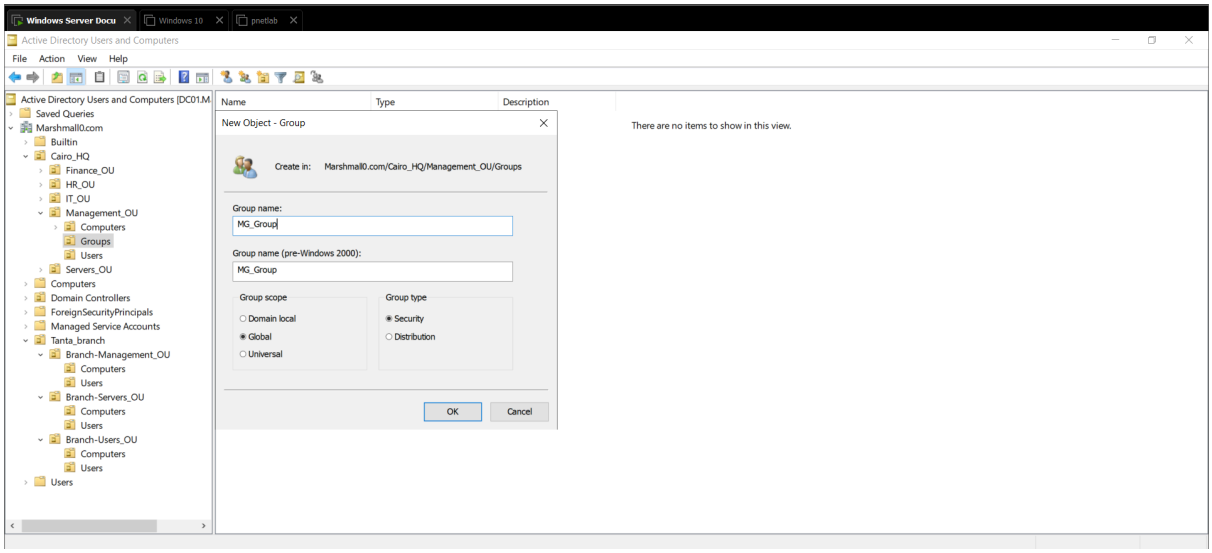
Users and security groups were mapped directly to the organizational structure to facilitate permissions and GPO enforcement:

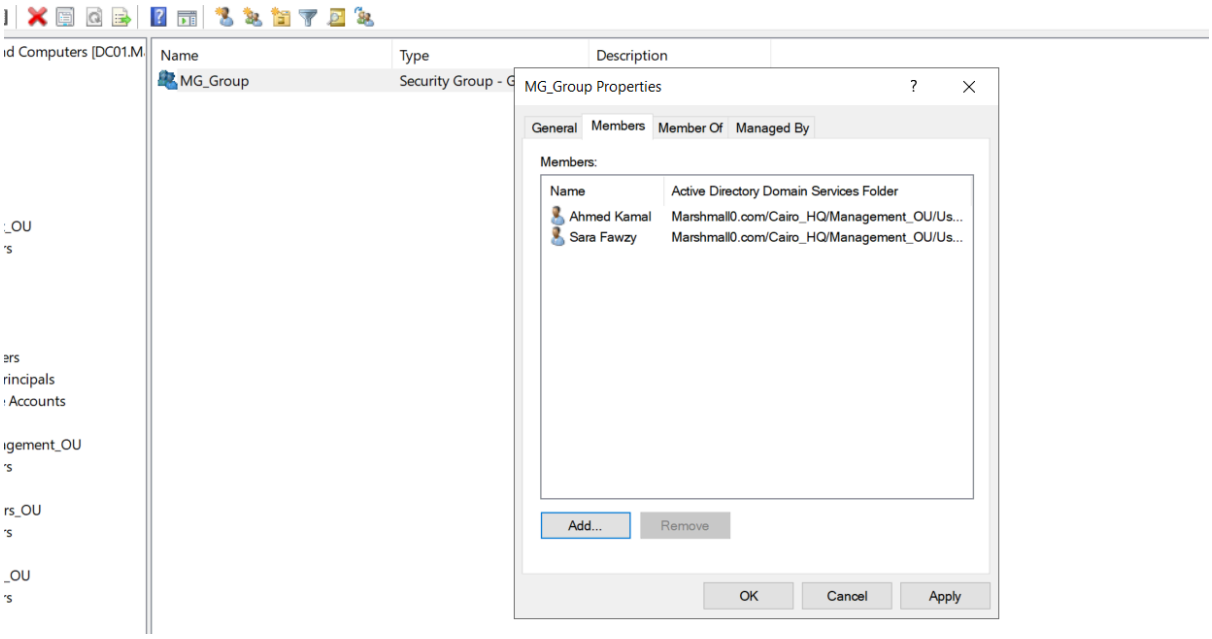
Main OU	Security Group Name	User 1	User 2
Management_OU	MG_Group	Ahmed.Kamal	Sara.Fawzy
HR_OU	HR_Group	Mona.Sayed	Tamer.Hassan
Finance_OU	FN_Group	Khaled.Mostafa	Nour.Ibrahim
IT_OU	IT_Group	Mahmoud.Ali	Yasmine.Galal


Active Directory Users and Computers
File Action View Help
Active Directory Users and Computers [DC01.M
Saved Queries
Marshmall0.com
Builtin
Cairo_HQ
Finance_OU
HR_OU
IT_OU
Management_OU
Computers
Groups
Users
Servers_OU
Computers
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accounts
Tanta_branch
Branch-Management_OU
Computers
Users
Branch-Servers_OU
Computers
Users
Branch-Users_OU
Computers
Users
Users
Name
Type
Description
There are no items to show in this view.
New Object - User
Create in: farshmall0.com/Cairo_HQ/Management_OU/Users
First name: Ahmed Kamal
Initials:
Last name:
Full name: Ahmed Kamal
User logon name:
ahmed.kamal
@Marshmall0.com
User logon name (pre-Windows 2000):
MARSHMALL0\
ahmed.kamal
< Back
Next >
Cancel


Windows Server Docu
Windows 10
pnetlab
Active Directory Users and Computers
File Action View Help
Active Directory Users and Computers [DC01.M
Saved Queries
Marshmall0.com
Builtin
Cairo_HQ
Finance_OU
HR_OU
IT_OU
Management_OU
Computers
Groups
Users
Servers_OU
Computers
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accounts
Tanta_branch
Branch-Management_OU
Computers
Users
Branch-Servers_OU
Computers
Users
Branch-Users_OU
Computers
Users
Users
Name
Type
Description
There are no items to show in this view.
New Object - User
Create in: Marshmall0.com/Cairo_HQ/Management_OU/User
Password:
Confirm
User must change password at next logon
User cannot change password
Password never expires
Account is disabled
< Back
Next >
Cancel


Active Directory Users and Computers
File Action View Help
Active Directory Users and Computers [DC01.M
Saved Queries
Marshmall0.com
Builtin
Cairo_HQ
Finance_OU
HR_OU
IT_OU
Management_OU
Computers
Groups
Users
Name
Type
Description
Ahmed Kamal
User
Sara Fawzy
User



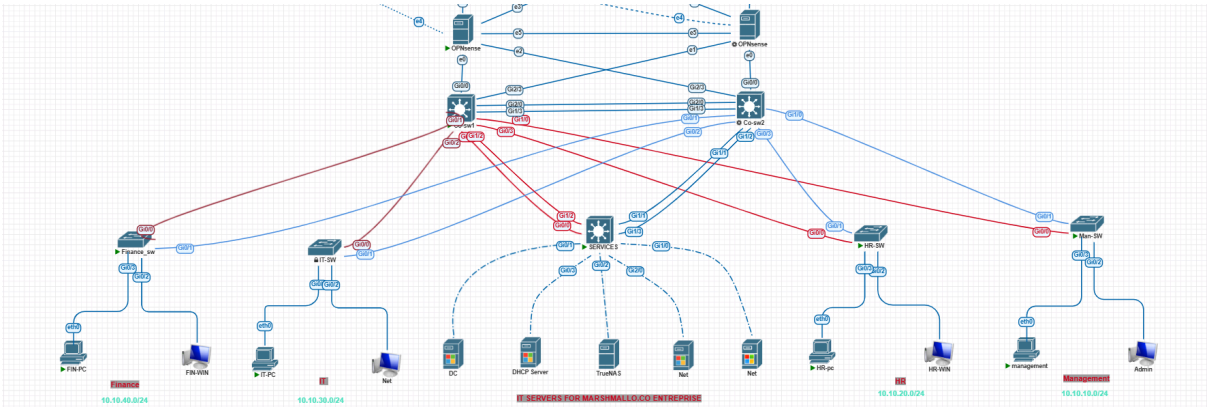


DHCP Server Infrastructure & VLAN Scope Management

1. Theoretical Concept: Multiple IP Addresses & DHCP Architecture

- **Multiple IP Addresses Concept:** A technique that allows assigning more than one IP address to a single Network Interface Card (NIC) or server, empowering the server to respond to multiple distinct network addresses simultaneously. It is utilized to segregate services on a single server (e.g., separating DNS and DHCP functionalities), improving administrative organization and security.
- **DHCP Scope Strategy per Department:** Dynamic Host Configuration Protocol (DHCP) scopes were meticulously engineered to automatically serve the distinct organizational units (Management, HR, IT, and Finance) across their corresponding VLAN subnets. A dedicated testing scope was also implemented to validate infrastructure configurations prior to full production deployment.

2. Practical Implementation: DHCP Server Deployment & Configuration



Phase A: Server Authorization & Core Configuration

1. **DHCP Authorization:**
   - Right-click the DHCP server name in the DHCP Management Console and select **Authorize**.
   - Verify that the status icon transitions to green, confirming active domain integration and authorization within Active Directory.
2. **Creating Scopes:**
   - Right-click **IPv4** and select **New Scope** to launch the configuration wizard.
   - Define the Scope Name (e.g., MG_Group - VLAN 10) and assign the designated subnet IP address range, subnet mask (e.g., 255.255.255.0), and prefix length (24).

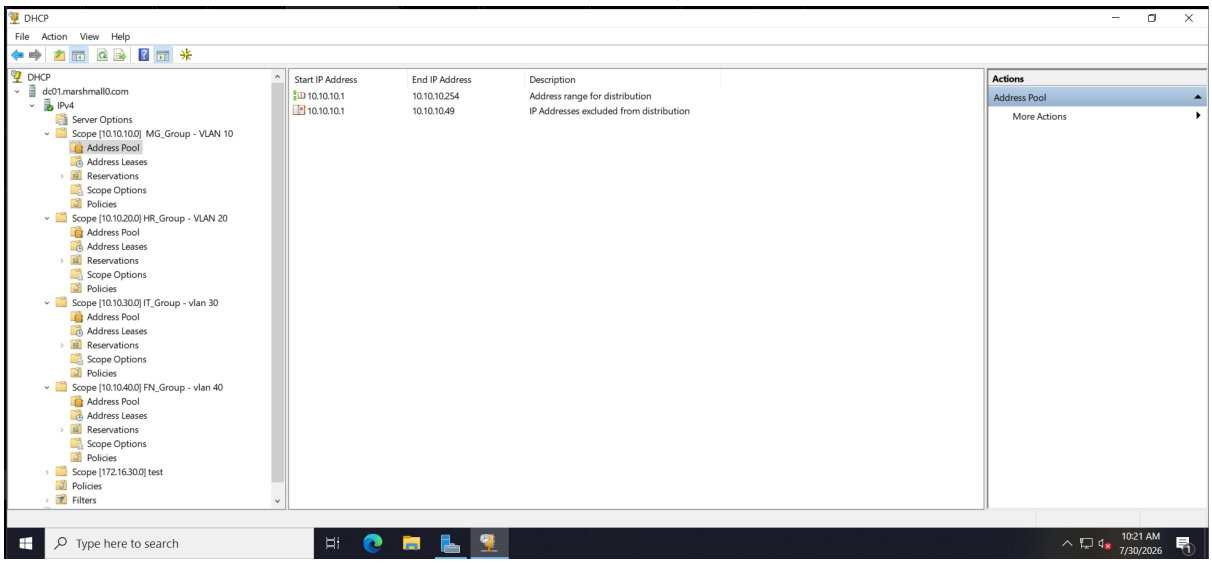
Phase B: Advanced Scope Parameters

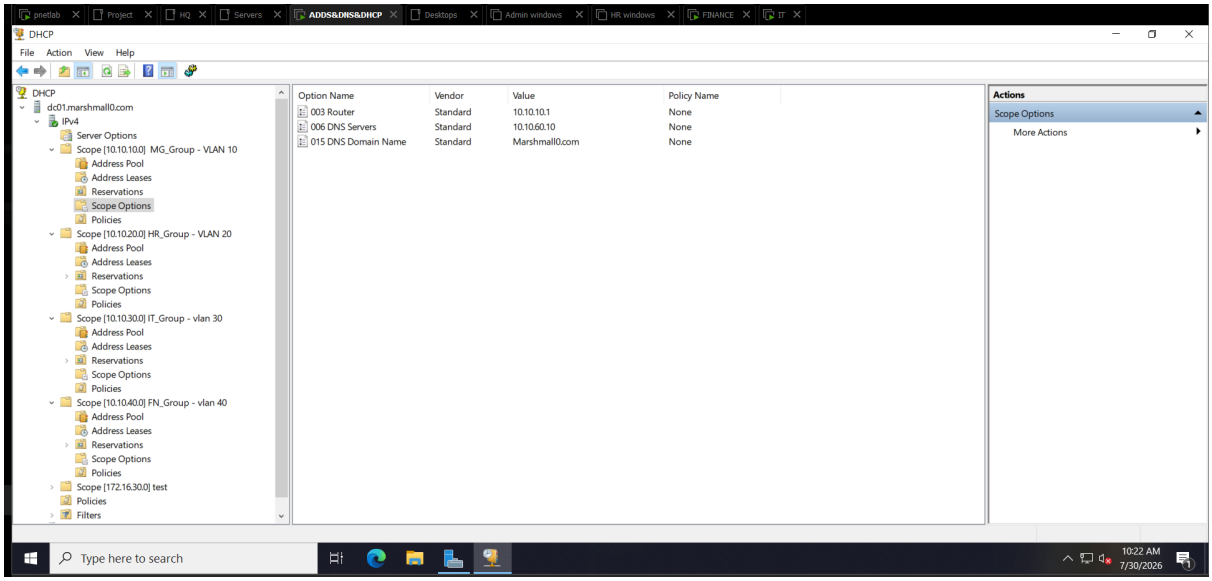
- **Exclusions:** Configured exclusion ranges (e.g., 10.10.10.1 to 10.10.10.49) to reserve static addresses for critical network gateways, core servers, and administrative equipment.
- **Lease Duration:** Retained the default enterprise standard of 8 days for dynamic IP allocations.
- **DHCP Options Configuration:**
  - **Option 003 (Router / Default Gateway):** Assigned the respective SVI gateway IP for each VLAN (e.g., 10.10.10.1).
  - **Option 006 (DNS Servers):** Provided the centralized DNS server IP address across all scopes to ensure proper name resolution.

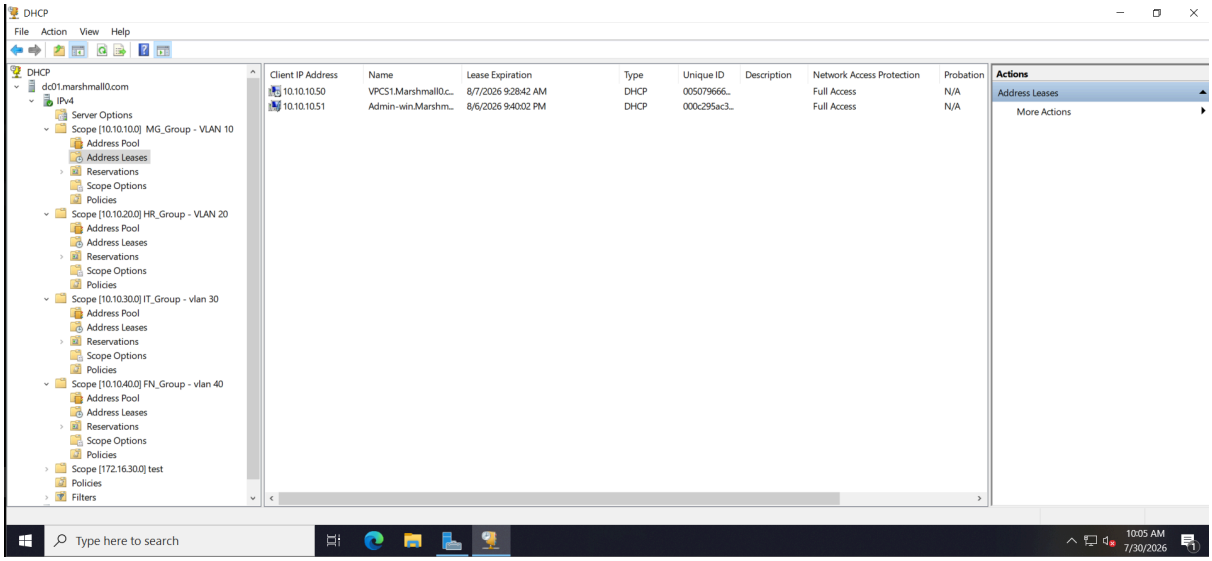
3. Network Planning & DHCP Scope Summary Table

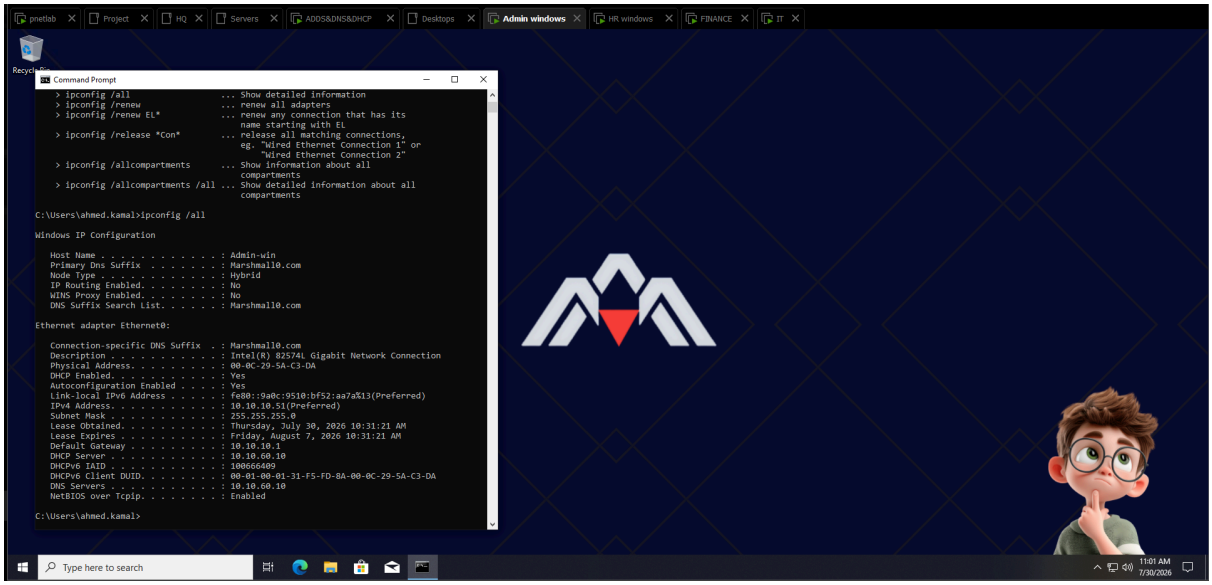
Scope / Department	Network ID	Pool Range	Gateway (Option 003)	DNS Server (Option 006)	Exclusion Range
VLAN 10 (MG_Group)	10.10.10.0/24	10.10.10.50 - 200	10.10.10.1	10.10.60.10	10.10.10.1 - 10.10.10.4

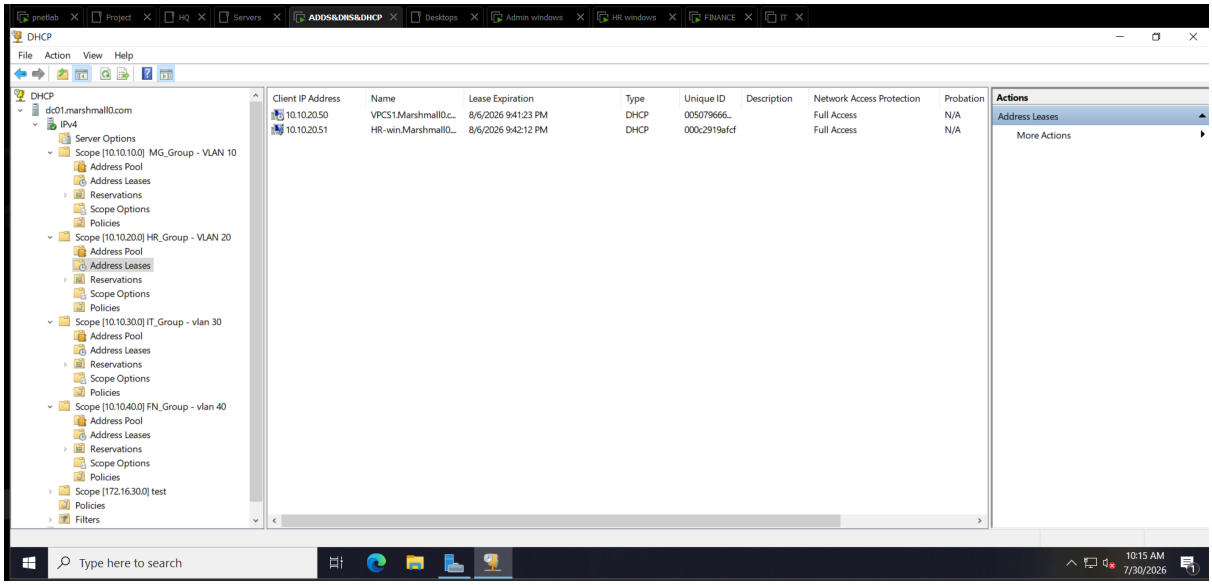
					9
VLAN 20 (HR_Group)	10.10.20.0/24	10.10.20.50 - 200	10.10.20.1	10.10.60.10	10.10.20.1 - 10.10.20.49
VLAN 30 (IT_Group)	10.10.30.0/24	10.10.30.50 - 200	10.10.30.1	10.10.60.10	10.10.30.1 - 10.10.30.49
VLAN 40 (FN_Group)	10.10.40.0/24	10.10.40.50 - 200	10.10.40.1	10.10.60.10	10.10.40.1 - 10.10.40.49

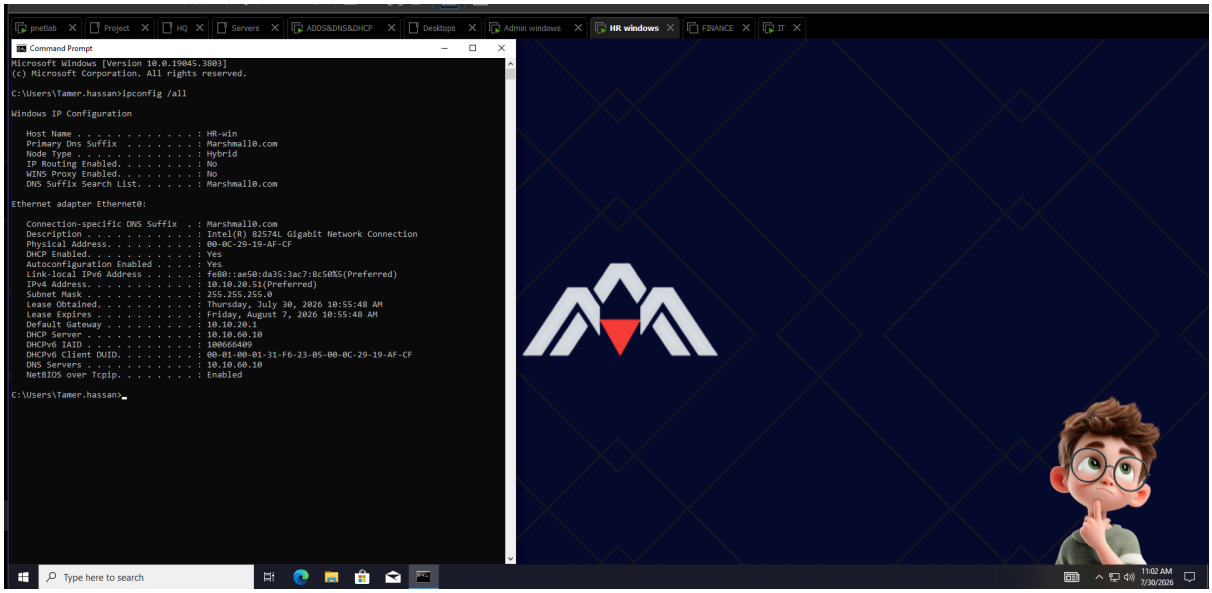


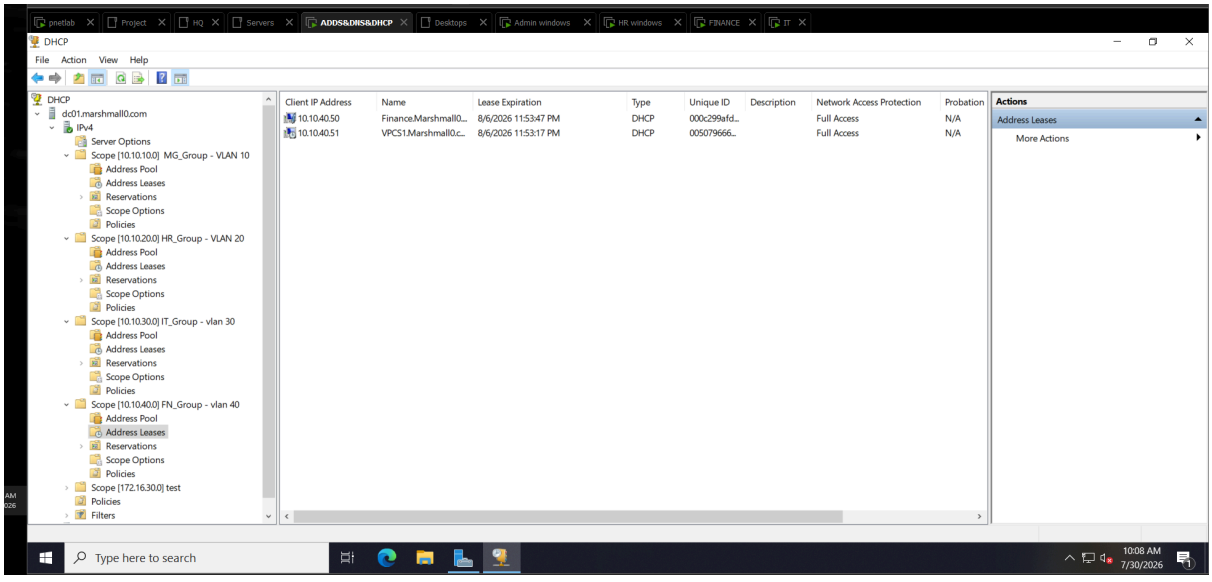

DHCP
File Action View Help
DHCP
dc01.marshmall0.com
IPv4
Server Options
Scope [10.10.10.0] MG_Group - VLAN 10
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.20.0] HR_Group - VLAN 20
Scope [10.10.30.0] IT_Group - vlan 30
Scope [10.10.40.0] FN_Group - vlan 40
Scope [172.16.30.0] test
Filters
Option Name
Vendor
Value
Policy Name
003 Router
Standard
10.10.10.1
None
006 DNS Servers
10.10.60.10
015 DNS Domain Name
Marshmall0.com
Actions
Scope Options
More Actions
Type here to search
10:22 AM
7/30/2026

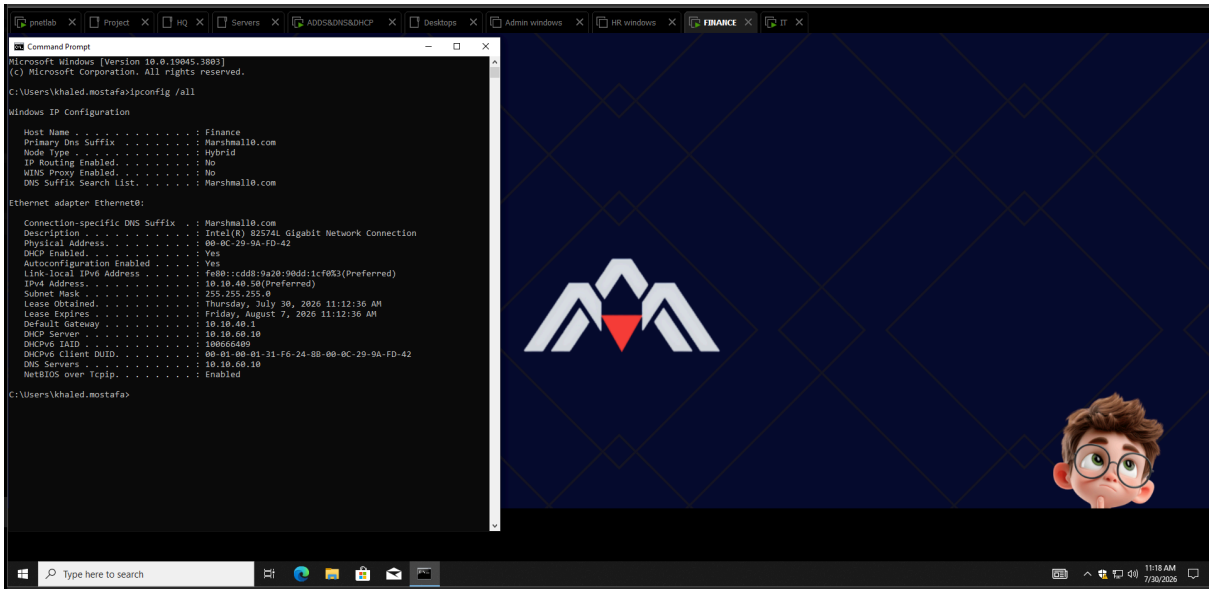

DHCP
File Action View Help
Client IP Address
Name
Lease Expiration
Type
Unique ID
Description
Network Access Protection
Probation
10.10.10.50
VPCS1.Marshmall0.c...
8/7/2026 9:28:42 AM
DHCP
00507966...
Full Access
N/A
10.10.10.51
Admin-win.Marshm...
8/6/2026 9:40:02 PM
00c295ac3...
Actions
Address Leases
More Actions
Type here to search
10:05 AM
7/30/2026

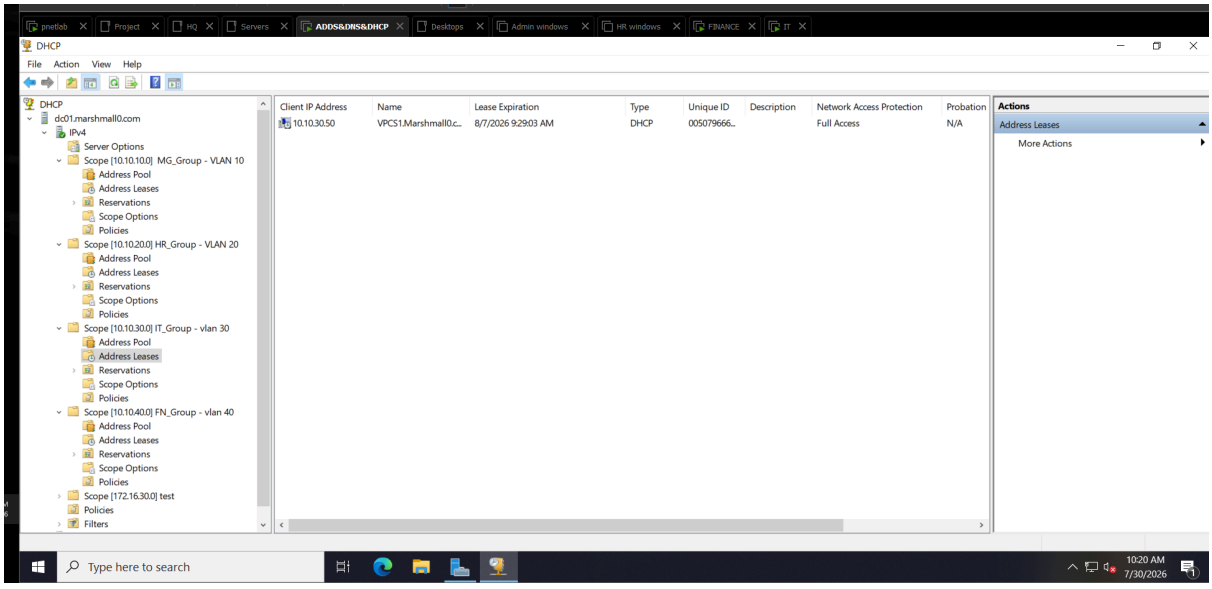

Command Prompt
> ipconfig /all ... Show detailed information
> ipconfig /renew ... renew all adapters
> ipconfig /renew EL* ... renew any connection that has its name starting with EL
> ipconfig /release *Con* ... release all matching connections, eg. "Wired Ethernet Connection 1" or "Wired Ethernet Connection 2"
> ipconfig /allcompartments ... Show information about all compartments
> ipconfig /allcompartments /all ... Show detailed information about all compartments
C:\Users\ahmed.kamal>ipconfig /all
Windows IP Configuration
Host Name . . . . . : Admin-win
Primary Dns Suffix . . . . . : Marshmall0.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : Marshmall0.com
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . : Marshmall0.com
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-5A-C3-DA
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::9a0c:9510:bf52:aa7a%13(Preferred)
IPv4 Address. . . . . : 10.10.10.51(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, July 30, 2026 10:31:21 AM
Lease Expires . . . . . : Friday, August 7, 2026 10:31:21 AM
Default Gateway . . . . . : 10.10.10.1
DHCP Server . . . . . : 10.10.60.10
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-F5-FD-8A-00-0C-29-5A-C3-DA
DNS Servers . . . . . : 10.10.60.10
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\ahmed.kamal>
Type here to search
11:01 AM
7/30/2026

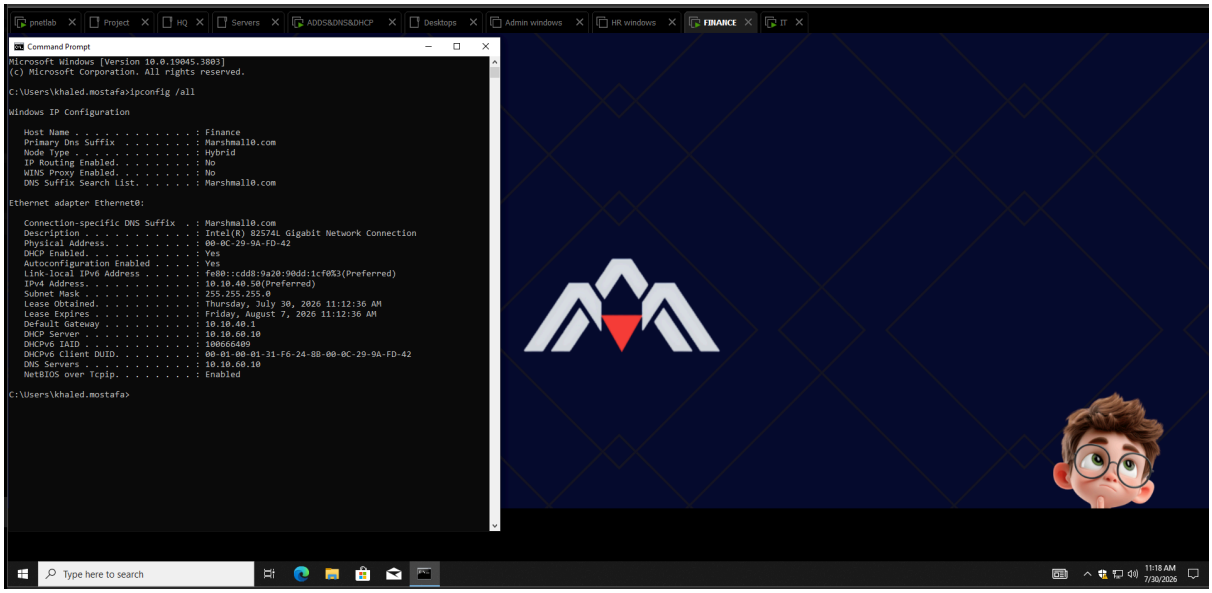

DHCP
File Action View Help
DHCP
dc01.marshmall0.com
IPv4
Server Options
Scope [10.10.10.0] MG_Group - VLAN 10
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.20.0] HR_Group - VLAN 20
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.30.0] IT_Group - vlan 30
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.40.0] FN_Group - vlan 40
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [172.16.30.0] test
Policies
Filters
Client IP Address
Name
Lease Expiration
Type
Unique ID
Description
Network Access Protection
Probation
10.10.20.50
VPCS1.Marshmall0.c...
8/6/2026 9:41:23 PM
DHCP
005079666...
Full Access
N/A
10.10.20.51
HR-win.Marshmall0...
8/6/2026 9:42:12 PM
DHCP
000c2919afcf
Full Access
N/A
Actions
Address Leases
More Actions
Type here to search
10:15 AM
7/30/2026


Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Tamer.hassan>ipconfig /all
Windows IP Configuration
Host Name . . . . . : HR-win
Primary Dns Suffix . . . . . : Marshmall0.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : Marshmall0.com
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . : Marshmall0.com
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-19-AF-CF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::ae50:da35:3ac7:8c50%5(Preferred)
IPv4 Address. . . . . : 10.10.20.51(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, July 30, 2026 10:55:48 AM
Lease Expires . . . . . : Friday, August 7, 2026 10:55:48 AM
Default Gateway . . . . . : 10.10.20.1
DHCP Server . . . . . : 10.10.60.10
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-F6-23-05-00-0C-29-19-AF-CF
DNS Servers . . . . . : 10.10.60.10
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\Tamer.hassan>
Type here to search
11:02 AM
7/30/2026


DHCP
File Action View Help
DHCP
dc01.marshmall0.com
IPv4
Server Options
Scope [10.10.10.0] MG_Group - VLAN 10
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.20.0] HR_Group - VLAN 20
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.30.0] IT_Group - vlan 30
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.40.0] FN_Group - vlan 40
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [172.16.30.0] test
Policies
Filters
Client IP Address
Name
Lease Expiration
Type
Unique ID
Description
Network Access Protection
Probation
10.10.40.50
Finance.Marshmall0...
8/6/2026 11:53:47 PM
DHCP
000c299afd...
Full Access
N/A
10.10.40.51
VPCS1.Marshmall0.c...
8/6/2026 11:53:17 PM
DHCP
005079666...
Full Access
N/A
Actions
Address Leases
More Actions
Type here to search
10:08 AM
7/30/2026


pnetlab
Project
HQ
Servers
ADDS&DNS&DHCP
Desktops
Admin windows
HR windows
FINANCE
IT
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:\Users\khaled.mostafa>ipconfig /all
Windows IP Configuration
Host Name . . . . . : Finance
Primary Dns Suffix . . . . . : Marshmall0.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : Marshmall0.com
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . : Marshmall0.com
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-9A-FD-42
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::ccd8:9a20:90dd:1cf0%3(Preferred)
IPv4 Address. . . . . : 10.10.40.50(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, July 30, 2026 11:12:36 AM
Lease Expires . . . . . : Friday, August 7, 2026 11:12:36 AM
Default Gateway . . . . . : 10.10.40.1
DHCP Server . . . . . : 10.10.60.10
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-F6-24-8B-00-0C-29-9A-FD-42
DNS Servers . . . . . : 10.10.60.10
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\khaled.mostafa>
Type here to search
11:18 AM
7/30/2026


pnetlab
Project
HQ
Servers
ADDS&DNS&DHCP
Desktops
Admin windows
HR windows
FINANCE
IT
DHCP
File Action View Help
DHCP
dc01.marshmall0.com
IPv4
Server Options
Scope [10.10.10.0] MG_Group - VLAN 10
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.20.0] HR_Group - VLAN 20
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.30.0] IT_Group - vlan 30
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [10.10.40.0] FN_Group - vlan 40
Address Pool
Address Leases
Reservations
Scope Options
Policies
Scope [172.16.30.0] test
Policies
Filters
Client IP Address
Name
Lease Expiration
Type
Unique ID
Description
Network Access Protection
Probation
10.10.30.50
VPCS1.Marshmall0.c...
8/7/2026 9:29:03 AM
DHCP
005079666...
Full Access
N/A
Actions
Address Leases
More Actions
Type here to search
10:20 AM
7/30/2026


pnetlab
Project
HQ
Servers
ADDS&DNS&DHCP
Desktops
Admin windows
HR windows
FINANCE
IT
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:\Users\khaled.mostafa>ipconfig /all
Windows IP Configuration
Host Name . . . . . : Finance
Primary Dns Suffix . . . . . : Marshmall0.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : Marshmall0.com
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . : Marshmall0.com
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-9A-FD-42
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::ccd8:9a20:90dd:1cf0%3(Preferred)
IPv4 Address. . . . . : 10.10.40.50(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, July 30, 2026 11:12:36 AM
Lease Expires . . . . . : Friday, August 7, 2026 11:12:36 AM
Default Gateway . . . . . : 10.10.40.1
DHCP Server . . . . . : 10.10.60.10
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-F6-24-8B-00-0C-29-9A-FD-42
DNS Servers . . . . . : 10.10.60.10
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\khaled.mostafa>
Type here to search
11:18 AM
7/30/2026

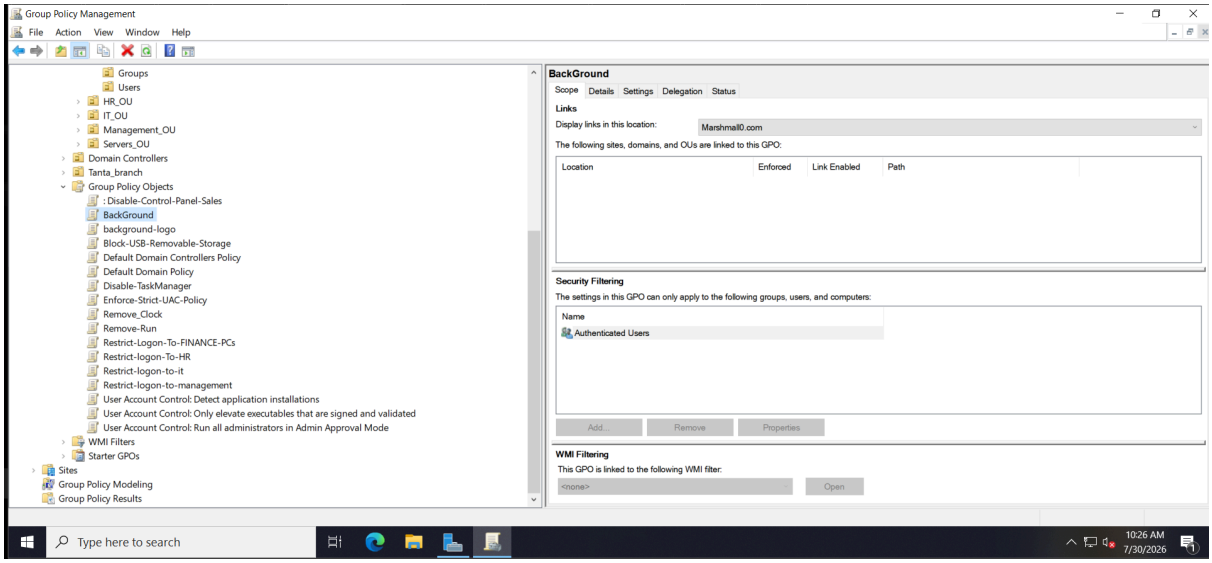
4. Advanced Enterprise DHCP Features & Security Controls

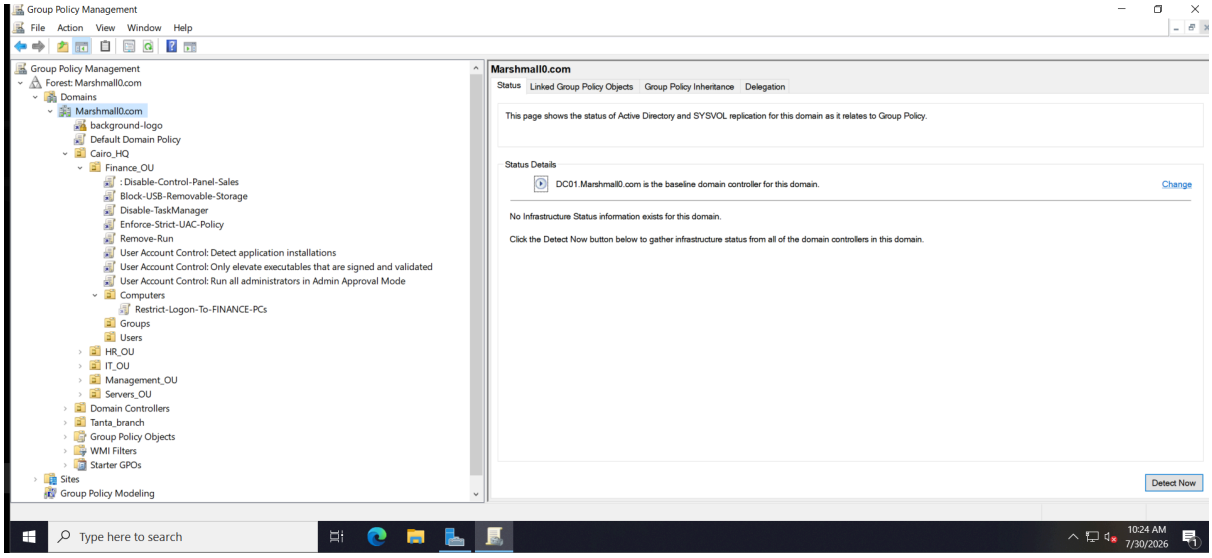
- **Superscopes:**
  - *Significance:* Enables grouping multiple logical scopes under a single administrative umbrella. Utilized when an existing subnet exhaustion occurs or when merging multiple subnets on the same physical broadcast domain.
- **Server Options vs. Scope Options:**
  - *Server Options:* Applied globally across all scopes (e.g., enterprise-wide DNS and domain names).
  - *Scope Options:* Tailored specifically to individual departmental subnets (such as unique gateway configurations), taking precedence over global options when overlapping.
- **DHCP Filters (Allow & Deny Lists):**
  - *Allow Filters:* Implemented for high-security environments to restrict IP lease allocations strictly to pre-approved MAC addresses, blocking unauthorized devices automatically.
  - *Deny Filters:* Deployed to blacklist specific compromised or non-compliant network endpoints while allowing standard corporate traffic to flow unhindered.

Group Policies (GPOs) & Automated Account Lockout Alert System

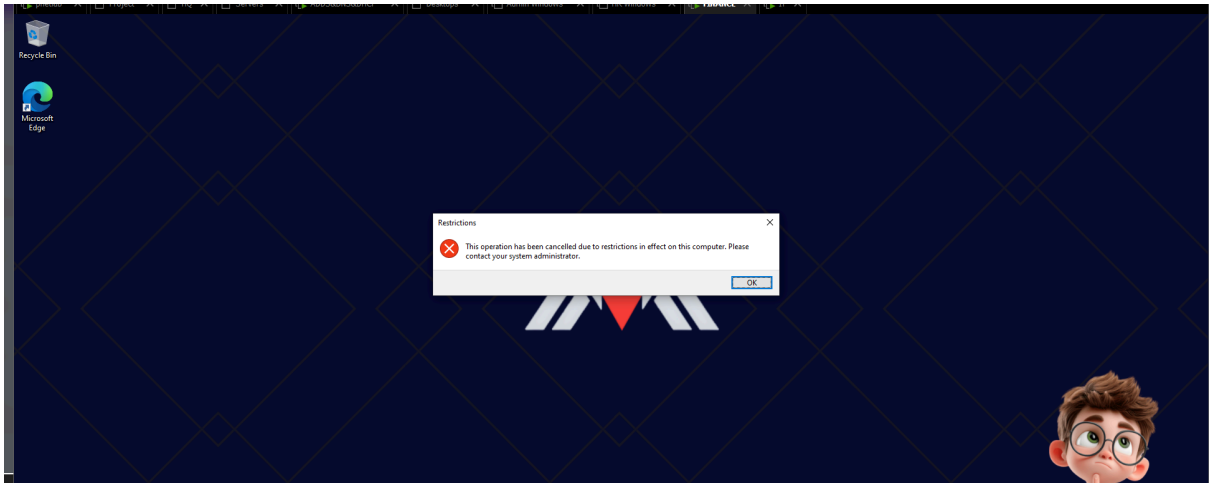
1. Departmental Group Policy Objects (GPOs)

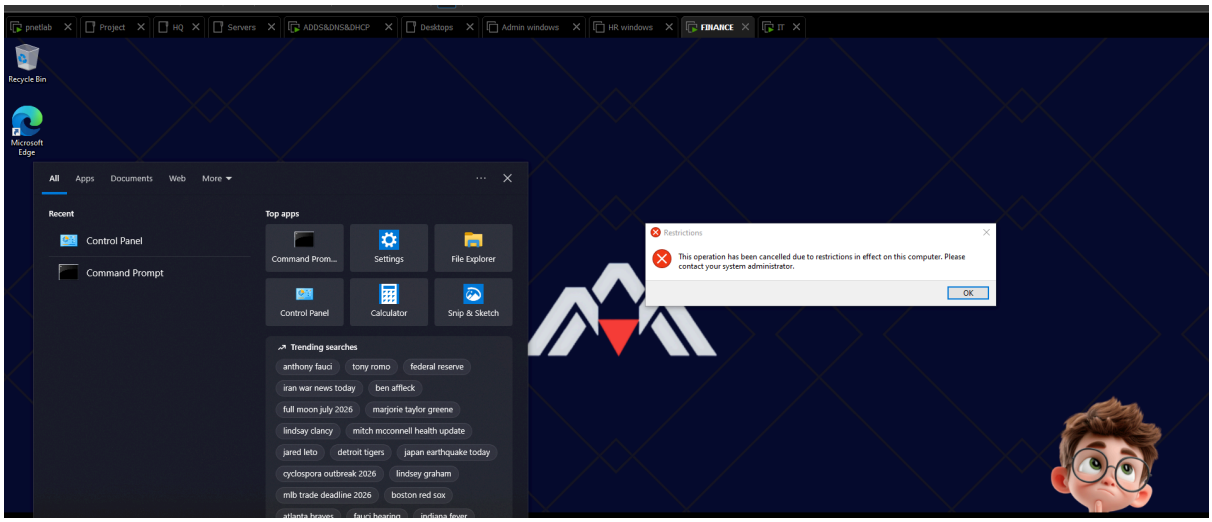
- **Overview:** Tailored Group Policies were deployed across the active departmental OUs (Management, HR, IT, and Finance) to enforce centralized security baselines, access restrictions, and customized desktop environments across client endpoints.
- **Significance & Value:** Ensures strict compliance with enterprise security requirements, automates configuration management, and restricts unauthorized user modifications across sensitive departmental subnets.

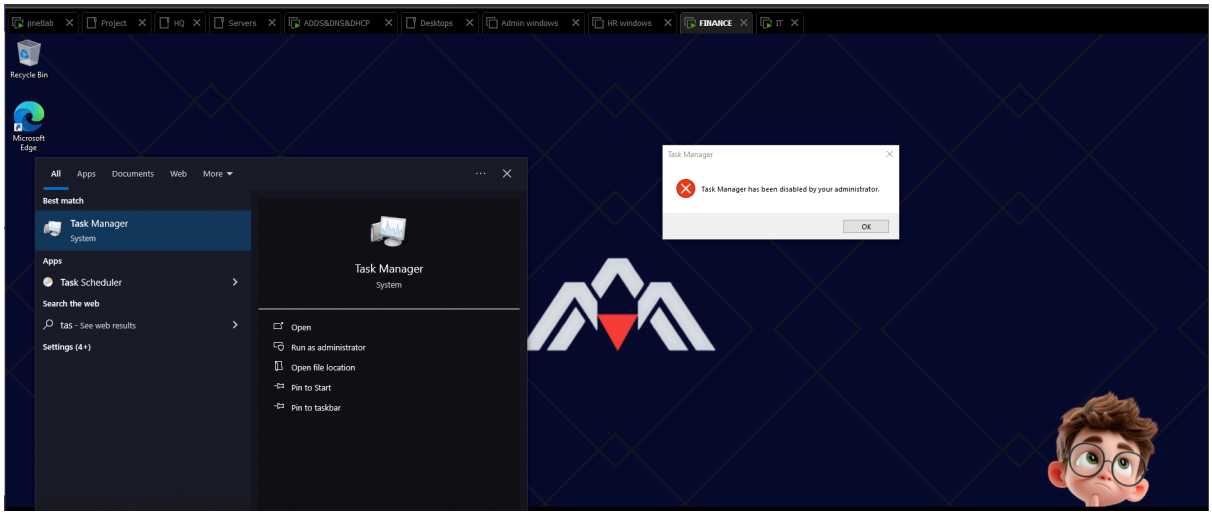


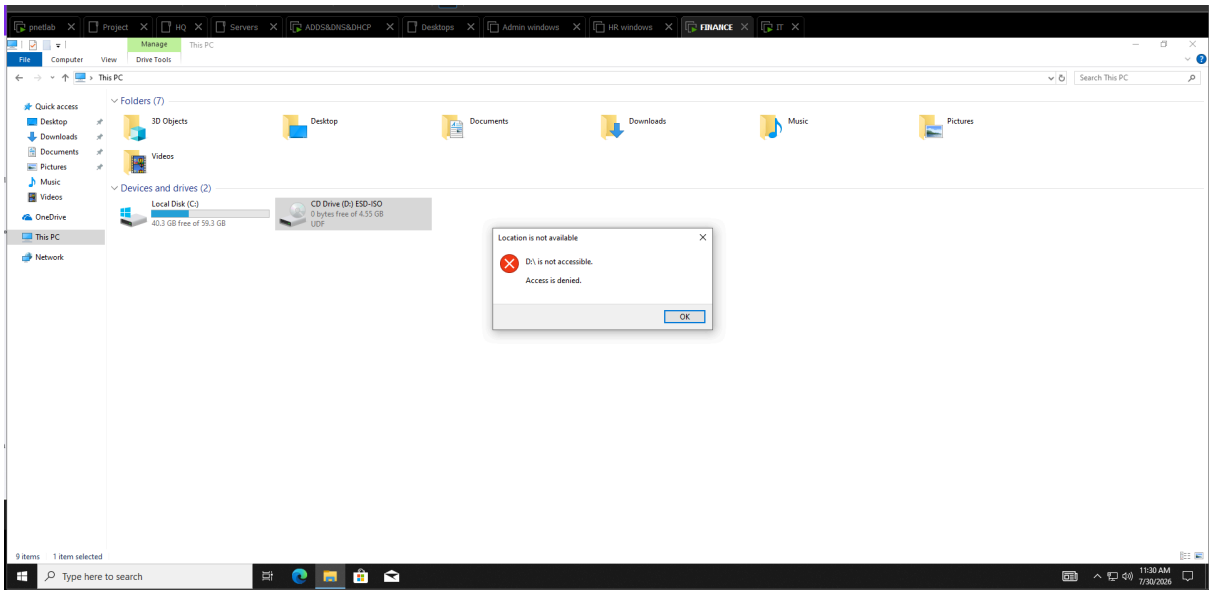


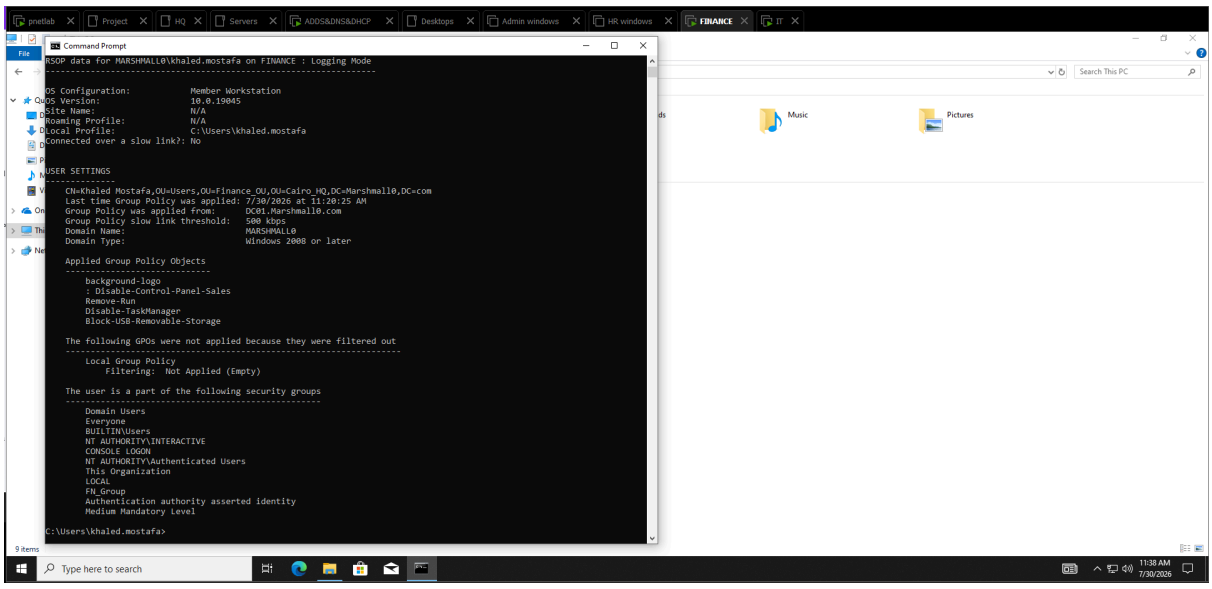
gpo for finance



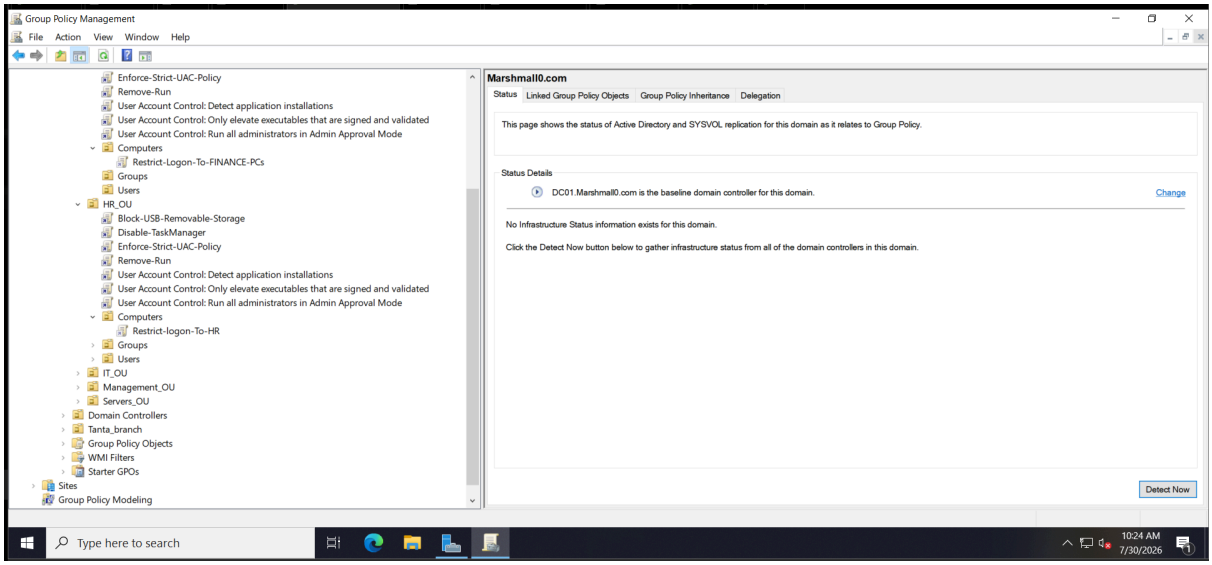



Task Manager
Task Manager has been disabled by your administrator.
OK
All
Apps
Documents
Web
More
Best match
Task Manager
System
Apps
Task Scheduler
Search the web
tas - See web results
Settings (4+)
Open
Run as administrator
Open file location
Pin to Start
Pin to taskbar

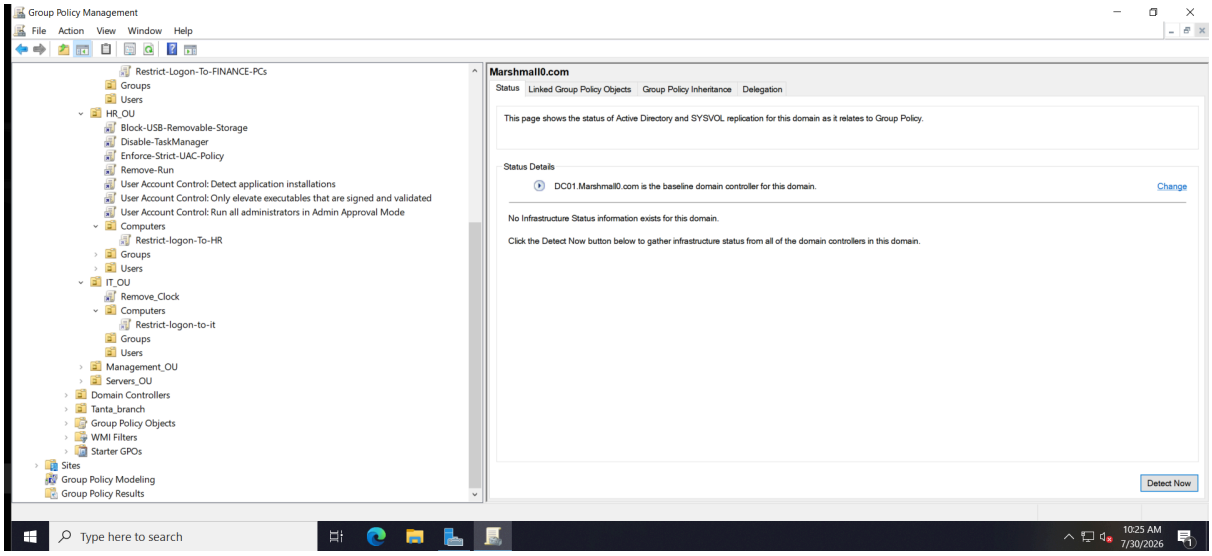

Folders (7)
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Devices and drives (2)
Local Disk (C:)
CD Drive (D:) ESD-ISO
Location is not available
D:\ is not accessible.
Access is denied.
OK


Command Prompt
RSOP data for MARSHMALL0\khaled.mostafa on FINANCE : Logging Mode
OS Configuration: Member Workstation
OS Version: 10.0.19045
Site Name: N/A
Roaming Profile: N/A
Local Profile: C:\Users\khaled.mostafa
Connected over a slow link?: No
USER SETTINGS
CN=Khaled Mostafa,OU=Users,OU=Finance_OU,OU=Cairo_HQ,DC=Marshmall0,DC=com
Last time Group Policy was applied: 7/30/2026 at 11:20:25 AM
Group Policy was applied from: DC01.Marshmall0.com
Group Policy slow link threshold: 500 kbps
Domain Name: MARSHMALL0
Domain Type: Windows 2008 or later
Applied Group Policy Objects
background-logo
: Disable-Control-Panel-Sales
Remove-Run
Disable-TaskManager
Block-USB-Removable-Storage
The following GPOs were not applied because they were filtered out
Local Group Policy
Filtering: Not Applied (Empty)
The user is a part of the following security groups
Domain Users
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
FN_Group
Authentication authority asserted identity
Medium Mandatory Level
C:\Users\khaled.mostafa>

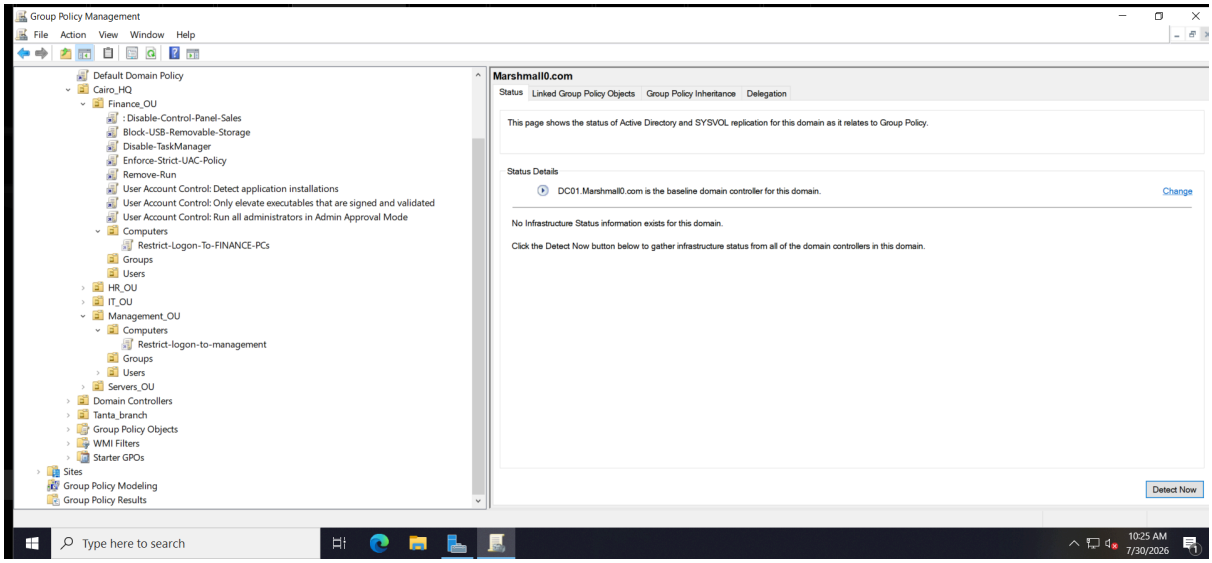
gp for HR



gp for IT



gpo for management



Automated Email Alert for Account Lockout

Using Event Log Trigger

When a user account is locked out, the server logs an event in the Event Viewer under **Event ID 4740**.

The steps simply involve:

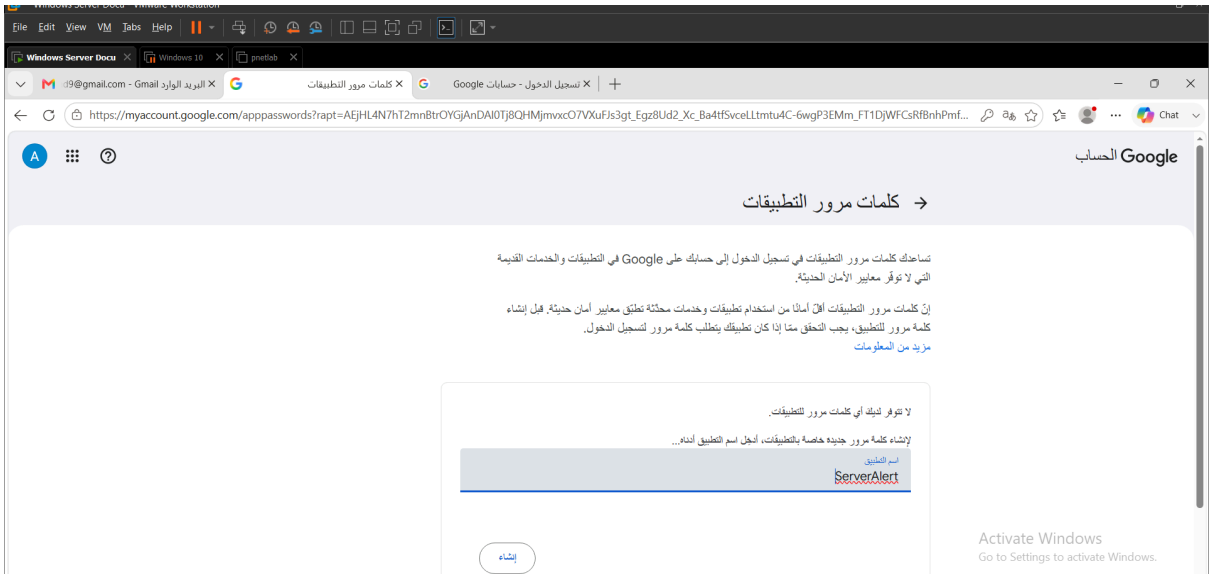
1. **Writing a PowerShell Script:** We write a simple script designed to monitor the Event Viewer, and if Event ID 4740 appears, it sends an email notification to the administrator containing the locked username.
2. **Using Task Scheduler:**
   - Open **Task Scheduler** on the server.
   - Create a **New Task**.
   - Under the **Triggers** tab, select **On an event**.
   - Choose **Log: Security** and **Event ID: 4740**.
   - Under the **Actions** tab, choose to start a program (which is our PowerShell script).

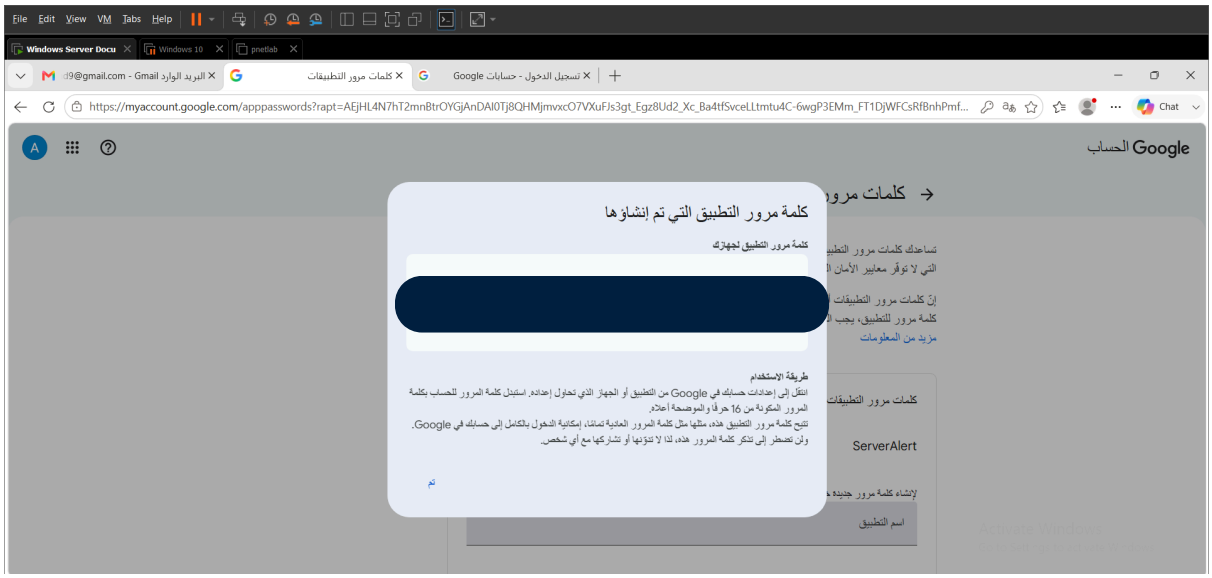
---

1. Creating a Google App Password:

- Go to your Google account security settings and generate an **App Password**.
- This will provide you with a **16-character code**.
- These 16 characters will then be used within the PowerShell script for

SMTP authentication.



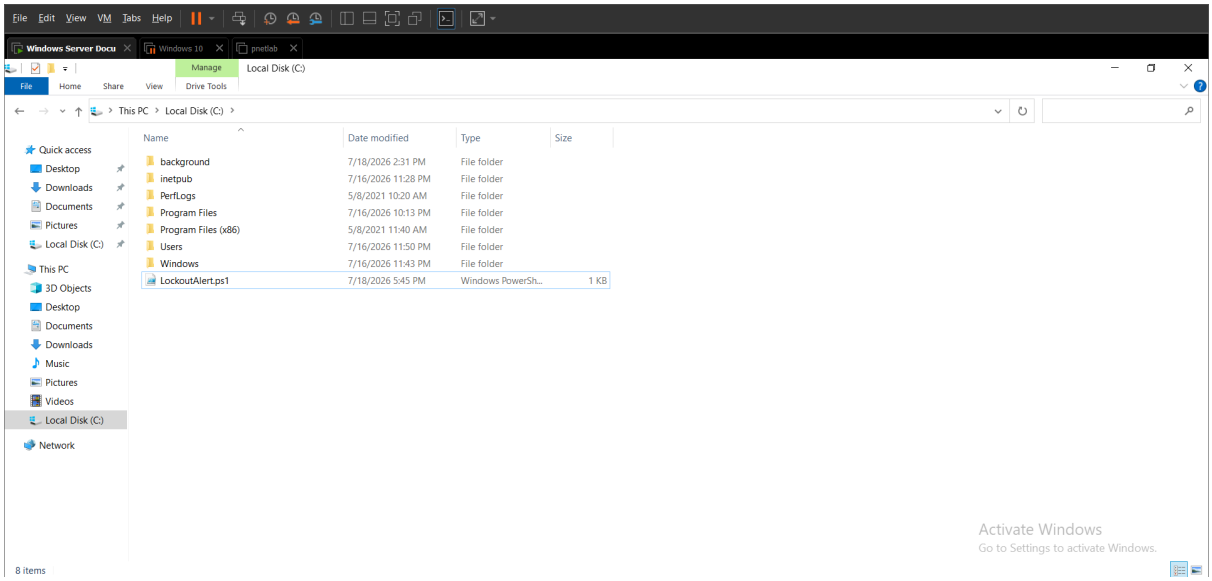


Have ChatGPT create this message for you, and we will add the sender's email, the recipient's email, and the 16-character app password.

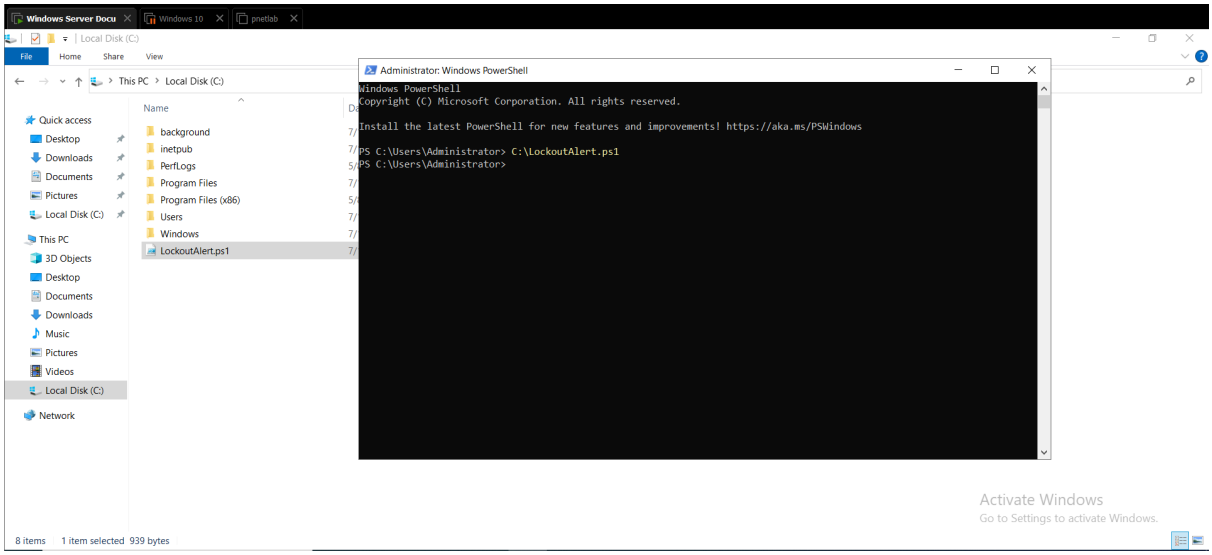
```
# This script retrieves the last locked-out user and sends an alert email
$Event = Get-WinEvent -LogName Security -FilterXPath "*
[System[EventID=4740]]" -MaxEvents 1 | Select-Object -
ExpandProperty Properties
$UserName = $Event[0].Value
$ComputerName = $Event[1].Value
$Time = (Get-Date).ToString()
$Body = "Alert: The user account: $UserName has been locked out
```

```
 from computer: $ComputerName at $Time  "
# Email Configuration
$EmailParams = @{
    To         = "YOUR_ADMIN_EMAIL@marshmallO.com"
    From       = "SERVER_ALERT@marshmallO.com"
    Subject    = "Critical: Account Lockout Alert!"
    Body       = $Body
    SmtpServer = "YOUR_SMTP_SERVER_ADDRESS"
    Port       = 25 # Commonly 25, 587, or 465 depending on your
provider
}
Send-MailMessage @EmailParams
```

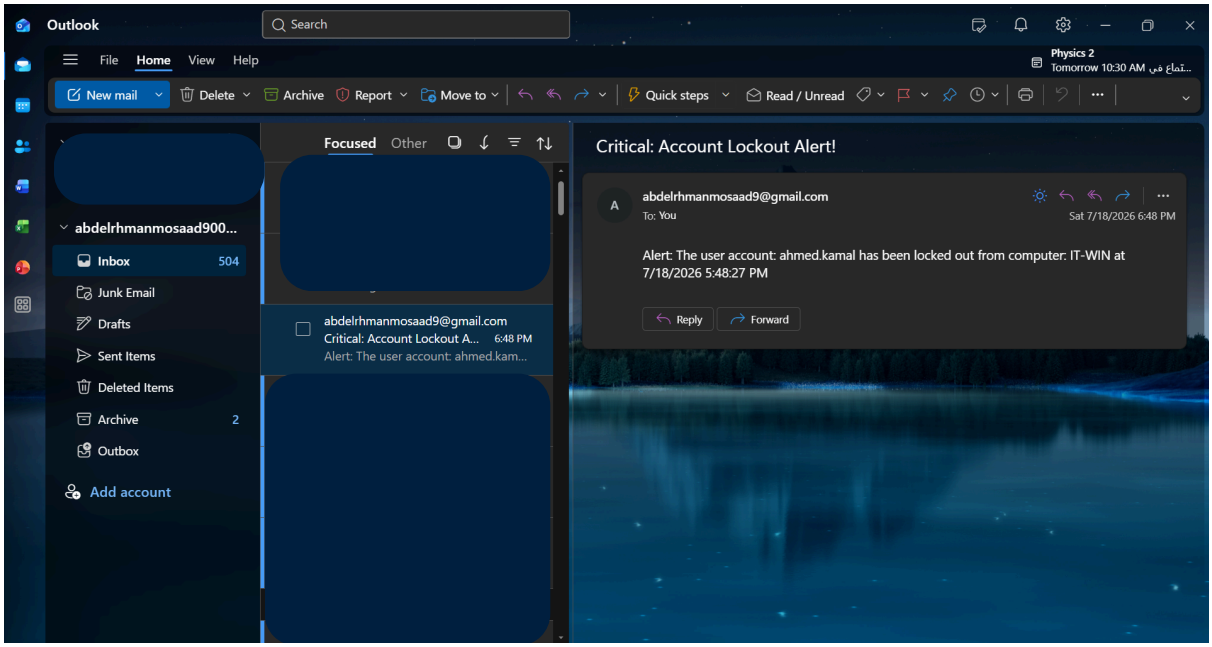
...and then put it in a text file >>>>>>>> and it is important to make the extension .ps1



"And to make sure it's working right now, go to PowerShell and run as administrator, then type the path of the .ps1 file."



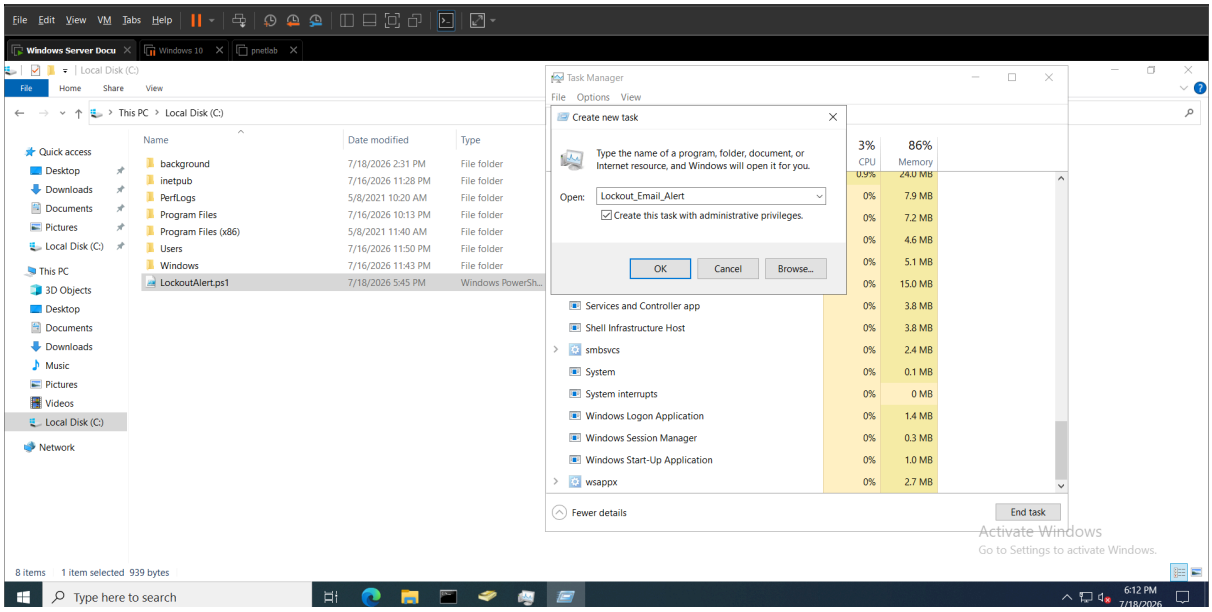
You will find that the email has been sent to you.



---

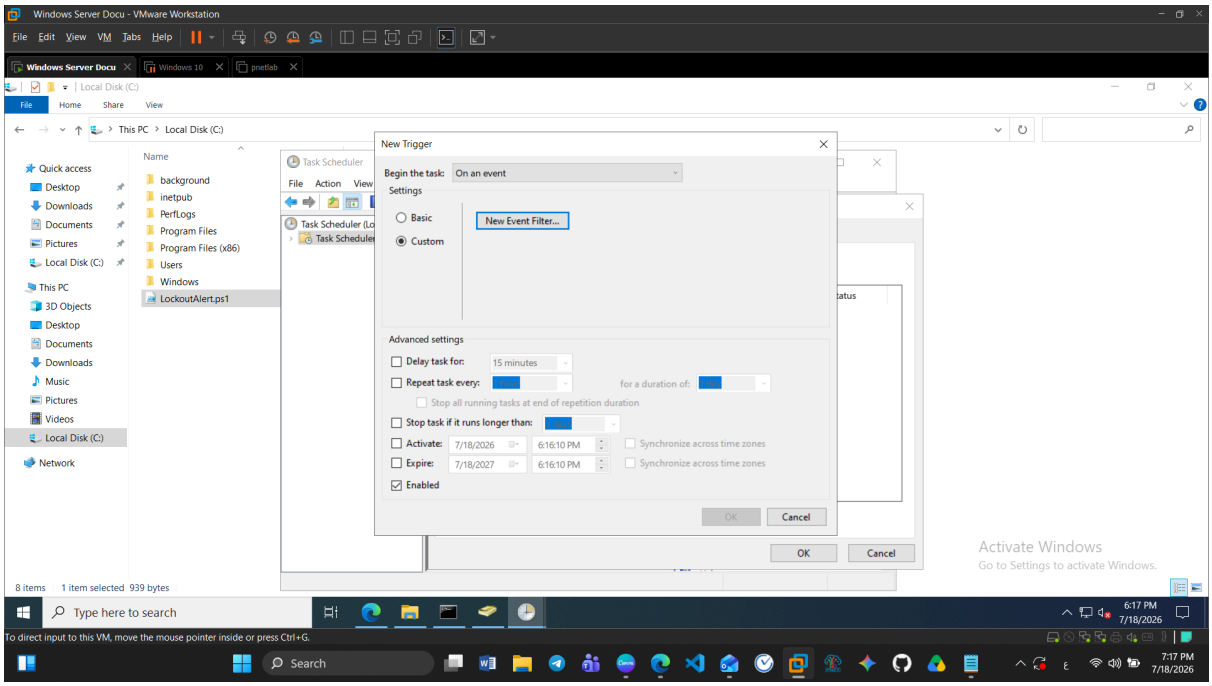
2. Task Scheduler Setup

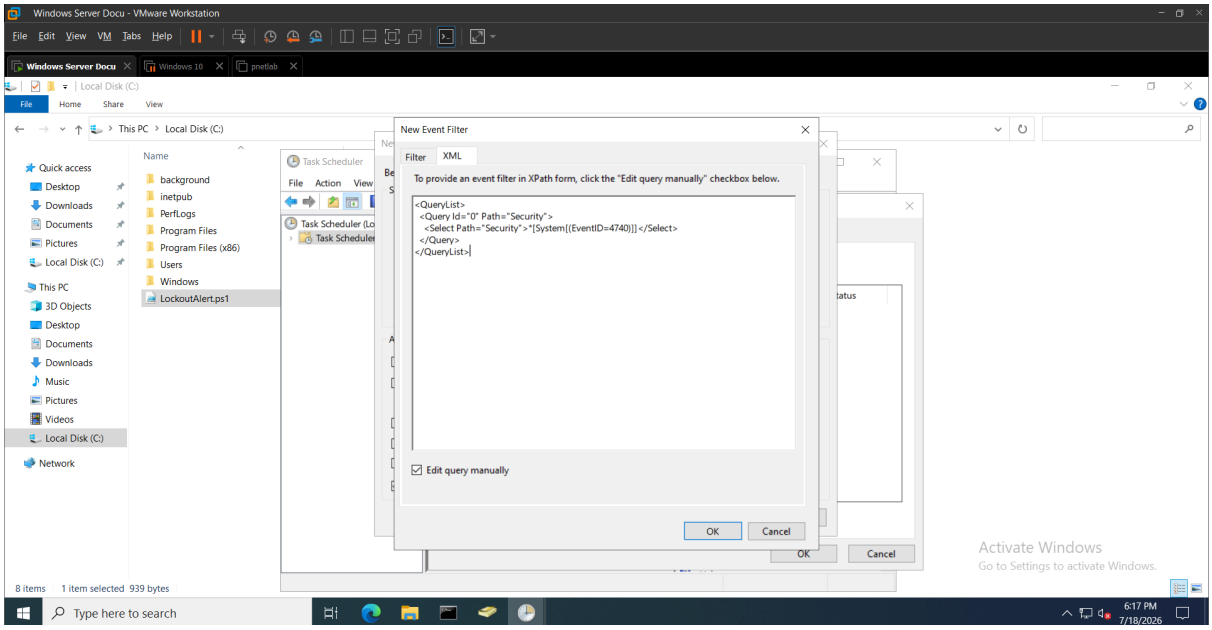
- Open **Task Scheduler**: from Administrative Tools.
- Create a new task: Right-click on **Task Scheduler Library** and select **Create Task**.
- **General tab:**
  - Name the task: Lockout_Email_Alert.
  - Select **Run whether user is logged on or not**.
  - Enable the option **Run with highest privileges**.



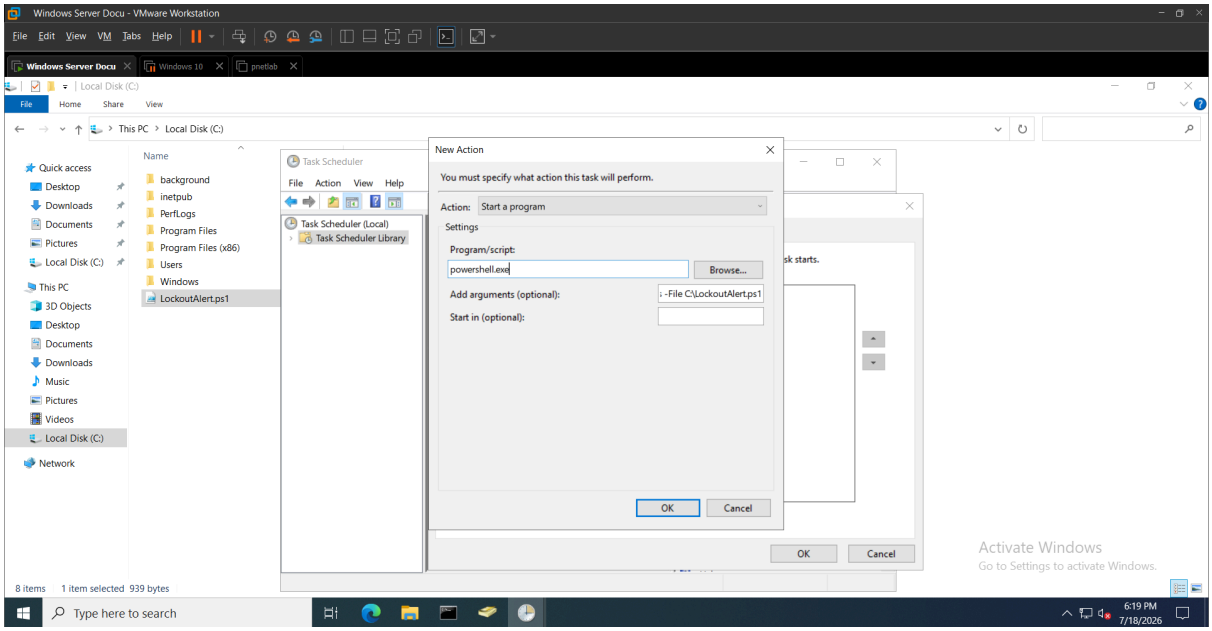
1. **Triggers tab:**
2. Click **New** -> choose **On an event**.
3. Choose **Custom** -> **New Event Filter**.
4. Click on the **XML** tab and enable the option **Edit query manually**.
5. Clear everything and paste the following code:

```
<QueryList>
  <Query Id="0" Path="Security">
    <Select Path="Security">*[System[(EventID=4740)]]</Select>
  </Query>
</QueryList>
```

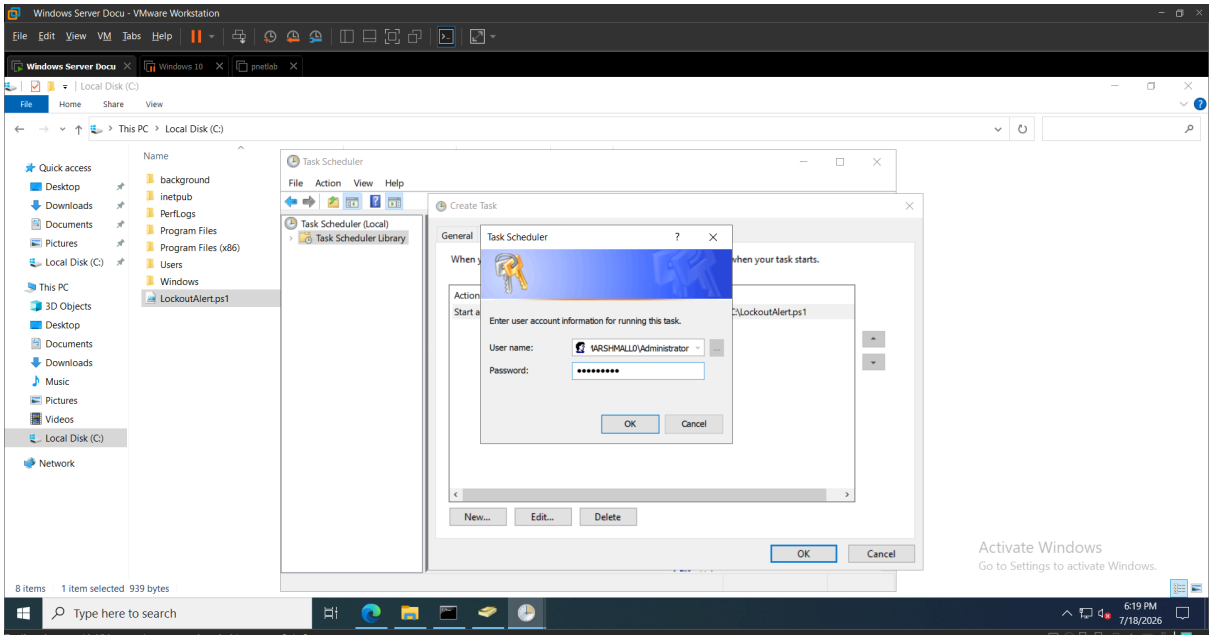




1. **Actions tab:**
2. Click **New** -> choose **Start a program**.
3. In the **Program/script** field, type: powershell.exe.
4. In the **Add arguments** field, type: -ExecutionPolicy Bypass -File C:\LockoutAlert.ps1.

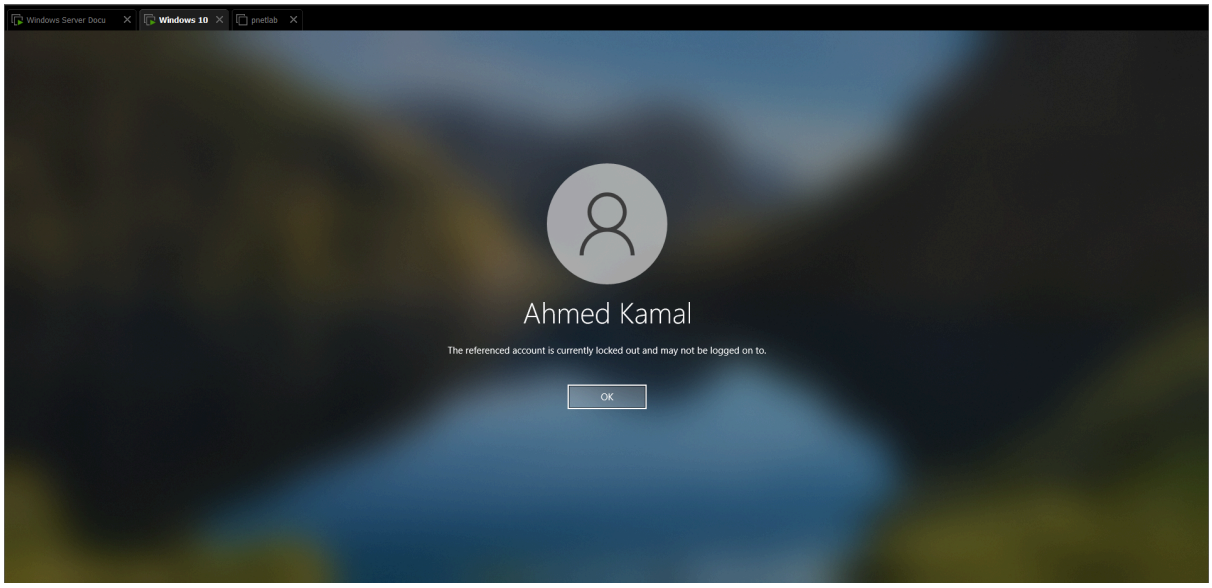


- **Final Step:**
- Click **OK** at the bottom of the main window.
- The server will prompt you to enter the "Username and Password" of your device (Administrator) to save and install this task; enter them and click **OK**.
  -

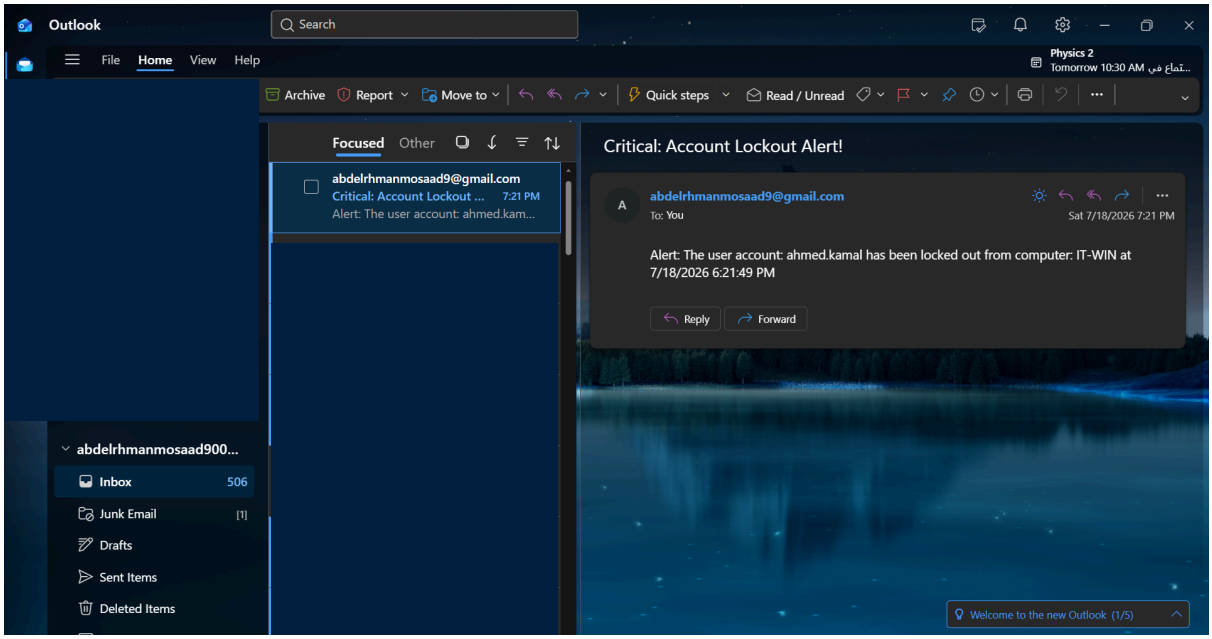


result

Type the password incorrectly until it outputs that message for you.



You will find that the email has been sent to you immediately.



TrueNAS Integration with Windows Database Server

In modern enterprise data center architecture, decoupling database compute resources from storage infrastructure is a critical strategy for achieving high availability, scalability, and optimal performance. This project implements a centralized Storage Area Network (SAN) using TrueNAS as the target storage pool provider, connected via the iSCSI (Internet Small Computer Systems Interface) protocol to a Windows Server acting as a dedicated Database Server.

Key Business & Technical Benefits:

- Resource Optimization: By offloading heavy transactional database files (.mdf, .ldf) to an external TrueNAS SAN, local server disk capacities are conserved, preventing performance bottlenecks associated with internal drive wear and tear.
- Performance and Throughput: iSCSI encapsulation over TCP/IP allows block-level data access, granting the database engine native-disk read/write speeds, which are essential for high-frequency transaction processing systems (OLTP).
- Centralized Management & Scalability: TrueNAS provides advanced enterprise data management features, including ZFS Zvol provisioning, snapshots, and data integrity verification, allowing IT administrators to expand storage dynamically without downtime on the database server.

- High Availability & Fault Isolation: Separating storage and database layers ensures that storage failures or maintenance can be handled independently from application compute nodes.

2. Core Architectural Components

The implementation relies on a client-server storage architecture comprising a Target and an Initiator:

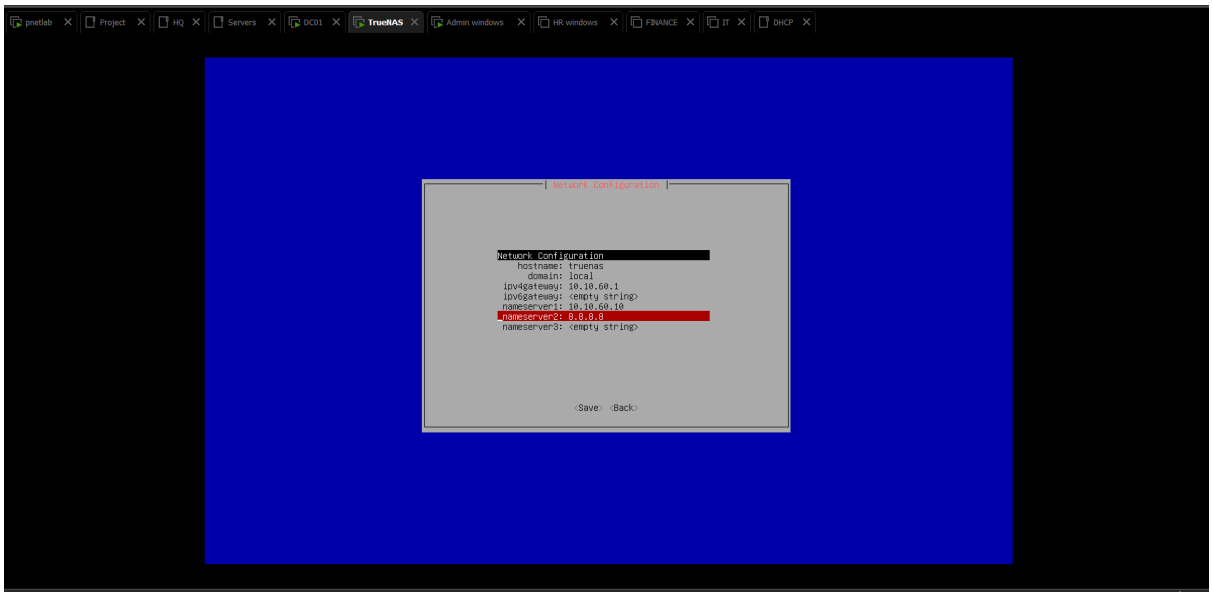
A. The Target (TrueNAS Core / Enterprise)

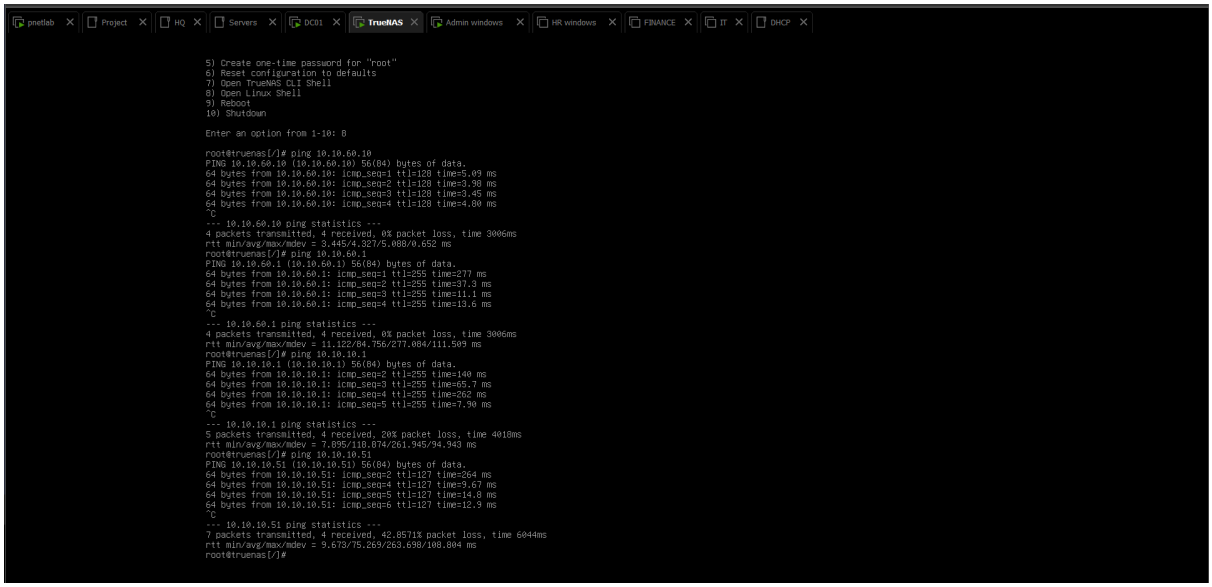
- Role: Acts as the centralized storage repository
- Function: It provisions raw storage blocks via ZFS volumes (Zvols) and exposes them over the network using the iSCSI protocol. It manages block-level read/write operations, data redundancy, and access controls for authorized clients.

B. The Initiator (Windows Server)

- Role: Acts as the client application node (Database Server).
- Function: It establishes a session over the network with the TrueNAS iSCSI Target, mounts the remote block storage, initializes it, and presents it to the Windows operating system as a physical local hard disk. The database management system then utilizes this volume to store transactional data securely and efficiently.

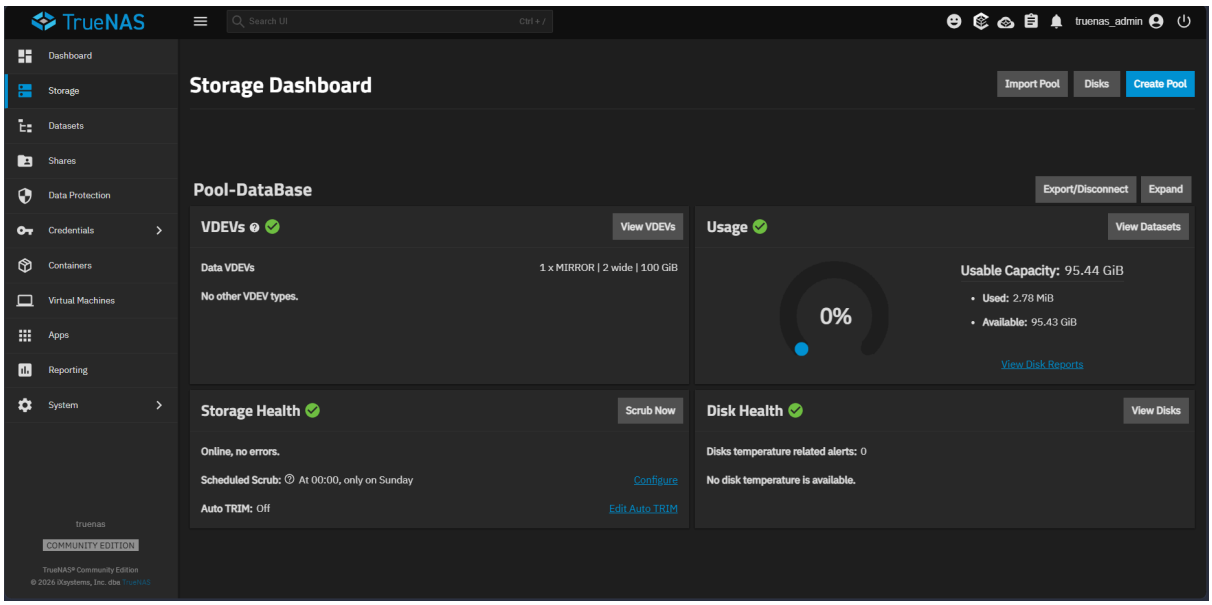
3. Step-by-Step Implementation Guide

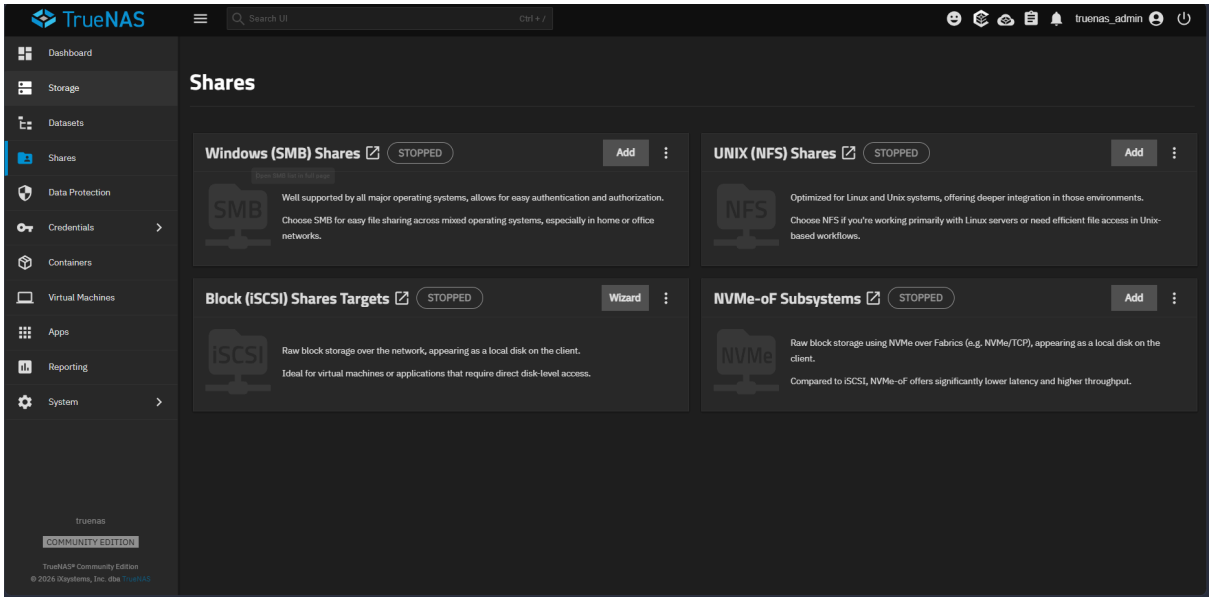




Phase 1: TrueNAS Pool and Zvol Configuration

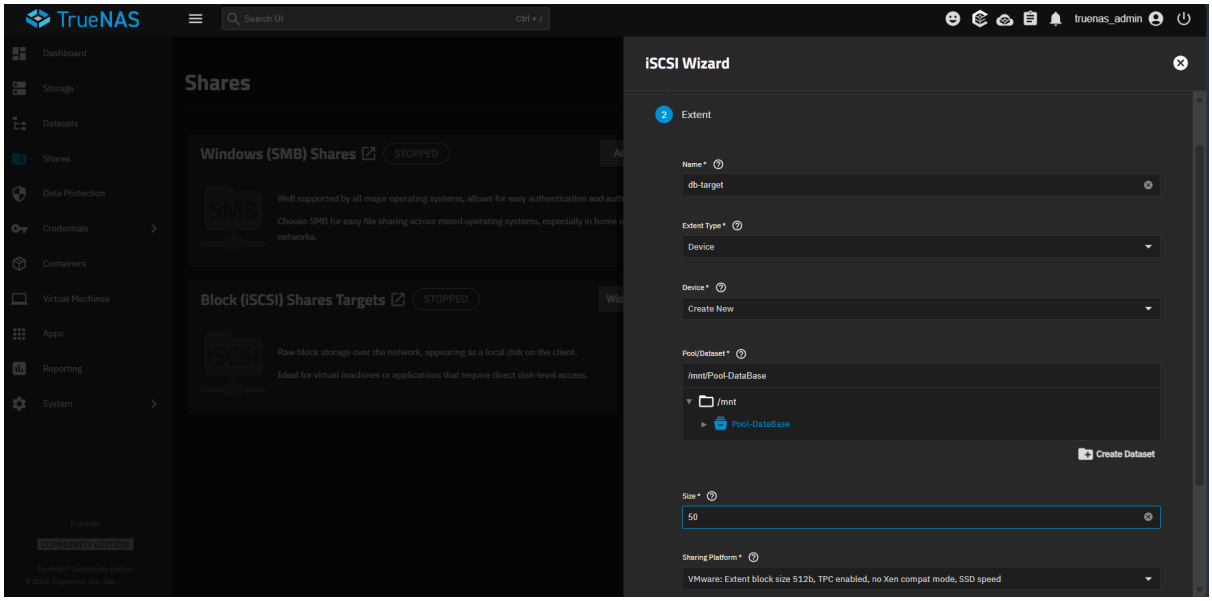
1. Navigate to Datasets: Open the TrueNAS web interface, select Storage from the left-hand navigation menu, and click on Datasets (or click the three dots menu next to your existing Pool and select Add Zvol).
2. Configure Zvol Parameters: In the Zvol creation window:
   - Name: Enter a descriptive name for the Zvol (e.g., db-zvol).
   - Size: Specify the capacity to allocate and expose to the Windows Server (e.g., 30 GiB).
3. Save Configuration: Click Save to create the storage block device.

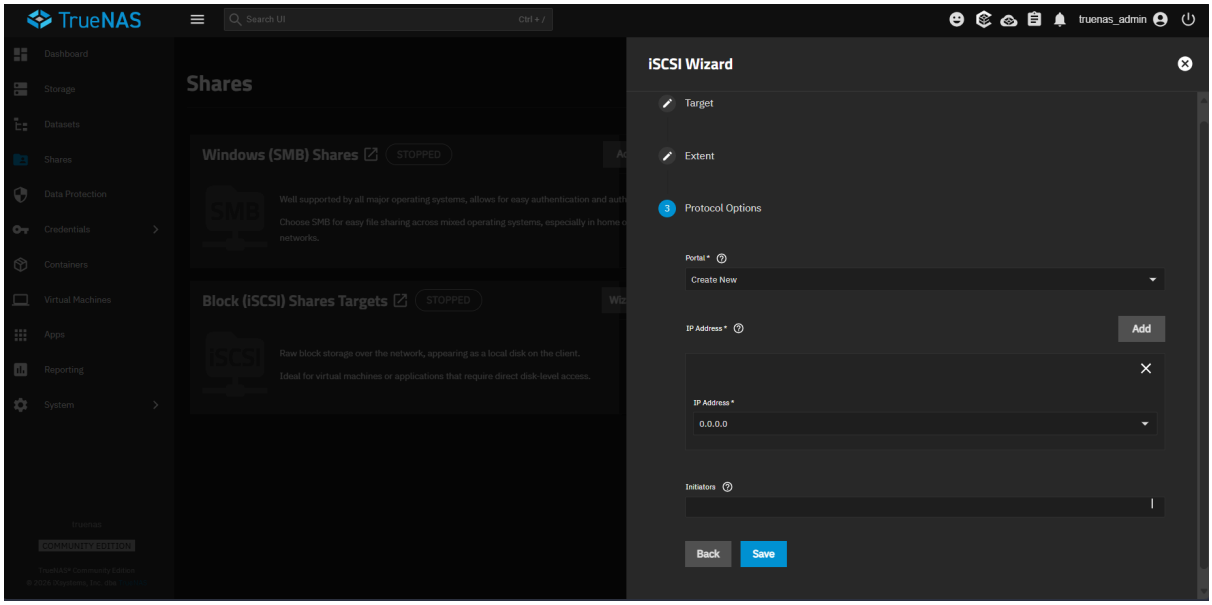


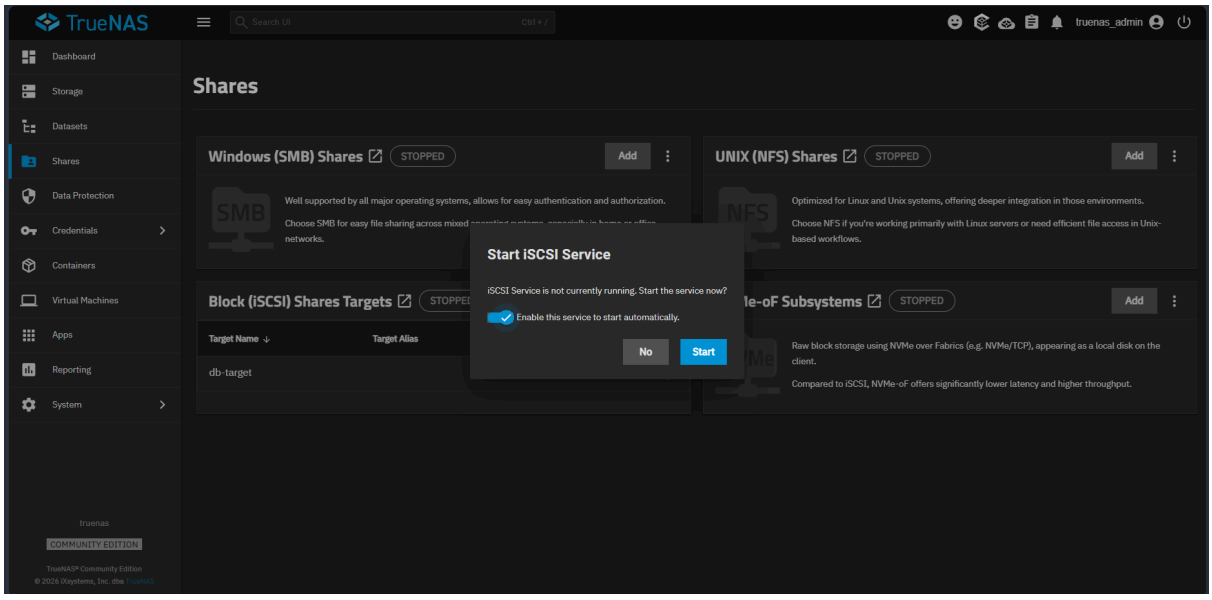


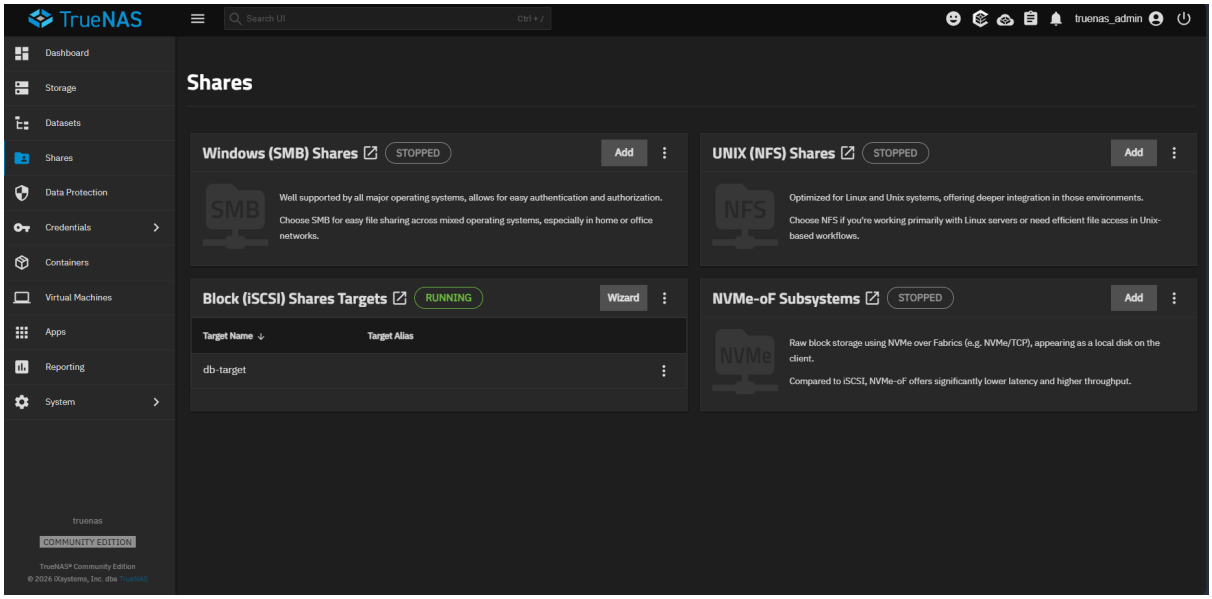
Phase 2: TrueNAS iSCSI Target Setup and Publishing

1. Access iSCSI Settings: From the TrueNAS side menu, select Sharing and click on Block Shares (iSCSI).
2. Run the Wizard: Click the Wizard button to launch the streamlined iSCSI setup helper.
3. Create Target: In the Target step, keep the option set to Create New and provide a descriptive name for the target (e.g., db-target).
4. Configure Extent: In the Extent step, select the extent type as Device, choose your created Pool or Zvol (e.g., Pool-DataBas), and define the required size (e.g., 30 or 50 GiB).
5. Protocol & Portal Configuration: In the Protocol Options step, create a new Portal, set the IP Address to 0.0.0.0 (listening on all interfaces), and click Add.
6. Define Initiators: Leave the Initiators field blank to allow unrestricted communication within the lab/testing environment.
7. Save and Complete: Click Save to finalize the configuration and create the iSCSI Target.
8. Start the iSCSI Service: Go to the Services menu from the left panel, locate the iSCSI service, toggle its state to Running, and ensure Start Automatically is enabled.


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Shares
Windows (SMB) Shares
STOPPED
Block (iSCSI) Shares Targets
STOPPED
iSCSI Wizard
2 Extent
Name *
db-target
Extent Type *
Device
Device *
Create New
Pool/Dataset *
/mnt/Pool-DataBase
/mnt
Pool-DataBase
Create Dataset
Size *
50
Sharing Platform *
VMware: Extent block size 512b, TPC enabled, no Xen compat mode, SSD speed


TrueNAS
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Windows (SMB) Shares
STOPPED
Block (iSCSI) Shares Targets
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iSCSI Wizard
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Extent
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Create New
IP Address *
Add
IP Address *
0.0.0.0
Initiators
Back
Save


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Credentials
Containers
Virtual Machines
Apps
Reporting
System
Shares
Windows (SMB) Shares
STOPPED
Add
Well supported by all major operating systems, allows for easy authentication and authorization.
Choose SMB for easy file sharing across mixed
networks.
UNIX (NFS) Shares
STOPPED
Add
Optimized for Linux and Unix systems, offering deeper integration in those environments.
Choose NFS if you're working primarily with Linux servers or need efficient file access in Unix-based workflows.
Start iSCSI Service
iSCSI Service is not currently running. Start the service now?
Enable this service to start automatically.
No
Start
Block (iSCSI) Shares Targets
Target Name
Target Alias
db-target
e-oF Subsystems
STOPPED
Add
Raw block storage using NVMe over Fabrics (e.g. NVMe/TCP), appearing as a local disk on the client.
Compared to iSCSI, NVMe-oF offers significantly lower latency and higher throughput.
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COMMUNITY EDITION



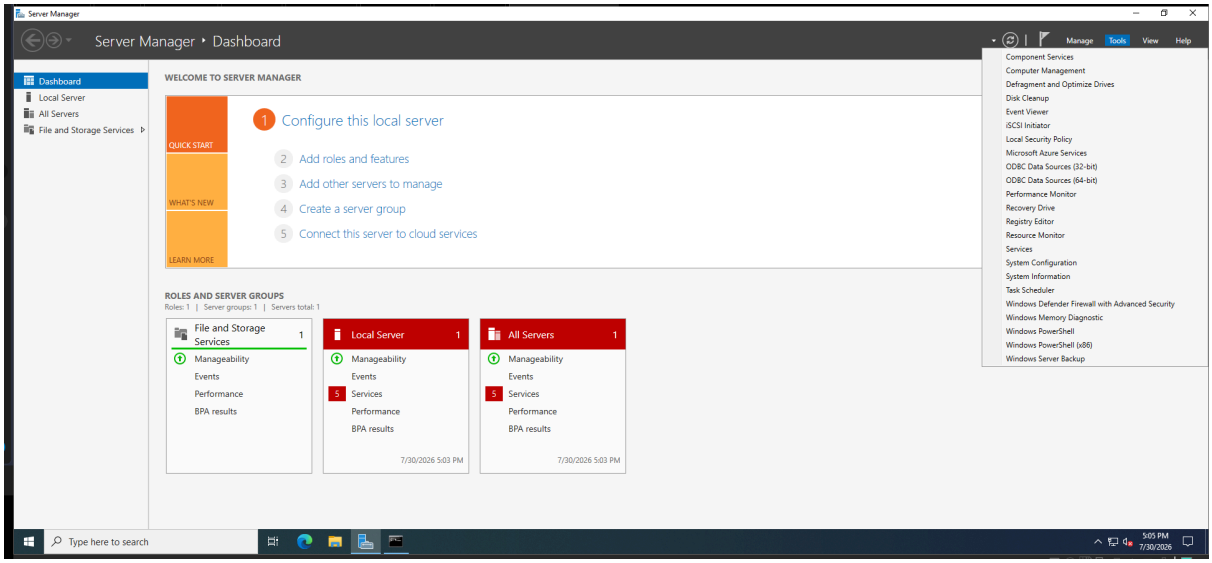
Phase 3: Windows Server iSCSI Initiator Configuration

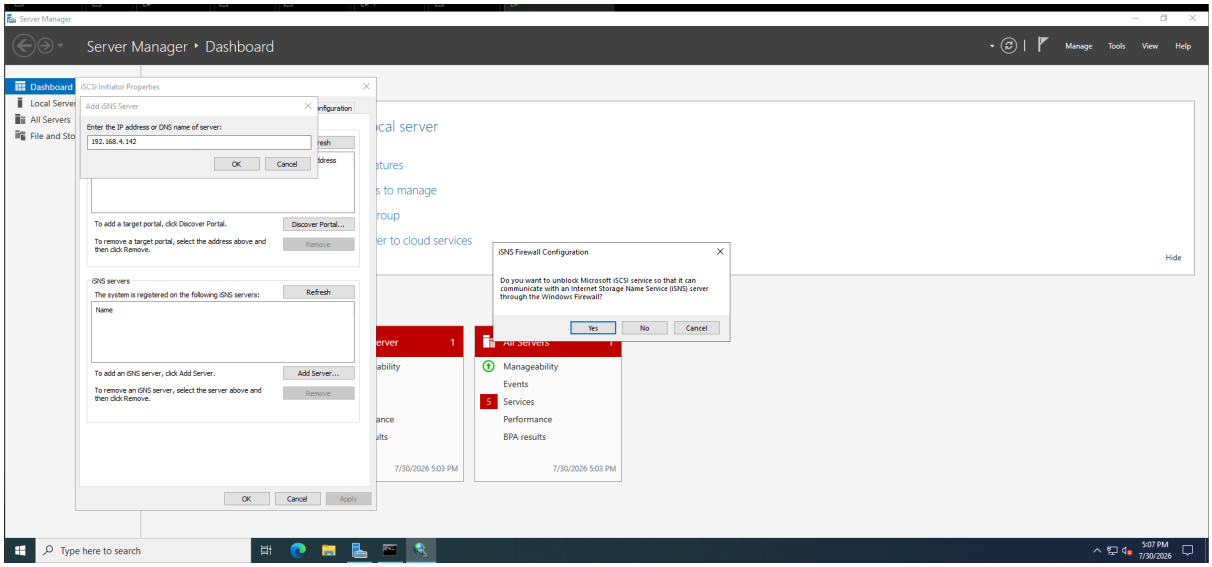
Step 1: Initialize the iSCSI Service

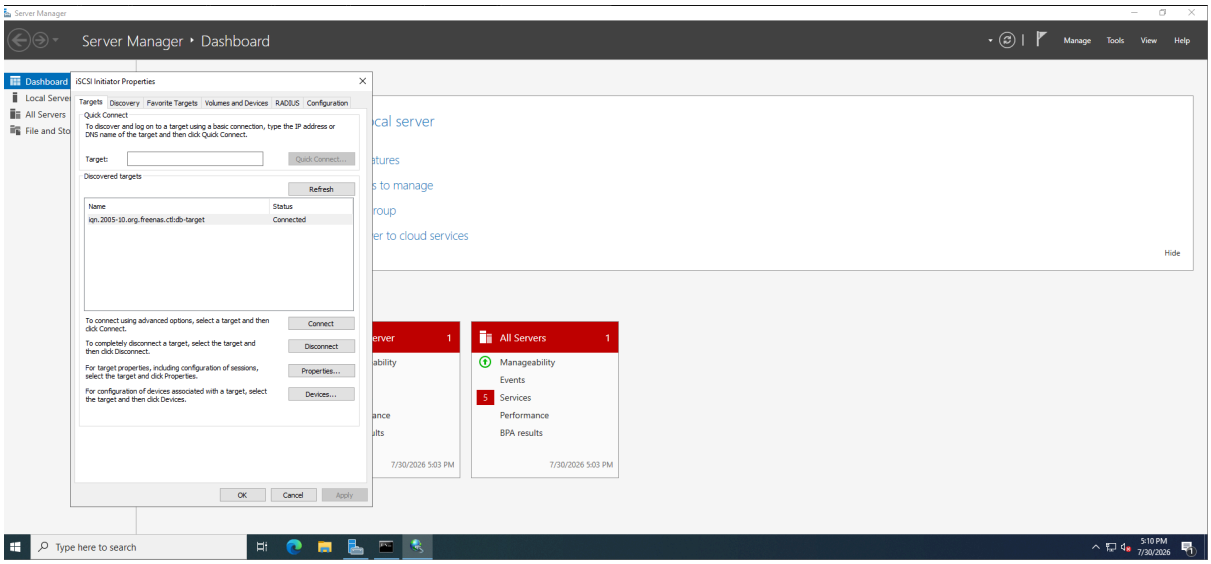
1. Open Server Manager, click on the Tools menu in the top right, and select iSCSI Initiator.
2. If prompted to start the Microsoft iSCSI service automatically on system boot, click Yes.

Step 2: Discovery and Connection

1. In the iSCSI Initiator Properties window, navigate to the Discovery tab.
2. Click Discover Portal..., enter the IP address of the TrueNAS server (e.g., 192.168.4.142), and click OK.
3. Switch to the Targets tab.
4. Under the Quick Connect section, verify or enter the TrueNAS IP address and click Quick Connect.
5. Ensure the status of the target changes to Connected (if prompted, click Connect and check Add to favorite targets), then click Done and OK.

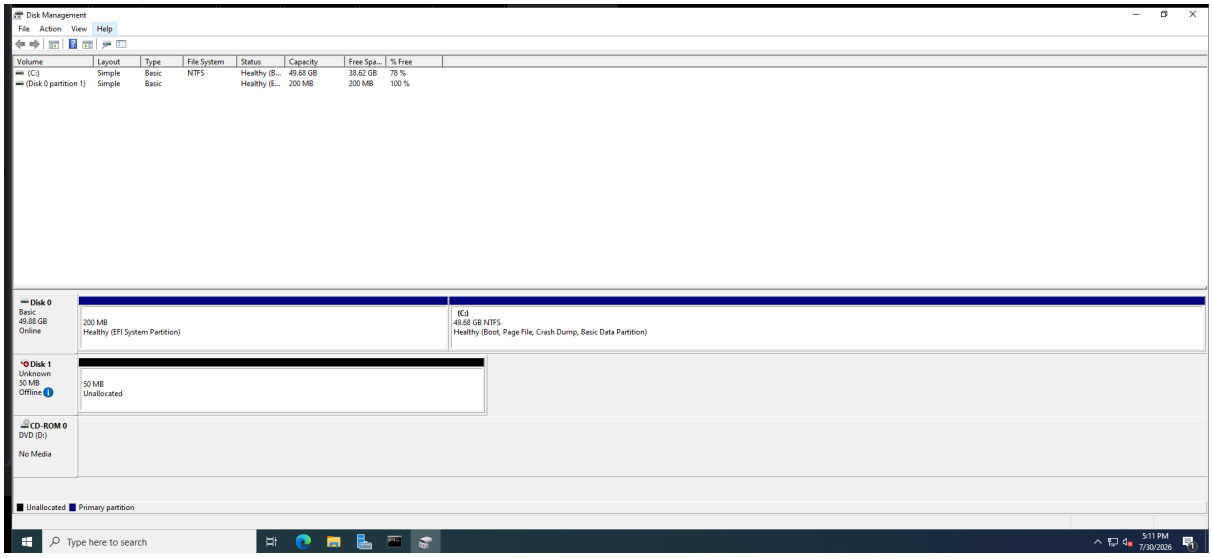


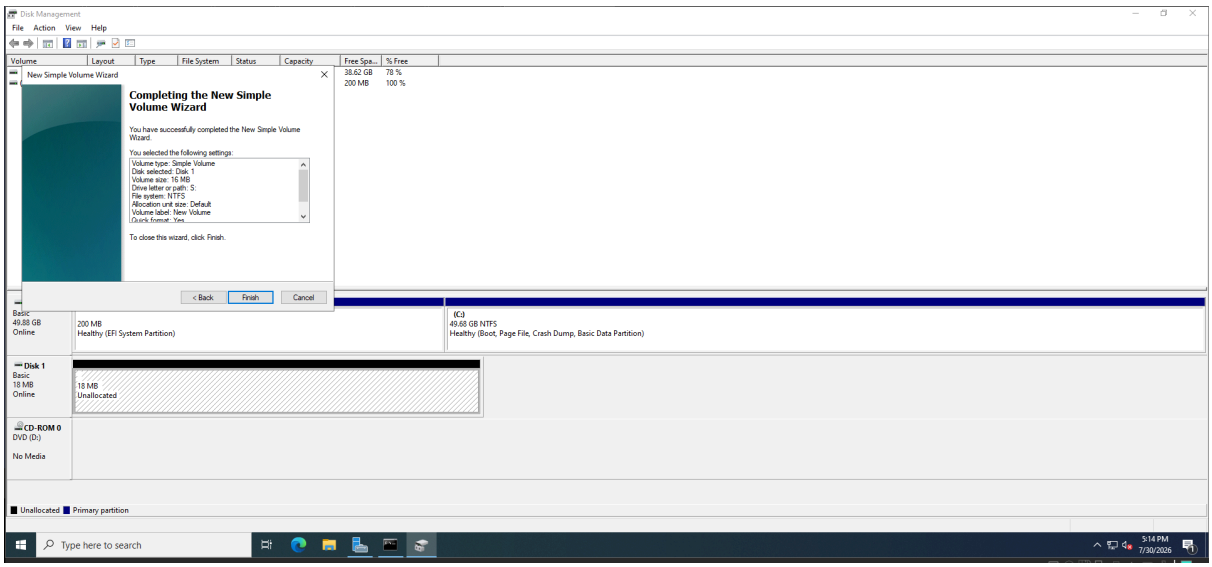


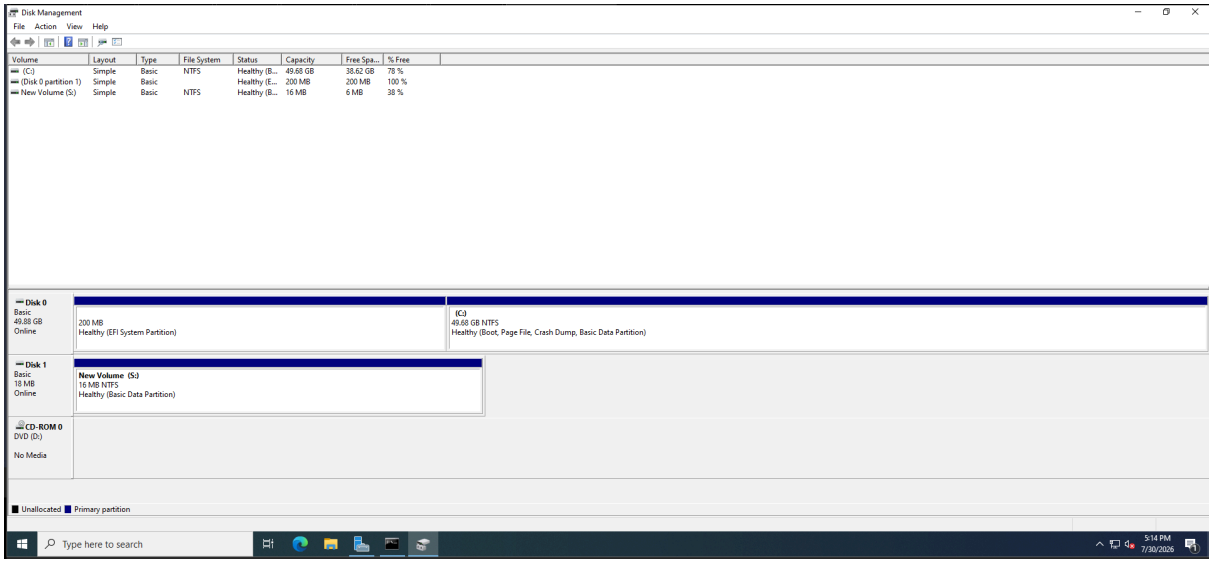


Phase 4: Disk Initialization and Partitioning in Windows (Disk Management)

1. Right-click the Windows Start button (Windows + X) and select Disk Management.
2. From the top menu, click Action and select Rescan Disks.
3. Locate the newly discovered raw disk at the bottom pane (displayed as Disk 1, showing as Offline). Right-click the disk label and select Online.
4. Right-click the disk label again and select Initialize Disk, choose the GPT (GUID Partition Table) partition style, and click OK.
5. Right-click the unallocated black space representing the volume and select New Simple Volume.
6. Click Next through the wizard steps until reaching the Assign a drive letter screen, choose an appropriate drive letter, click Next, and proceed through to Finish to format the partition.

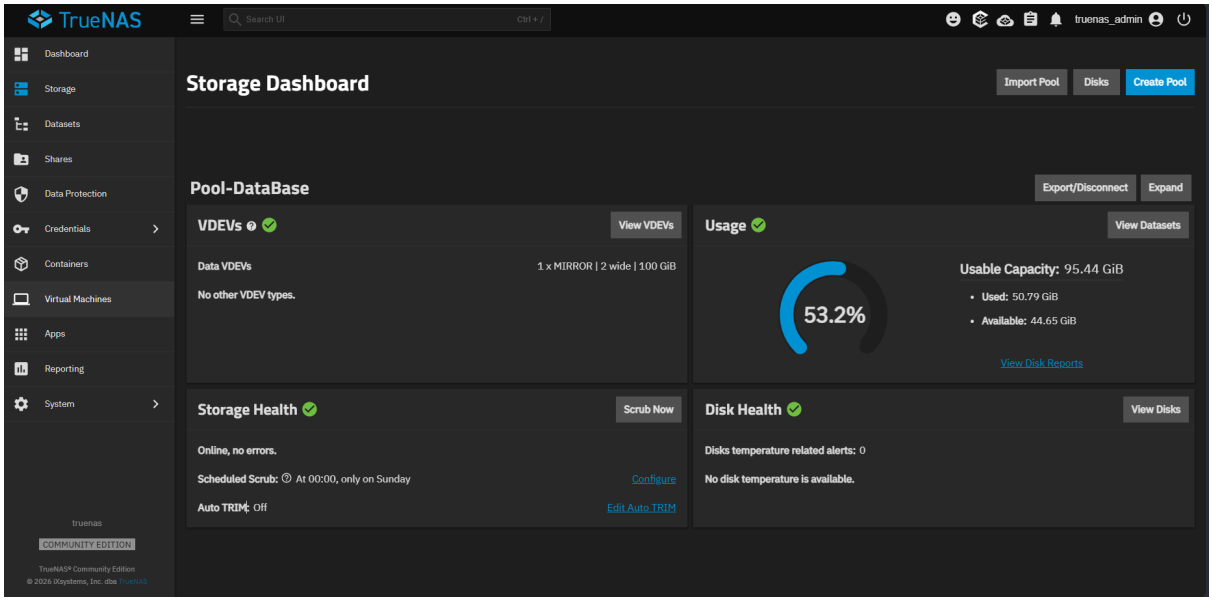


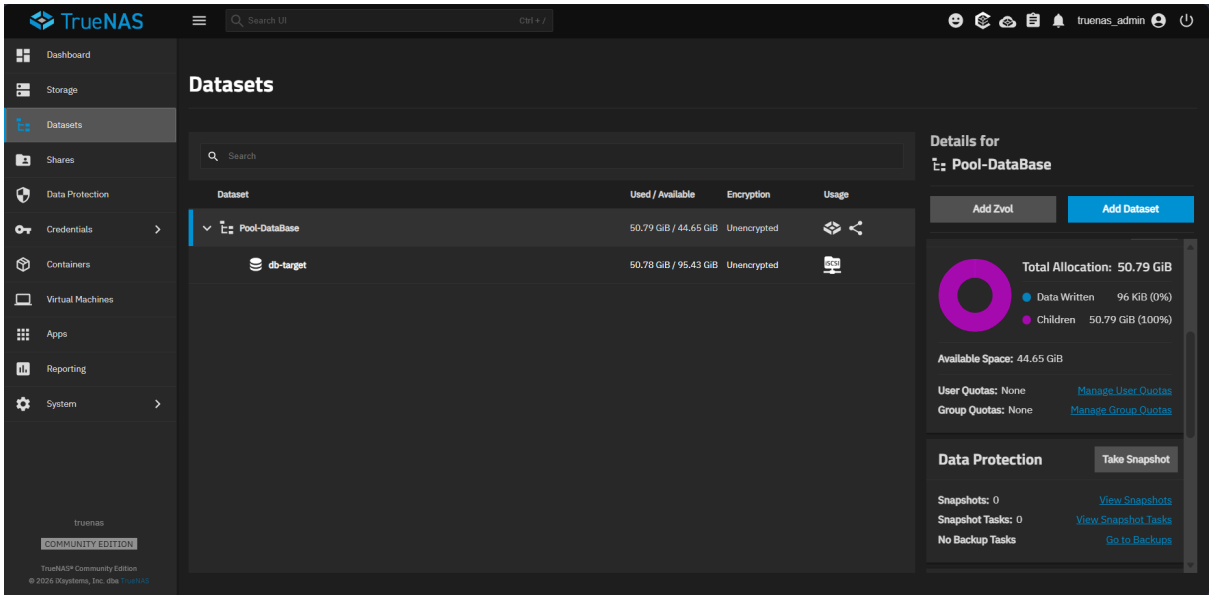




explanation of what TrueNAS allows you to monitor and provide from those interfaces:

- **Capacity and Usage Monitoring: The dashboard shows that the total usable capacity of the pool named Pool-DataBas is 95.44 GiB (utilizing a 1 x MIRROR | 2 wide configuration). Out of this, 50.79 GiB is currently used, accounting for a 53.2% utilization rate, leaving 44.65 GiB available.**
- **Zvol (Block Storage) Management: The Datasets section displays the provisioned Zvol named db-target taking up 50.78 GiB, acting as the block storage disk ready to be connected to the database server.**
- **Storage and Disk Health: TrueNAS provides continuous health tracking to ensure the drives operate without errors (Online, no errors), alongside scheduled maintenance features like data scrubbing to protect data integrity.**
- **Performance and Resource Reporting: The system provides live, detailed graphs tracking resource utilization such as CPU performance and data read/write rates, enabling administrators to monitor server performance and instantly identify potential bottlenecks.**


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System
Storage Dashboard
Import Pool
Disks
Create Pool
Pool-DataBase
Export/Disconnect
Expand
VDEVs
View VDEVs
Data VDEVs
1 x MIRROR | 2 wide | 100 GiB
No other VDEV types.
Usage
View Datasets
53.2%
Usable Capacity: 95.44 GiB
Used: 50.79 GiB
Available: 44.65 GiB
View Disk Reports
Storage Health
Scrub Now
Online, no errors.
Scheduled Scrub: At 00:00, only on Sunday
Configure
Auto TRIM: Off
Edit Auto TRIM
Disk Health
View Disks
Disks temperature related alerts: 0
No disk temperature is available.
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Apps
Reporting
System
Datasets
Search
Dataset
Used / Available
Encryption
Usage
Pool-DataBase
50.79 GiB / 44.65 GiB
Unencrypted
db-target
50.78 GiB / 95.43 GiB
Unencrypted
Details for
Pool-DataBase
Add Zvol
Add Dataset
Total Allocation: 50.79 GiB
Data Written 96 KiB (0%)
Children 50.79 GiB (100%)
Available Space: 44.65 GiB
User Quotas: None
Manage User Quotas
Group Quotas: None
Manage Group Quotas
Data Protection
Take Snapshot
Snapshots: 0
View Snapshots
Snapshot Tasks: 0
View Snapshot Tasks
No Backup Tasks
Go to Backups
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50%
Reset
Reporting
CPU
Auto Refresh
CPU
Exporters
CPU Usage
%CPU
Start: 2026-07-30 07:37:06
End: 2026-07-30 08:37:06
CPU Temperature
Celsius
Start: 2026-07-30 07:37:06
End: 2026-07-30 08:37:06
System Load Average
Load
Start: 2026-07-30 07:37:06
End: 2026-07-30 08:37:06

Enterprise File Server Architecture & Implementation Guide

1. Conceptual Overview

In an enterprise IT environment, managing files locally on individual user machines leads to data fragmentation, lack of backups, and security vulnerabilities. Implementing a centralized File Server integrated with Active Directory (Domain) solves these challenges by centralizing data storage and enforcement.

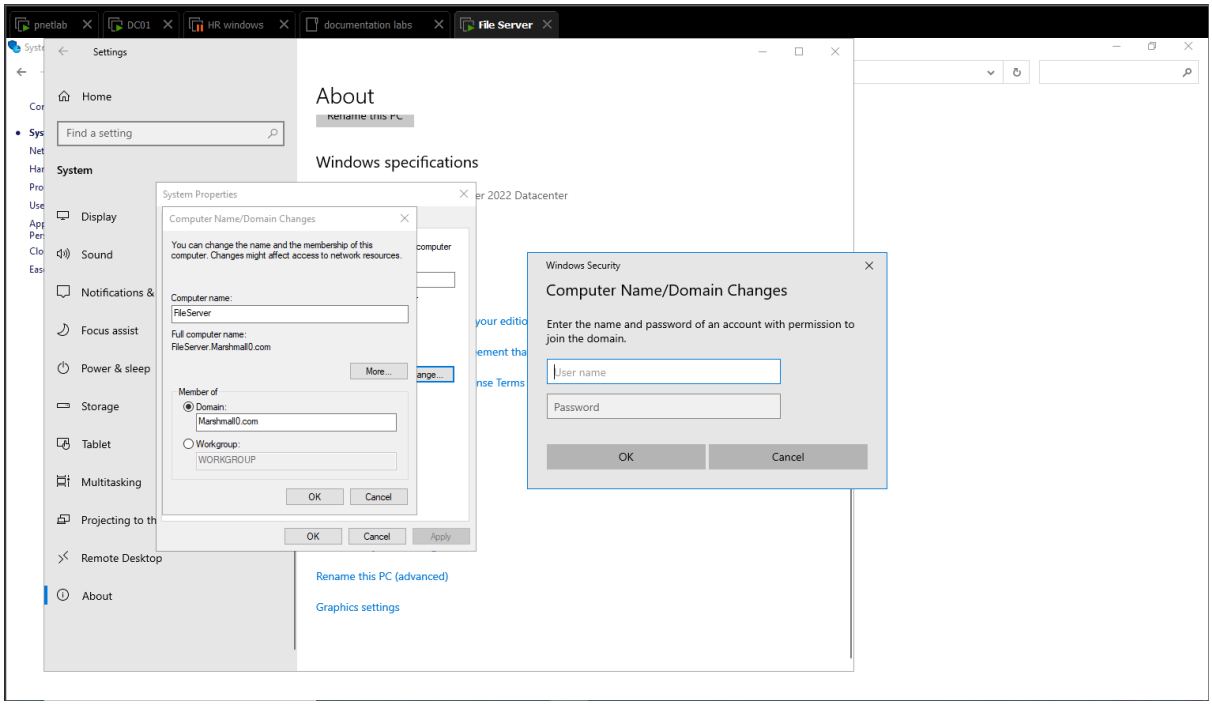
Core Concepts & Benefits:

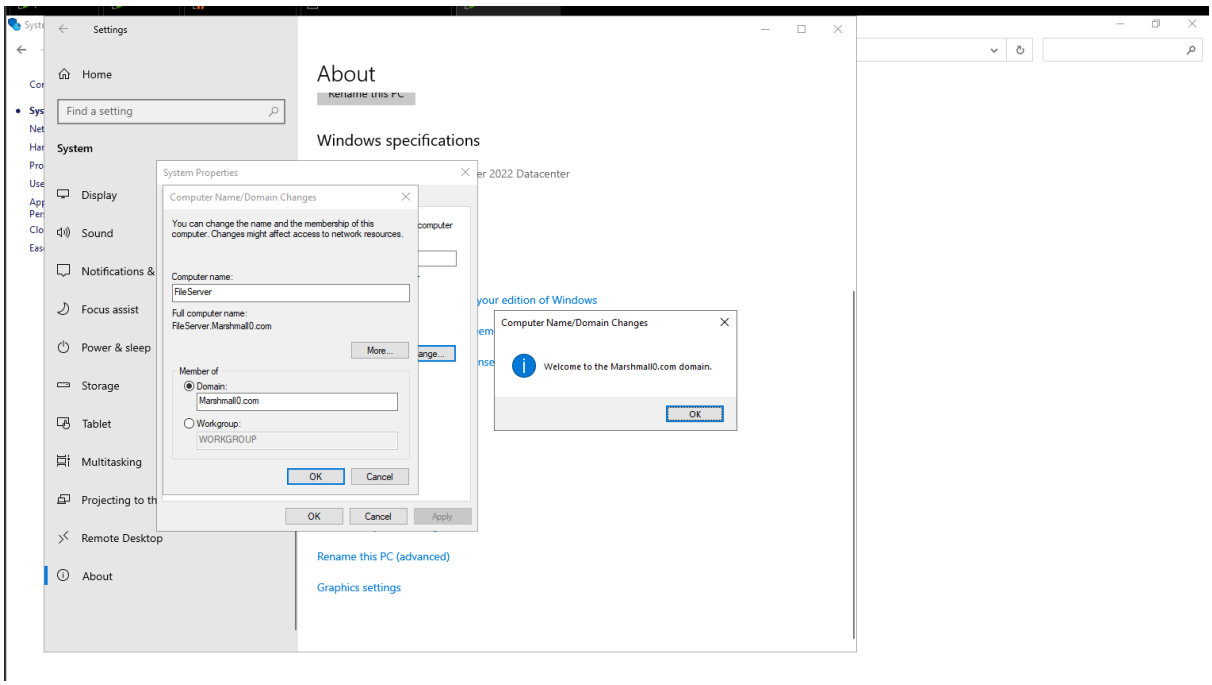
- **Domain Integration: Joining the File Server to the Active Directory domain (e.g., it.local) allows seamless authentication, security group management, and centralized access controls across the organization.**
- **Centralized Storage Repository: Storing shared resources on a dedicated server ensures uniform data governance, simplified disaster recovery, and consistent backup policies.**
- **Automated Deployment via Group Policy (GPP): Instead of requiring users or technicians to manually map network shares through the Run dialog or command line, Group Policy Preferences (Drive Maps) automatically provisions shared folders as a persistent network drive (e.g., Drive Z:) directly inside every user's This PC upon login.**
- **Centralized Management: Administrative modifications—such as updating share paths, changing security permissions, or altering drive mappings—are managed from a single point of control (DC01) and propagate instantly to all domain users without manual intervention.**

2. Practical Implementation Steps

Phase 1: Joining the File Server to the Domain

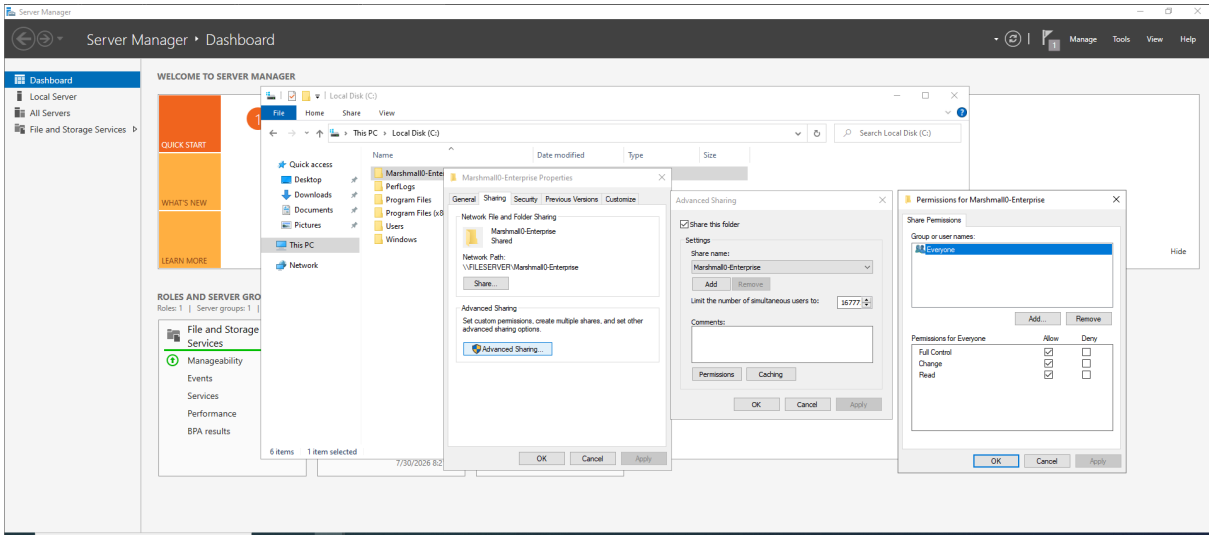
1. **Open Control Panel or System Properties on the File Server (or press Windows + Pause/Break).**
2. **Next to the computer name, click Change settings.**
3. **Under the Computer Name tab, click the Change... button.**
4. **Under the Member of section, select Domain and enter your domain name (e.g., it.local).**
5. **Click OK, and when prompted, enter the credentials for the domain Administrator account (from DC01).**
6. **Click OK and perform a system Restart on the server.**

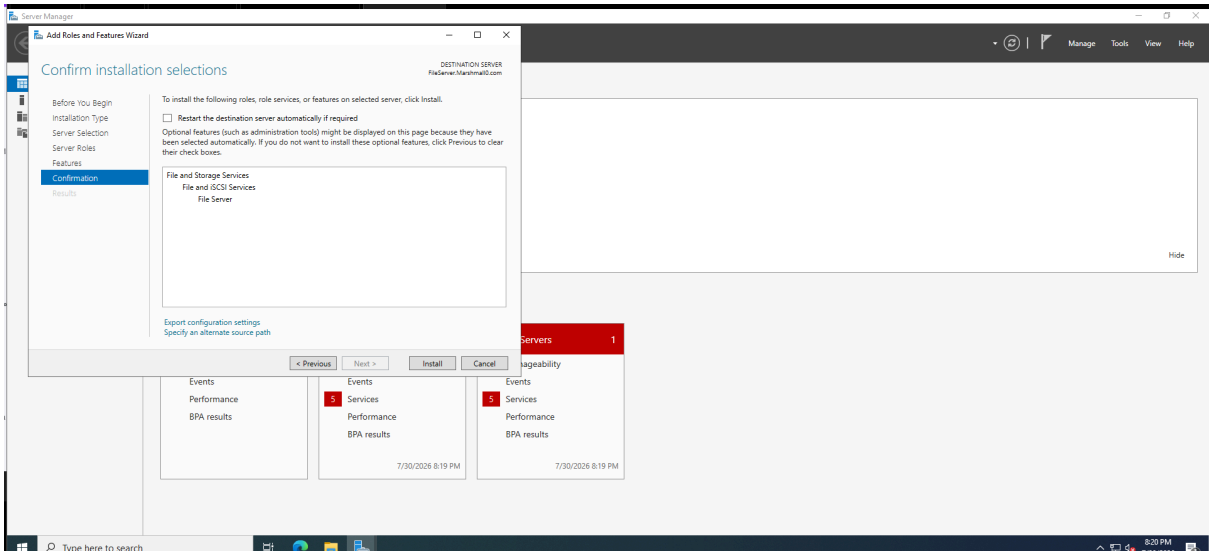




Phase 2: Installing the File Server Role

1. **Open Server Manager.**
2. **Click on Add roles and features.**
3. **Proceed by clicking Next until reaching the Server Roles page.**
4. **Expand File and Storage Services, and subsequently expand File and iSCSI Services.**
5. **Check the box for File Server (if not already enabled by default).**
6. **Continue clicking Next until you reach the Install confirmation page, click Install, and wait for the installation to finish.**





Phase 3: Creating and Sharing the Folder on the System Drive (C:)

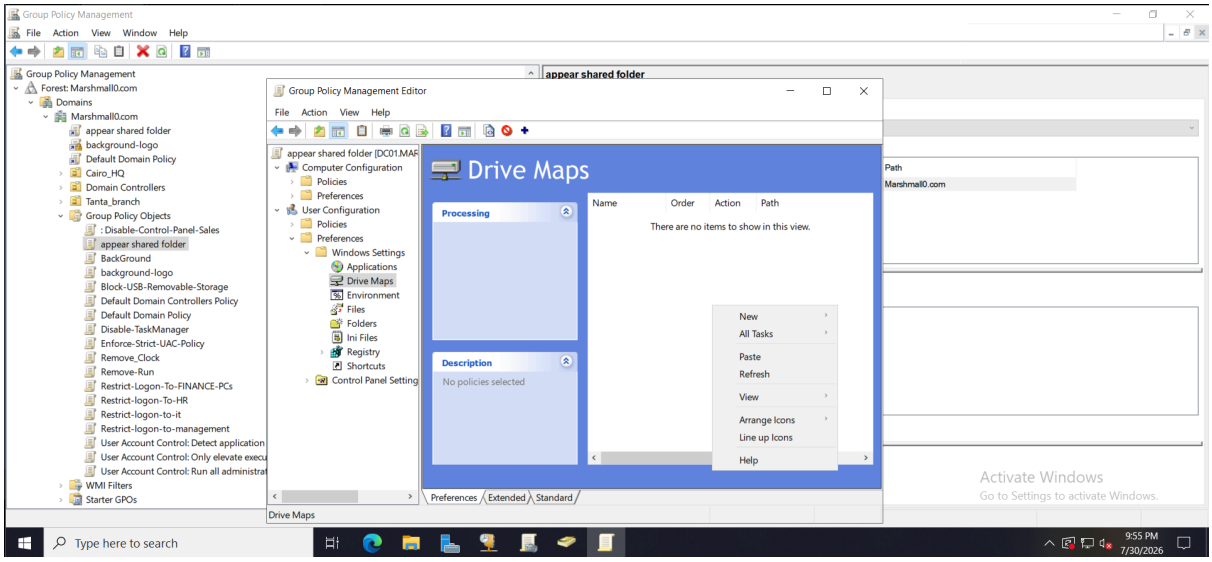
1. **Open This PC and navigate to the system drive (C:).**
2. **Create a New Folder and name it TestShare.**
3. **Right-click the folder and select Properties.**
4. **Navigate to the Sharing tab and click Advanced Sharing...**
5. **Check Share this folder, and in the Share name field, keep it as TestShare (or hide it using TestShare$).**
6. **Click the Permissions button, ensure the Everyone group is granted Full Control (to ensure quick, unhindered connectivity testing), and click OK then Apply.**
7. **(Optional) Switch to the Security tab to verify NTFS permissions, then click OK.**

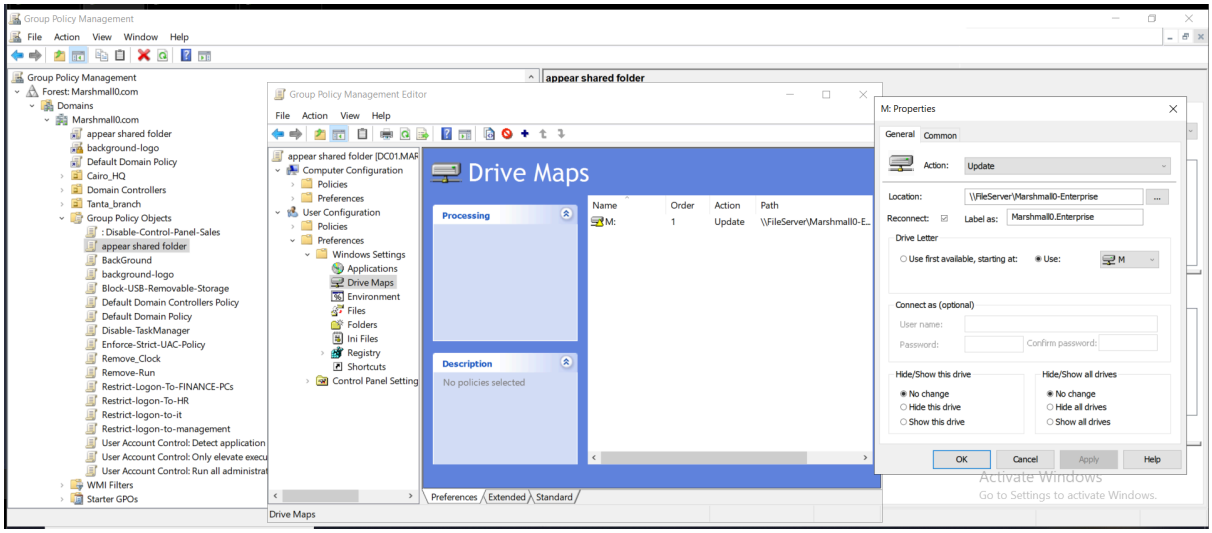
Phase 4: Automating Network Drive Mapping via Group Policy (From DC01)

1. **Open the Group Policy management tool from the menu: Group Policy Management (gpmc.msc).**
2. **Expand your domain, right-click the Default Domain Policy (or the specific OU containing your users), and select Edit.**
3. **Navigate through the following path in the Group Policy Management Editor:**

- **User Configuration**
- **$\rightarrow$ Preferences**
- **$\rightarrow$ Windows Settings**
- **$\rightarrow$ Drive Maps**

4. **Right-click anywhere in the empty space, select New $\rightarrow$ Mapped Drive.**
5. **In the configuration window (Common Tab):**
   - **Action: Choose Update (or Create).**
   - **Location: Enter the network share path using the format: \\DC01\TestShare.**
   - **Reconnect: Check the Reconnect option to ensure it remains persistently mapped.**
   - **Label as: Enter the display name visible to users, such as Company_Files or Shared Data.**
   - **Drive Letter: Choose a drive letter to display on client machines (e.g., Z:).**
6. **Click Apply then OK.**

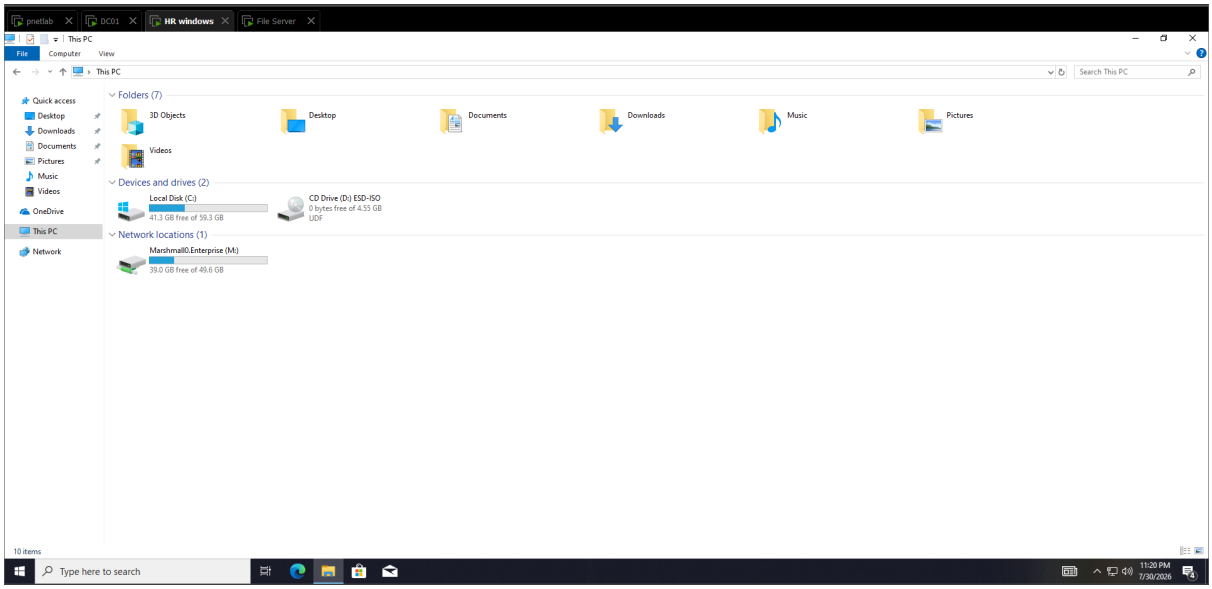




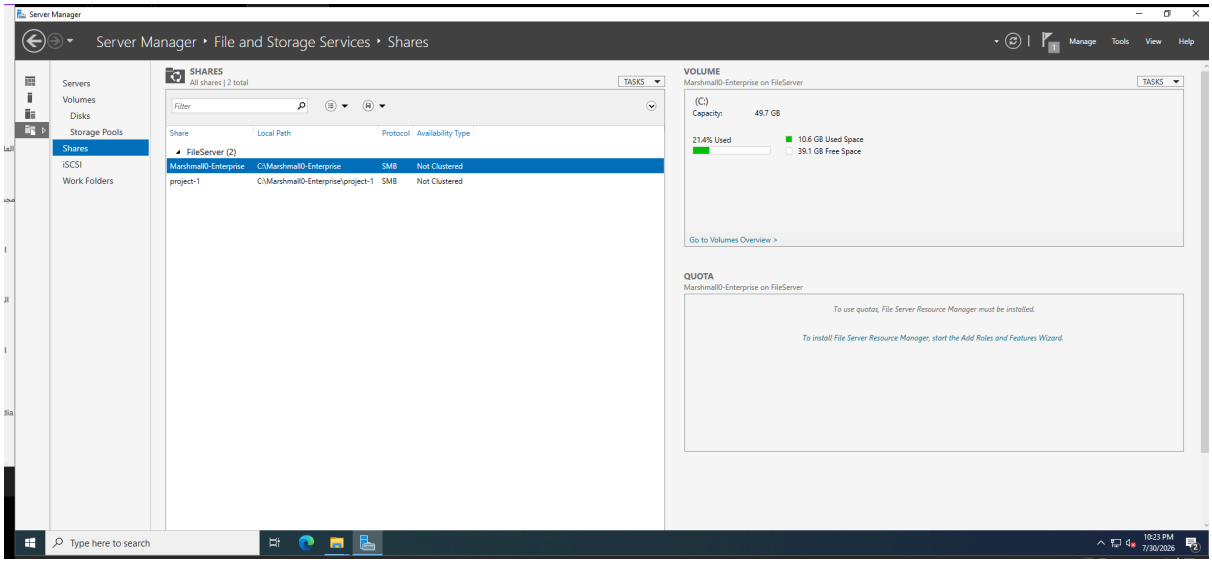
Phase 5: Verification & Testing

- **Seamless Access: Without typing paths or using the Run menu, whenever a domain user logs into their machine, they open This PC and immediately find a newly**

mapped network drive labeled with letter Z: containing all shared resources.



- **Central Control: Any modifications made to the storage structure or policies through the GPO apply instantly across all user environments without manual reconfiguration**



Enterprise Web Server Architecture & IIS Implementation Guide

1. Conceptual Overview

In an enterprise IT infrastructure, hosting web applications requires a robust, high-performance web server engine. Internet Information Services (IIS) serves as the native Microsoft web server role on Windows Server architectures.

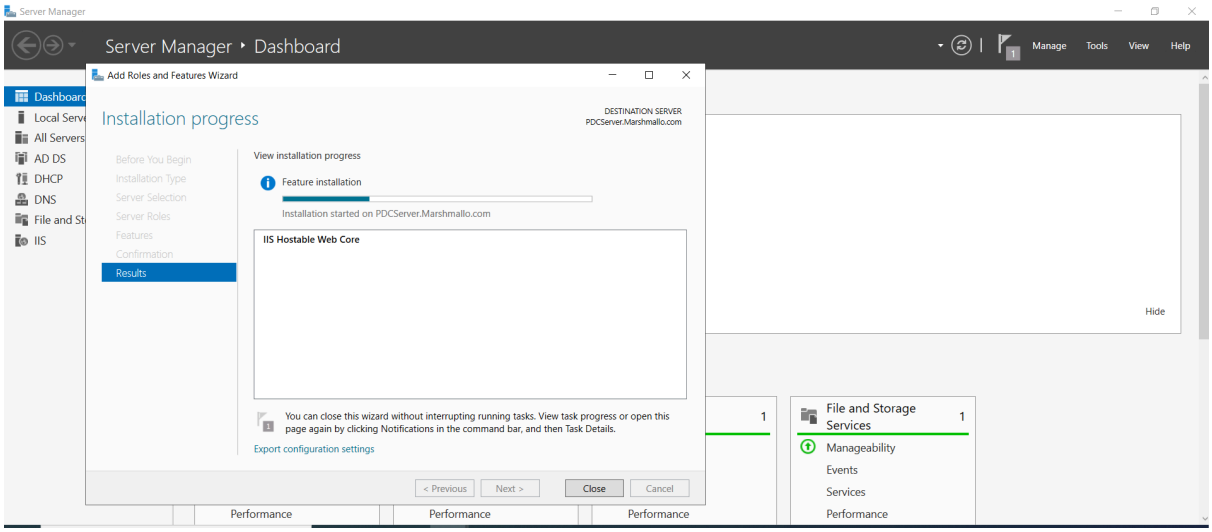
Core Concepts & Benefits:

- Request Processing Engine: IIS acts as the central engine responsible for listening to incoming HTTP/HTTPS requests from client browsers, processing them, and returning the appropriate web content (such as HTML pages, media, and application data).
- Modular Architecture: IIS supports various protocols, security configurations, and application pools, allowing administrators to isolate web applications and secure enterprise content.
- Domain & DNS Integration: By pairing IIS bindings with an internal DNS Server, domain names (e.g., www.marshmallO.com) are dynamically resolved to the server's IP address, allowing clients to access internal web resources seamlessly using human-readable URLs rather than raw IP addresses.

2. Practical Implementation Steps

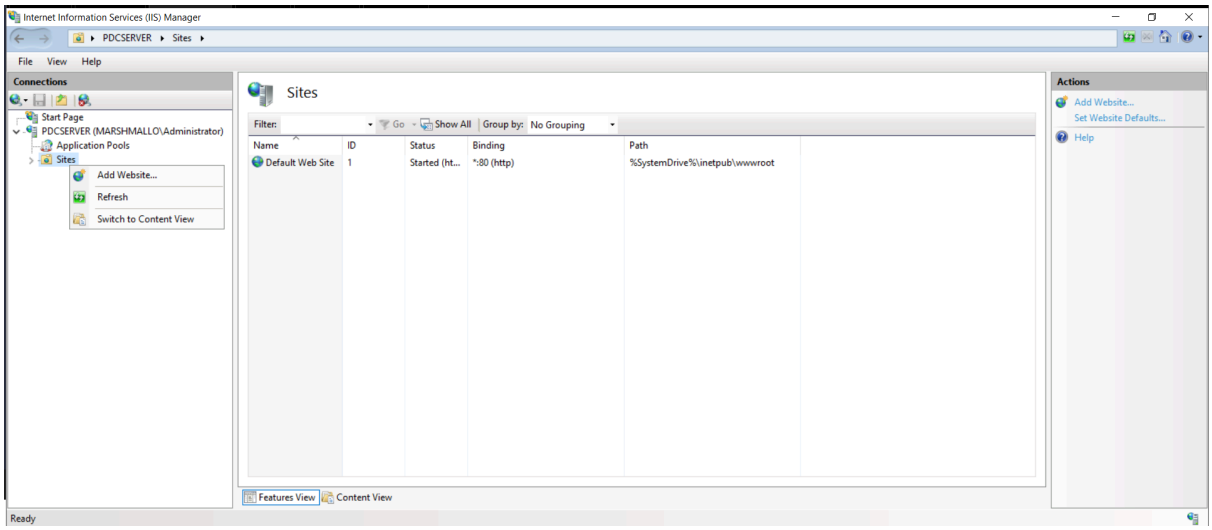
Phase 1: Installing the IIS Role

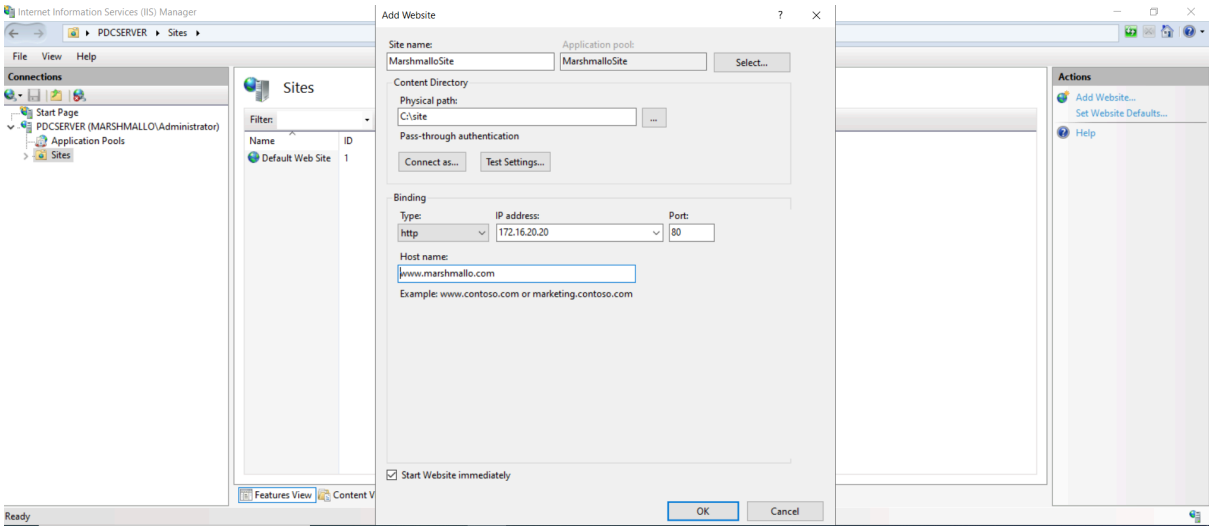
1. Open Server Manager on your server.
2. Click on Add roles and features and proceed by clicking Next until you reach the Server Roles page.
3. Locate the option Web Server (IIS) and check the box next to it, then click Add Features.
4. Continue clicking Next until you reach the Confirmation page, and click Install.
5. Wait for the installation to finish completely, then click Close.



3. Setting Up Website Files on the Server

1. Navigate to the default IIS storage directory on the server: C:\inetpub\wwwroot
2. Create a new folder inside wwwroot and name it: marshmallO-site
3. Place your logo image file inside this new folder, ensuring the filename is strictly in English without spaces (e.g., logo.png).
4. Open Notepad, paste your web page code (index.html), and save the file inside C:\inetpub\wwwroot\marshmallO-site with the exact name: index.html (making sure the extension does not save as .txt).



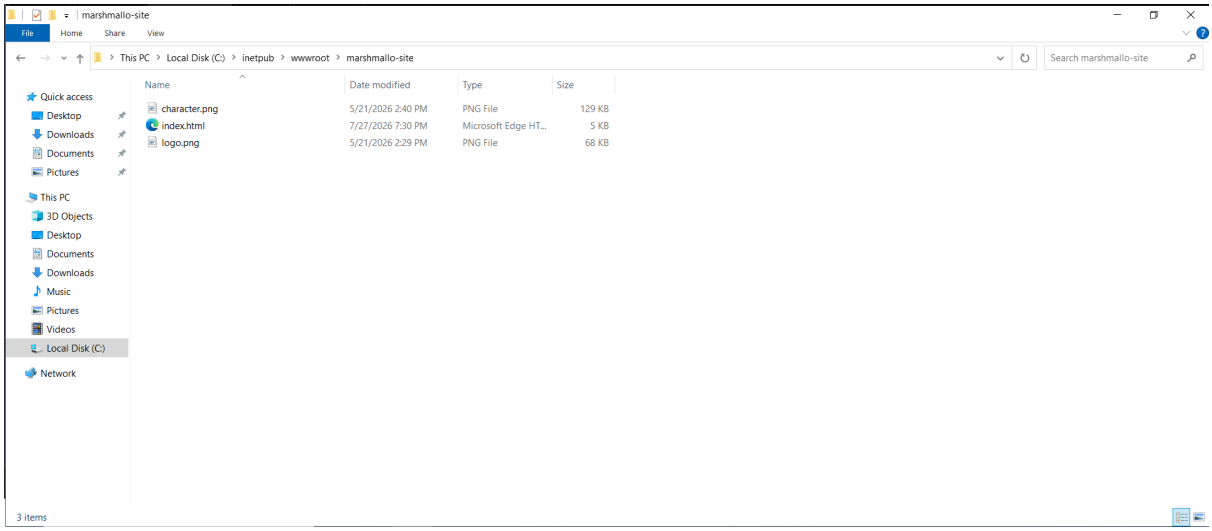


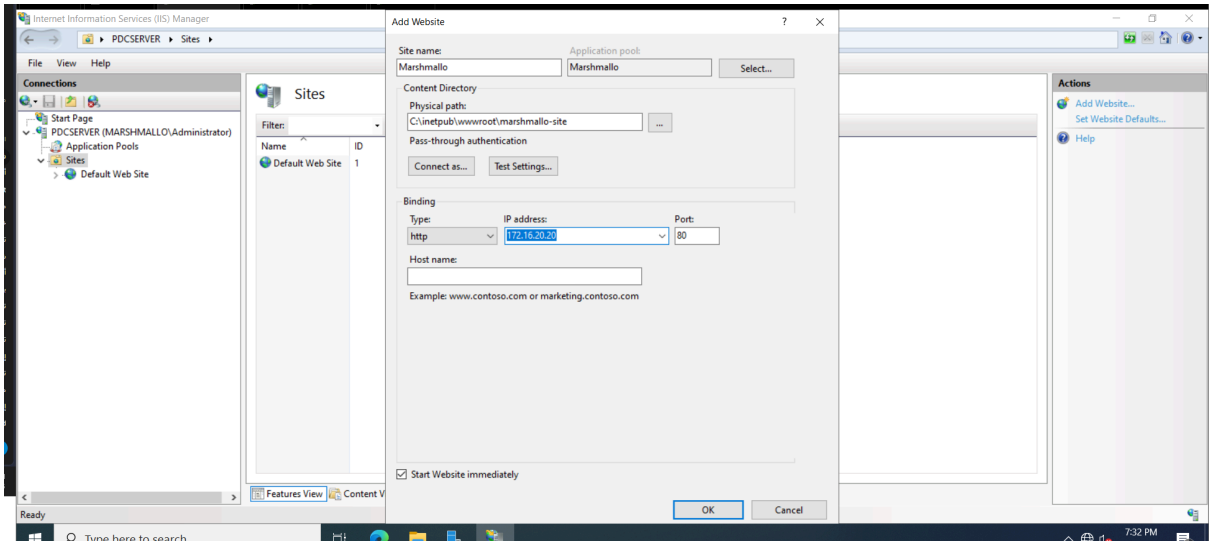
Phase 4: Configuring IIS and Modifying Site Bindings

1. From the Tools menu at the top of Server Manager, open Internet Information Services (IIS) Manager.
2. Expand your server node in the left-hand Connections pane, right-click Sites, and select Add Website...
3. Configure the site parameters as follows:
   - Site name: MarshmallO

- Physical path: Click the three dots (...) and select the folder created earlier: C:\inetpub\wwwroot\marshmallO-site
- Type: http
- IP address: All Unassigned (or your specific server IP)
- Port: 80 (or 8080 if port 80 is occupied by the default website)
- Host name: www.marshmallO.com

4. Click OK to save and create the website successfully.

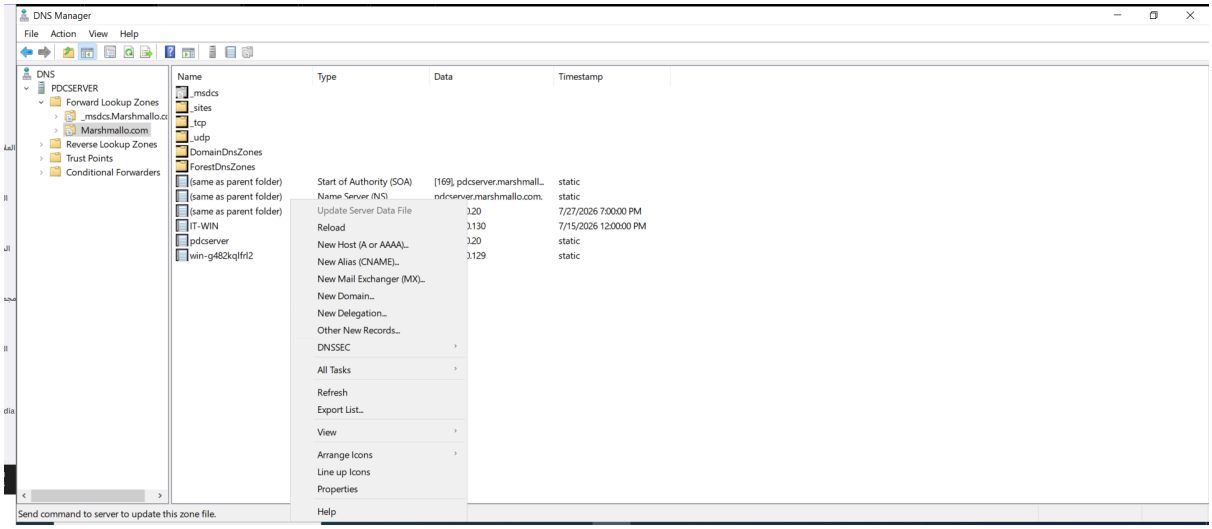


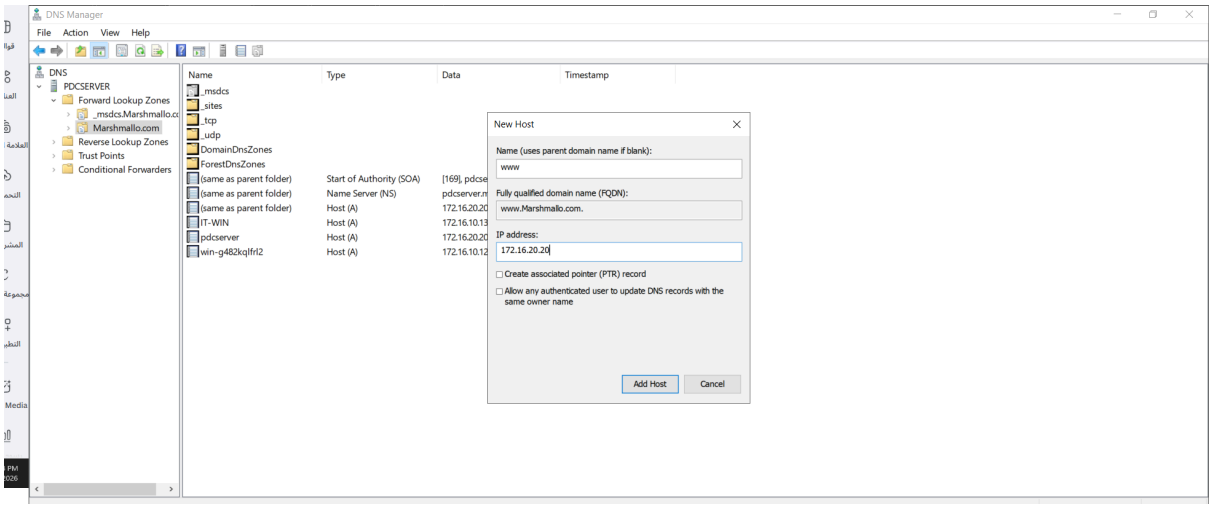


Phase 5: Configuring DNS Resolution on the DNS Server (DC01)

1. Open DNS Manager from Server Manager on the DNS server.
2. Expand your domain name under Forward Lookup Zones, right-click it, and select New Host (A or AAAA)...
3. In the Name field, enter: www (to bind it to www.marshmallO.com).
4. In the IP address field, enter the server's actual IP address (e.g., 172.16.20.20).

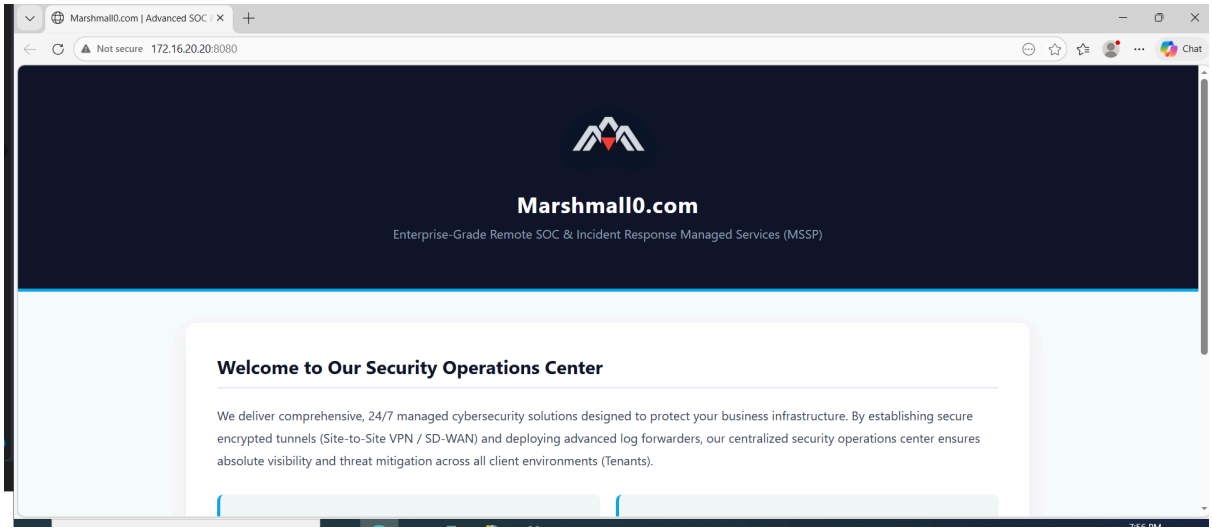
5. Click Add Host.

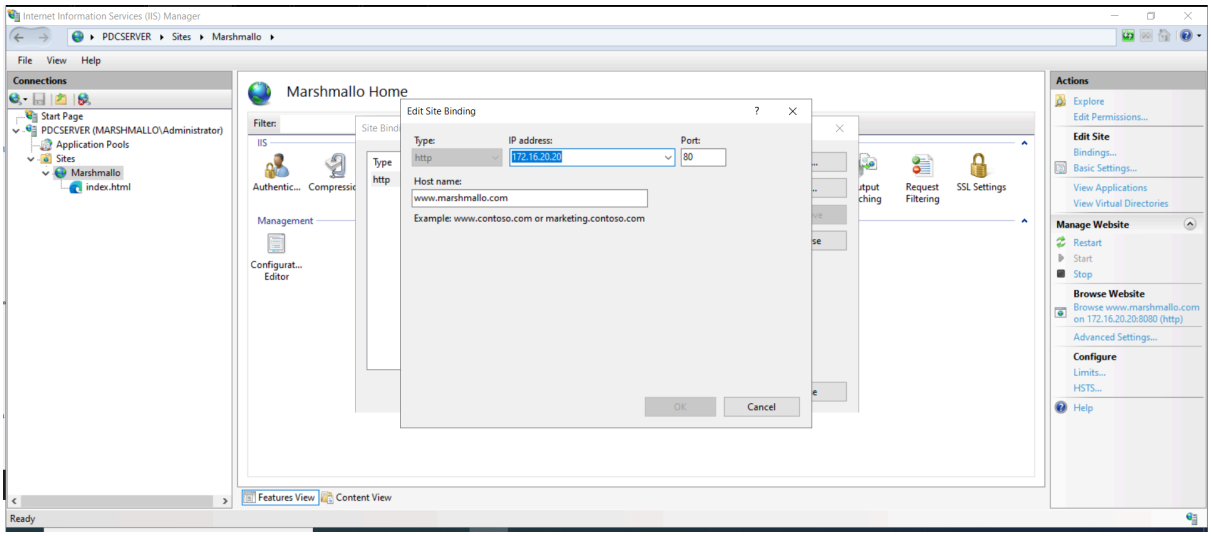


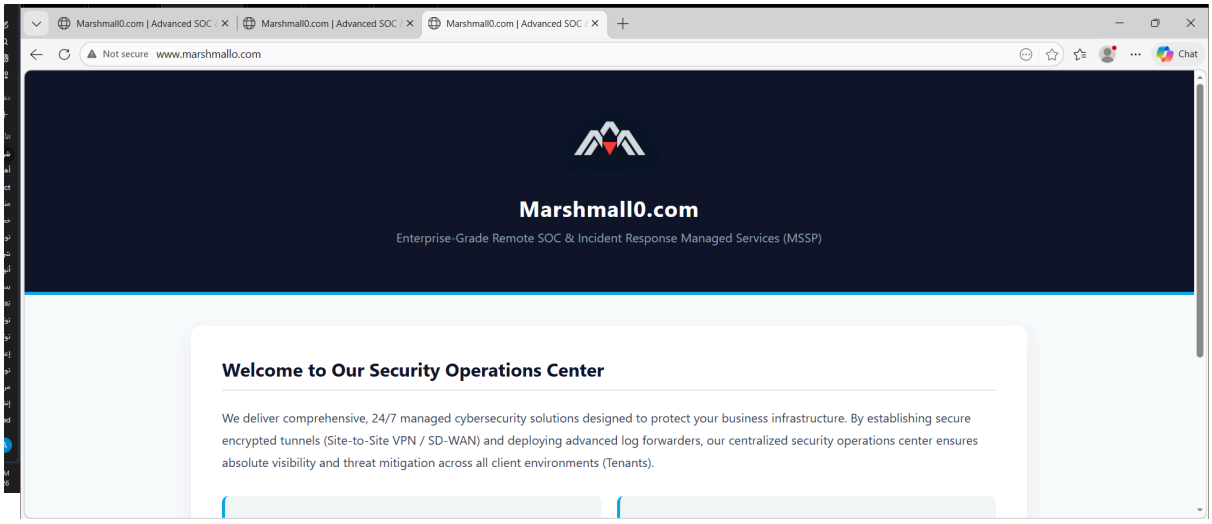


Phase 6: Verification & Testing

- Client Request Flow: When any client machine on the network opens a browser and types http://www.marshmall0.com:8080 (or http://www.marshmall0.com if bound to port 80), the client queries the internal DNS server for the IP address mapped to www.marshmall0.com.
- Resolution & Delivery: The DNS server replies with the server's IP address (172.16.20.20), routing the client directly to the IIS web server, which instantly renders the Marshmall0 website interface.







Importance of the Child Domain (General Overview)

- Decentralized Administration: Enables the division of a large network into smaller segments, allowing each branch or department to have its own

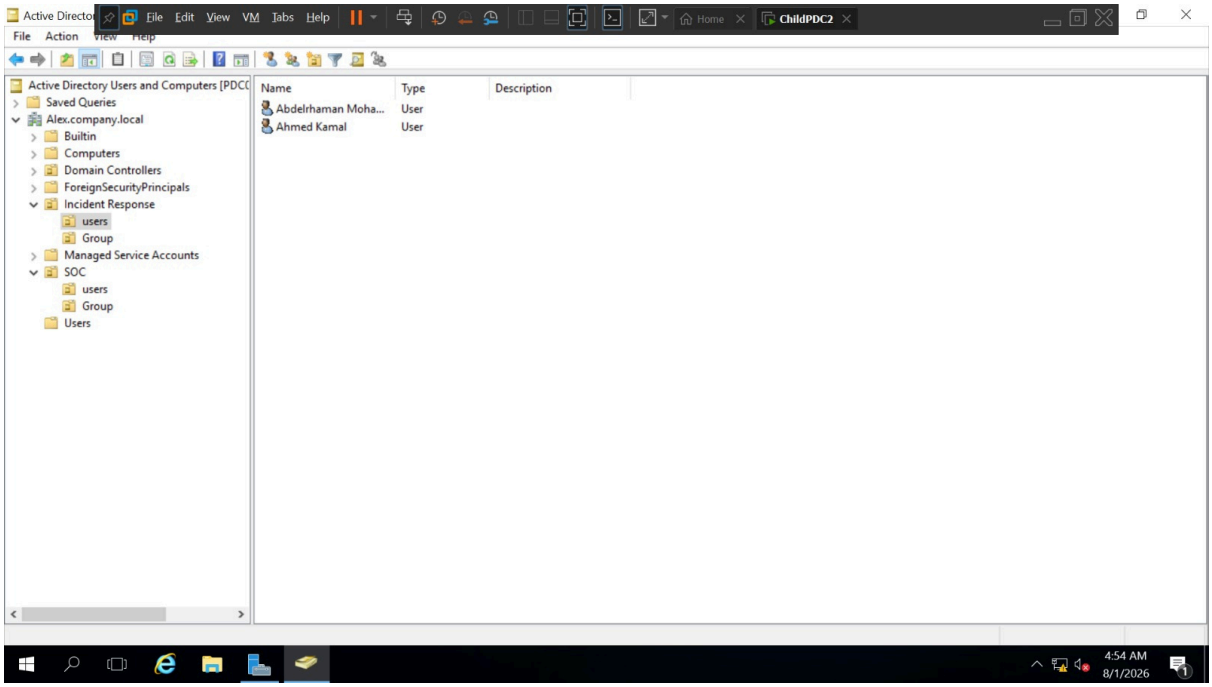
policies and permissions for managing users and devices without constantly relying on the main server (Parent Domain).
- Traffic Reduction: Lightens authentication pressure and network traffic on the primary server by handling client requests locally within the Child server.
- Organizational Structure Independence: Provides the ability to create a sub-domain name representing a specific geographical location or department (such as Alex.company.local) while maintaining linkage and trust with the parent organization (company.local).

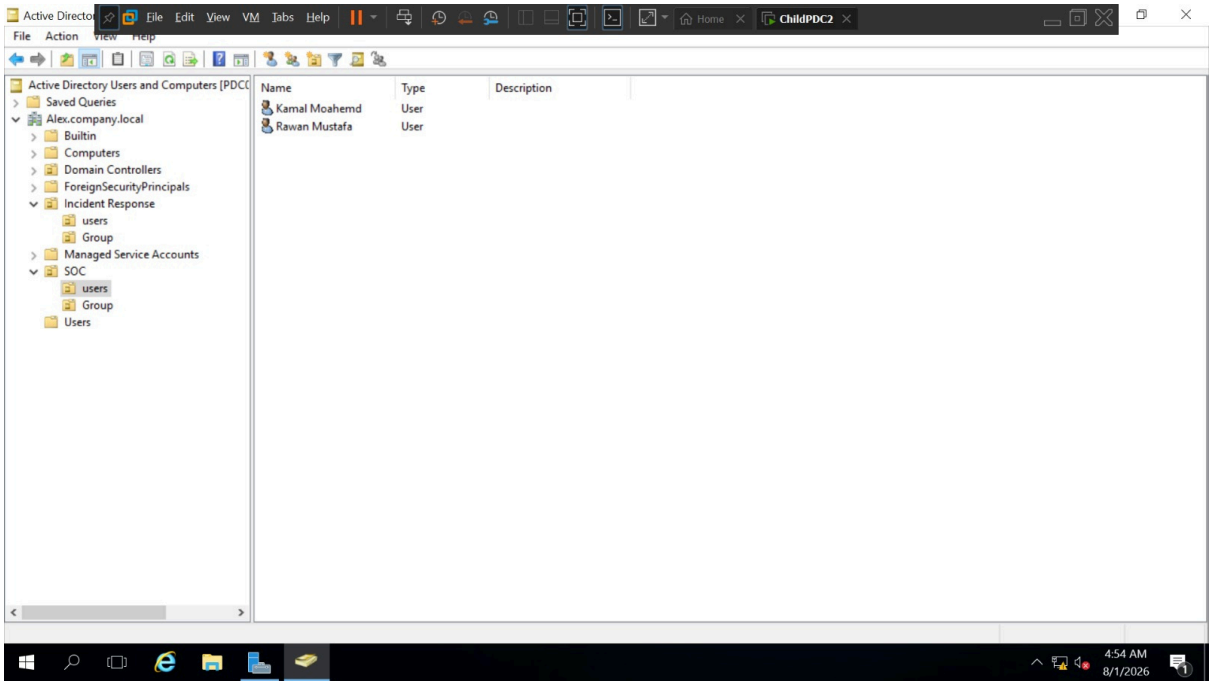
2. Importance of the Child Domain (Branch Level)

- Branch Operational Independence: Processes logon requests and services locally within the branch, ensuring business continuity and efficiency even if the connection to the main server encounters issues.
- Customized User Management: Organizes users from different departments (such as the Incident Response team and the SOC team) into dedicated Organizational Units (OUs) within the branch to simplify the enforcement of permissions and security policies.
- Integrated Local Services: Delivers essential network services (such as DNS and DHCP) directly within the branch to minimize latency.

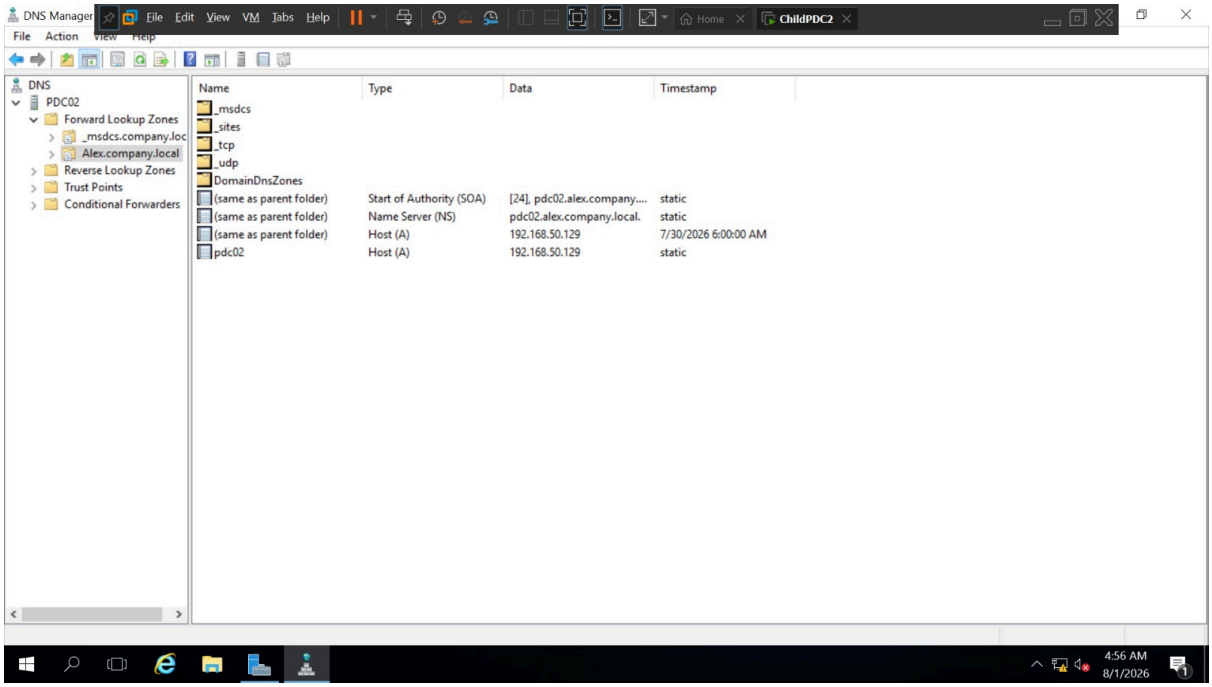
3. Practical Implementation Steps

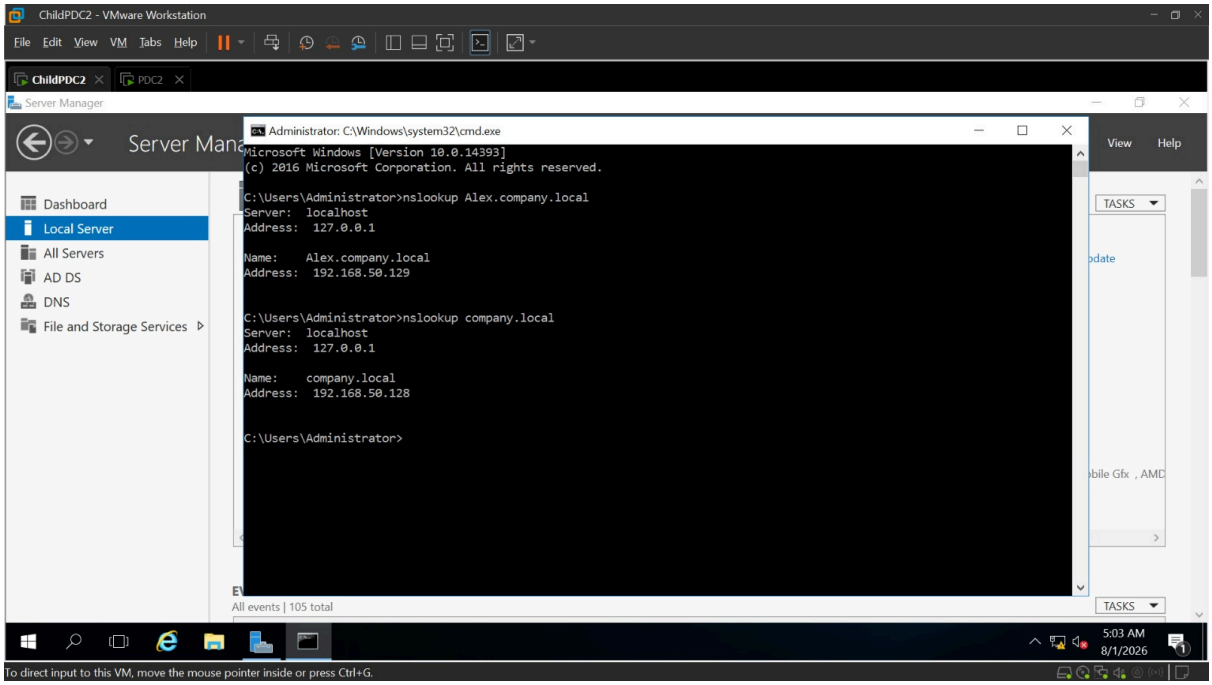
- Creating Organizational Units (OUs):
  - Organized departments and divided users into specialized OUs, such as Incident Response (containing users: Abdelrhman Mosaad and Ahmed Kamal) and SOC (containing users: Kamal Mohamed and Rawan Mustafa).

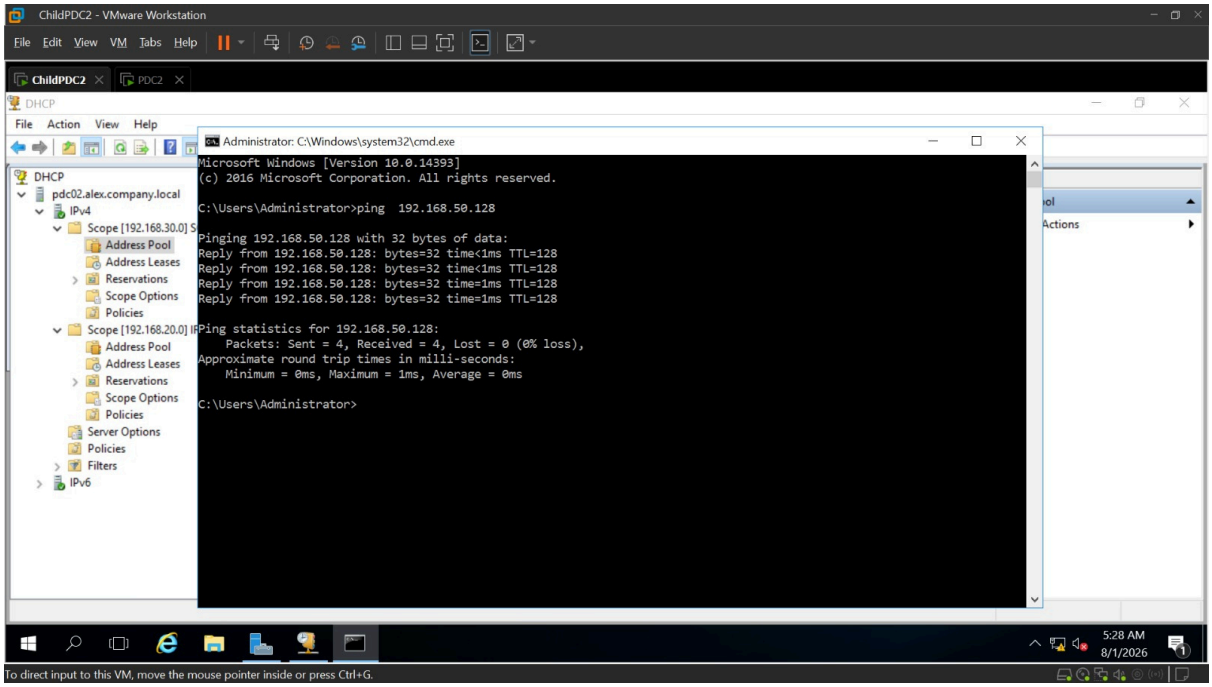




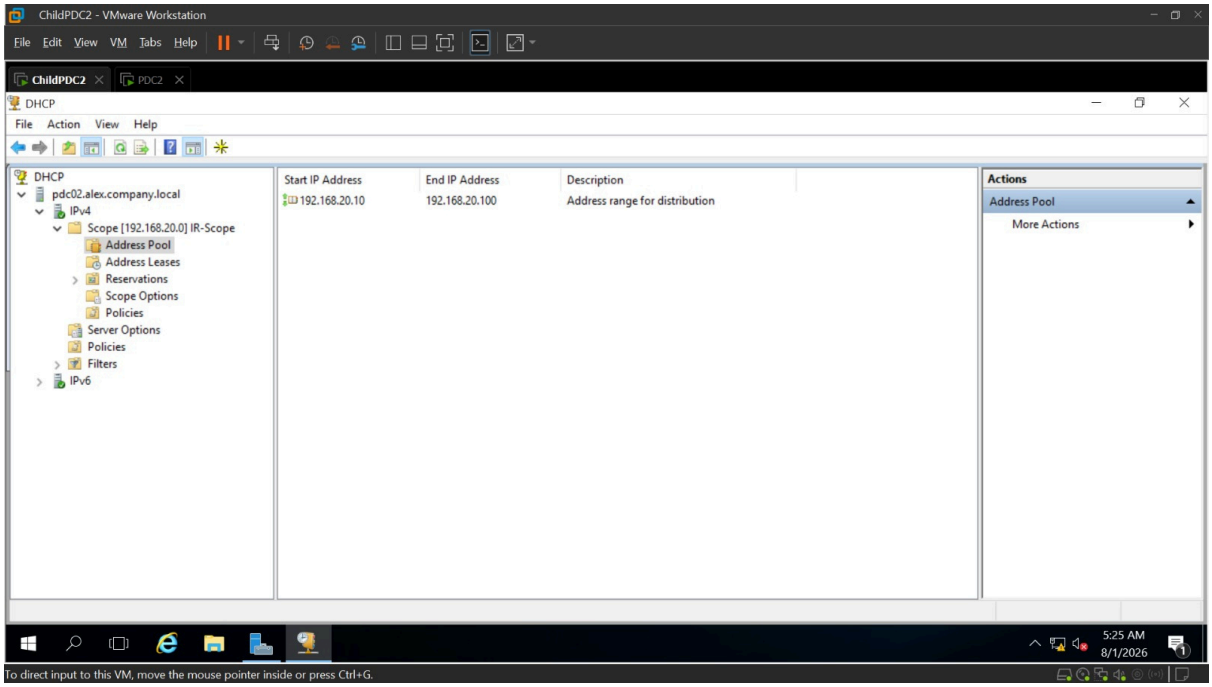
- DNS Manager:
  - Configured Forward Lookup Zones for the branch domain name resolution.

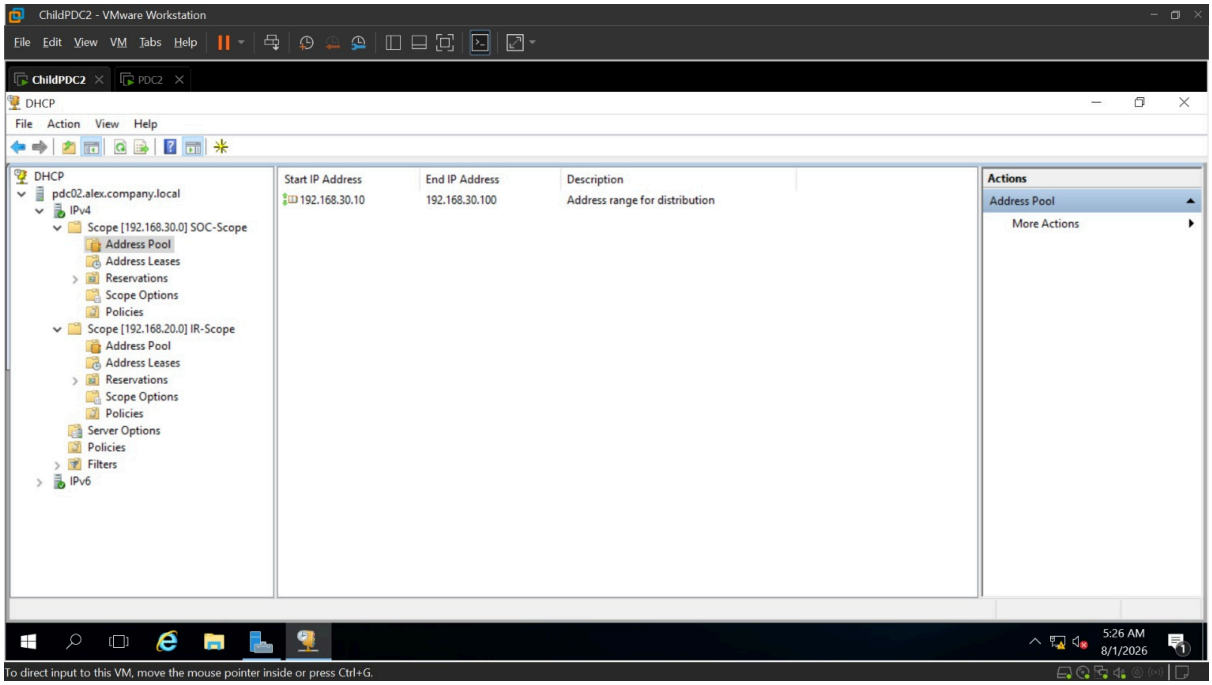

DNS Manager
File Edit View VM Tabs Help
ChildPDC2
File Action View Help
DNS
PDC02
Forward Lookup Zones
_msdcs.company.loc
Alex.company.local
Reverse Lookup Zones
Trust Points
Conditional Forwarders
Name
Type
Data
Timestamp
_msdcs
_sites
_tcp
_udp
DomainDnsZones
(same as parent folder)
Start of Authority (SOA)
[24], pdc02.alex.company....
static
(same as parent folder)
Name Server (NS)
pdc02.alex.company.local.
static
(same as parent folder)
Host (A)
192.168.50.129
7/30/2026 6:00:00 AM
pdc02
Host (A)
192.168.50.129
static
4:56 AM
8/1/2026


ChildPDC2 - VMware Workstation
File Edit View VM Tabs Help
ChildPDC2
PDC2
Server Manager
Server Mana
Dashboard
Local Server
All Servers
AD DS
DNS
File and Storage Services
View
Help
TASKS
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Users\Administrator>nslookup Alex.company.local
Server: localhost
Address: 127.0.0.1
Name: Alex.company.local
Address: 192.168.50.129
C:\Users\Administrator>nslookup company.local
Server: localhost
Address: 127.0.0.1
Name: company.local
Address: 192.168.50.128
C:\Users\Administrator>
All events | 105 total
TASKS
5:03 AM
8/1/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



- DHCP server :
  - Established two primary scopes to organize the branch network:
    - IR-Scope distributing IP addresses ranging from 192.168.20.10 to 192.168.20.100.
    - SOC-Scope distributing IP addresses ranging from 192.168.30.10 to 192.168.30.100.
  - Configured the default gateway router for them ending with .1.
- Connectivity between Child and PDC:
  - Performed a Ping connectivity test from the branch server to the main server's IP address (192.168.50.128), verifying successful communication.





4. Capabilities and Interactions of the Child Domain

- Global Catalog & Trust (Viewing Users and Groups):
  - Thanks to the Global Catalog, users and devices in the Child domain can view and search for user accounts and groups located in the Parent Domain (and vice versa), as long as permissions and policies allow it.
- Accessing Shared Resources:
  - Users in the Child Domain can access shared files or servers located on the Parent Domain network, governed by proper access

permissions.

- DNS Name Resolution:
  - The Child Domain fully recognizes the Parent Domain's address and vice versa (as demonstrated via the nslookup test where the branch server successfully queried the main domain and resolved its name to its IP address), allowing branch devices to communicate with the primary server by name rather than IP alone.
- Cross-Domain Authentication:
  - When a user from the Child domain attempts to access a resource located in the Parent domain, the Parent domain trusts the security token coming from the Child domain and grants access without requiring a separate account to be created on the primary server.